B.Tech.

IN

COMPUTER SCIENCE AND ENGINEERING
with Specialization in Cyber Security

CURRICULUM
AND
SYLLABI

(Applicable from 2026 Admission onwards)

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY KOTTAYAM

Pala – 686635, KERALA, INDIA

Table of Contents

Programme Educational Objectives (PEOs) 4

Programme Outcomes (POs) and PSOs 5

Curriculum 6

Programme Structure 11

Detailed Syllabus 16

SEMESTER I 16

MAC111 Calculus and Linear Algebra [3-1-0-4] . . . 16
CSC111 Computer Programming and Problem Solving [3-1-0-4] . . . 16
ECC111 Electronic Devices and Digital Logic Design [3-1-0-4] . . . 17
HMC111 Communication Skills for the Digital Age [3-0-0-3] . . . 18
CSC112 Foundations of Computing Systems [2-0-0-2] . . . 19
CSL113 IT Workshop Lab [0-1-2-2] . . . 20
CSL111 Computer Programming Lab [0-0-2-1] . . . 20
ECL111 Electronic Devices and Digital Logic Design Lab [0-0-2-1] . . . 21

SEMESTER II 21

IMA111 Discrete Mathematics [3-1-0-4] . . . 22
CSC121 Data Structures [3-1-0-4] . . . 22
CSC122 Computer Organization [3-1-0-4] . . . 23
HMC121 Personality Development & Soft Skills [2-1-0-3] . . . 23
CAC121 Artificial Intelligence [3-0-0-3] . . . 24
CSC123 Object Oriented Programming [2-1-0-3] . . . 25
CSL121 Data Structures lab (C, C++) [0-0-2-1] . . . 26
CSL123 Object Oriented Programming Lab [0-0-2-1] . . . 27

SEMESTER III 27

MAC211 Probability, Statistics and Random Processes [3-1-0-4] . . . 27
HMC211 Engineering Management, Entrepreneurship Development & start ups [2-0-0-2] . . . 28
CSC211 Computer Networks [2-1-0-3] . . . 29
CSC212 Design and Analysis of Algorithms [2-1-0-3] . . . 30
CSC213 Theory of Computation [3-1-0-4] . . . 30
CSC214 Operating Systems [3-0-0-3] . . . 31
CSL212 Algorithm Design Lab [0-0-2-1] . . . 32
CSL211 Computer Networks Lab [0-0-2-1] . . . 32
CSL214 Operating Systems Lab [0-0-2-1] . . . 33

SEMESTER IV 33

MAC223 Mathematical Primitives for Cyber Security [3-1-0-4] . . . 33
HMC221 Engineering Economics, Finance & Business Analytics [2-0-0-2] . . . 34
CAC221 Machine Learning and Data Analysis [2-1-0-3] . . . 35
CSC222 Language Processors [2-1-0-3] . . . 36
CSC223 Database Management Systems [2-1-0-3] . . . 36
CSC224 Advanced Data Structures [2-1-0-3] . . . 37
CSL222 Language Processors Lab [0-0-2-1] . . . 37
CSL223 Database Management Systems Lab [0-0-2-1] . . . 38
CAL221 Machine Learning and Data Analysis Lab [0-0-2-1] . . . 39
CYC221 Introduction to Cybersecurity and Cybercrimes [3-0-0-3] . . . 40

SEMESTER V 40
CYC311 Modern Cryptography [3-0-0-3] . . . 40
CYC312 Secure Software Engineering and DevSecOps [2-1-0-3] . . . 41
CYC313 Cloud and Critical Infrastructure Security [3-0-0-3] . . . 42
CYC314 IoT and Cyber-Physical Systems Security [3-0-0-3] . . . 43
CYI311 Computer Forensics [0-1-2-2] . . . 44
CYI312 Artificial Intelligence for Cyber Defense [0-1-2-2] . . . 45
CYL311 Cloud Computing and IoT Systems Lab [0-0-2-1] . . . 46

SEMESTER VI 47
CYC321 Ethical Hacking [3-0-0-3] . . . 47
IOEC XXX Design Thinking and Innovation [2-1-0-3] . . . 48
CYI321 Threat Intelligence and Security Operations Centre(SOC) [0-1-2-2] . . . 49
CYI322 Smart Phone Forensics and Application Security [0-1-2-2] . . . 50
CYL321 Ethical Hacking Lab [0-0-2-1] . . . 50

SEMESTER VII 51
CYC411 Cyber Warfare,Adversarial Tactics and Resilience [2-1-0-3] . . . 52

Program Electives 53
CYE001 Cybersecurity Governance, Risk, and Compliance [2-1-0-3] . . . 53
CYE002 Criminal Psychology [3-0-0-3] . . . 53
CYE003 Secure and Privacy-Preserving Machine Learning [3-0-0-3] . . . 54
CYE004 Large Language Models Security and Prompt Engineering [3-0-0-3] . . . 55
CYE005 Distributed and Federated Learning Security [3-0-0-3] . . . 55
CYE006 Secure and Resilient UAV Systems [3-0-0-3] . . . 56
CYE007 Hardware and Embedded System Security [3-0-0-3] . . . 57
CYE008 Advanced Network Security and Zero Trust Architecture [3-0-0-3] . . . 58
CYE009 Next Generation Multimedia and Generative AI Security [3-0-0-3] . . . 59
CYE010 Cyber Law and Digital Ethics [3-0-0-3] . . . 60
CYE011 Secure and Trustworthy Artificial Intelligence [3-0-0-3] . . . 61
CYE012 Quantum Computing & Post Quantum Cryptography [3-0-0-3] . . . 62
CYE013 Blockchain and Web3 Security [3-0-0-3] . . . 63
CYE014 Lightweight Cryptograpy [3-0-0-3] . . . 63
CYE015 Secure Mobile Computing [3-0-0-3] . . . 65

Open Electives 66
CYO001 Financial Cyber Crimes and Fraud Analytics [2-1-0-3] . . . 66
CYO002 Malware Analysis and Reverse Engineering [3-0-0-3] . . . 67
CYO003 Dark Web Intelligence and Investigation [3-0-0-3] . . . 68
CYO004 Web Application Security [3-0-0-3] . . . 69
CYO005 Agentic AI for Cyber Security [3-0-0-3] . . . 70

Programme Educational Objectives (PEOs) of B.Tech. in COMPUTER SCIENCE AND ENGINEERING with Specialization in Cyber Security

PEO1	**Technical Leadership and Resilient Systems Architecting:** Graduates will establish themselves as successful cybersecurity professionals and technical leaders capable of conceptualizing and designing resilient systems and secure software architectures. Leveraging advanced investigative analysis and modern cybersecurity tools, they will apply adversarial tactics and cyber warfare principles to proactively harden complex ecosystems.
PEO2	**Ethical Governance, Continuous Innovation, and Professional Excellence:** Graduates will demonstrate professional excellence and accountability by integrating ethical principles, cyber laws, and governance, risk, and compliance (GRC) frameworks into technical and organizational leadership. They will engage in applied research and continuous learning to dynamically adapt to rapidly shifting security paradigms, seamlessly navigating next-generation challenges.
PEO3	**Human-Centric Security and Public Welfare:** Graduates will champion digital safety and societal well-being by developing inclusive, human-centric security architectures. They will leverage advanced investigative methodologies to mitigate digital threats against communities, ensuring the protection of vulnerable populations and fostering enduring public trust in technological ecosystems.

Programme Outcomes (POs) and Programme Specific Outcomes (PSOs) of B.Tech. in COMPUTER SCIENCE AND ENGINEERING with Specialization in Cyber Security

PO1	**Engineering and Mathematical Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and computing to conceptualize and solve complex problems in computer science and cybersecurity.
PO2	**Secure System Design and Development** Identify, formulate, and analyse complex security challenges using research literature and research-based methodologies to design resilient systems, secure software architectures, and robust network protocols that meet specific industry and societal needs.
PO3	**Modern Tool Usage and Investigative Analysis:** Create, select, and apply appropriate techniques, resources, and modern cybersecurity tools including advanced analytical, testing, and investigative utilities to analyse and neutralize advanced security threats.
PO4	**Professional Ethics, Governance, and Communication:** Apply ethical principles, legal mandates, and compliance frameworks to professional responsibilities, while communicating complex technical concepts effectively to both technical teams and organizational leadership.
PSO1	**Offensive Security, Forensics, and Threat Operations:** Demonstrate proficiency in executing the complete penetration testing lifecycle, conducting sophisticated digital and smartphone forensics, and managing Security Operations Centre (SOC) workflows leveraging Artificial Intelligence (AI) to proactively hunt, analyse, and neutralize Advanced Persistent Threats (APTs) and malware.
PSO2	**Next-Generation Security Architecture and Cyber Resilience:** Design, implement, and evaluate advanced security mechanisms for complex, emerging ecosystems spanning cyber-physical infrastructures, decentralized networks, and intelligent AI-driven platforms by applying adversarial tactics, threat emulation, and cyber warfare principles to proactively harden architectures.

CURRICULUM

Total credits for completing B.Tech in Computer Science And Engineering with Specialization in Cyber Security is 160. (In addition, 10 mandatory Activity Credits must be earned; these are not counted towards the CGPA.)

COURSE CATEGORIES AND CREDIT REQUIREMENTS
The structure of the B.Tech. programme shall have the following course categories:

Sl. No.	Course Category	Number of Courses	Minimum Credits
1	Institute Core (IC)	34	87
2	Programme Core (PC)	7 Theory + 6 Lab	32
3	Programme Electives (PE)	6	18
4	Open Electives (OE)	4	11
5	Projects (P)	2	12
		Total credits (counted for CGPA)	**160**
6	Activity Credits (AC) – *mandatory, not counted for CGPA*	–	10

COURSE REQUIREMENTS. The effort to be put in by the student is indicated in the tables as follows:
L: Lecture (one unit is of 50 minutes) **T**: Tutorial (one unit is of 50 minutes) **P**: Practical (one unit is of one hour)

1. Institute Core (IC)

(a) Mathematics

Sl.	Code	Course Title	L	T	P	Credits
1	MAC111	Calculus and Linear Algebra	3	1	0	4
2	MAC121	Discrete Mathematics	3	1	0	4
3	MAC211	Probability, Statistics, and Random Processes	3	1	0	4
4	MAC223	Mathematical Primitives for Cyber Secuirty	3	1	0	4
		Total	**12**	**4**	**0**	**16**

(b) Economics, Management & Entrepreneurship

Sl.	Code	Course Title	L	T	P	Credits
1	HMC211	Engg. Management, Entrepreneurship Development & Start-ups	2	0	0	2
2	HMC221	Engg. Economics, Finance & Business Analytics	2	0	0	2
		Total	**4**	**0**	**0**	**4**

(c) Communication Skills and Personality Development

Sl.	Code	Course Title	L	T	P	Credits
1	HMC111	Communication Skills for the Digital Age	2	1	0	3
2	HMC121	Personality Development & Soft Skills	3	0	0	3
		Total	**5**	**1**	**0**	**6**

(d) Engineering Foundation Courses

Sl.	Code	Course Title	L	T	P	Credits
1	CSC111	Computer Programming & Problem Solving	3	1	0	4
2	ECC111	Electronic Circuits and Digital Logic Design	3	1	0	4
3	CSL111	Computer Programming Lab	0	0	2	1
4	CSC1112	Foundations of Computing Systems	2	0	0	2
5	CSL113	IT Workshop Lab	0	1	2	2
6	ECL111	Electronic Circuits and Digital Logic Design Lab	0	0	2	1
7	CSC121	Data Structures	3	1	0	4
10	CSC122	Computer Organization	3	1	0	4
8	CAC121	Artificial Intelligence	3	0	0	3
11	CSC123	Object Oriented Programming	2	1	0	3
12	CSL121	Data Structures Lab (C, C++)	0	0	2	1
13	CSL123	Object Oriented Programming Lab	0	0	2	1
14	CSC211	Computer Networks	3	0	0	3
15	CSL212	Design and Analysis of Algorithms	2	1	0	3
16	CSL213	Theory of Computation	3	1	0	4
17	CSL214	Operating Systems	3	0	0	3
18	CSL212	Algorithm Design Lab	0	0	2	1
19	CSL211	Computer Networks Lab	0	0	2	1
20	CSL214	Operating Systems Lab	0	0	2	1
21	CAC221	Machine Learning and Data Analysis	2	1	0	3
22	CSC222	Language Processor	2	1	0	3
23	CSC223	Database Management Systems	2	1	0	3
24	CSC224	Advanced Data Structures	2	1	0	3
25	CSL222	Language Processor Lab	0	0	2	1
26	CSL223	Database Management Systems Lab	0	0	2	1
27	CAL222	Machine Learning and Data Analysis Lab	0	0	2	1
		Total	**38**	**12**	**22**	**61**

2. Programme Core (PC)

Sl.	Code	Course Title	L	T	P	Credits
1	CYC221	Introduction to Cybersecurity and Cybercrimes	3	0	0	3
2	CYC311	Modern Cryptography	3	1	0	4
3	CYC312	Secure Software Engineering and DevSecOps	2	1	0	3
4	CYC313	Cloud and Critical Infrastructure Security	3	0	0	3
5	CYC314	IoT and Cyber-Physical Systems Security	3	0	0	3
6	CYI311	Computer Forensics	0	1	2	2
7	CYI312	Artificial Intelligence for Cyber Defense	0	1	2	2
8	CYL311	Cloud Computing and IoT Systems Lab	0	0	2	1
9	CYC321	Ethical Hacking	3	0	0	3
10	CYI321	Threat Intelligence and Security Operations Centre(SOC)	0	1	2	2
11	CYI322	Smart Phone Forensics and Application Security	0	1	2	2
12	CYL321	Ethical Hacking Lab	0	0	2	1
13	CYC411	Cyber Warfare,Adversarial Operations and Defensive Architectures	2	1	0	3
		Total	20	6	12	32

3. List of Programme Electives (PE)

Sl.	Code	Course Title	L	T	P	Credits
1	CYE001	Cybersecurity Governance, Risk, and Compliance	3	0	0	3
2	CYE002	Criminal Psychology	3	0	0	3
3	CYE003	Secure and Privacy-Preserving Machine Learning	3	0	0	3
4	CYE004	Large Language Models Security and Prompt Engineering,	3	0	0	3
5	CYE005	Distributed and Federated Learning Security	3	0	0	3
6	CYE006	Secure and Resilient UAV Systems	3	0	0	3
7	CYE007	Hardware and Embedded System Security	3	0	0	3
8	CYE008	Advanced Network Security and Zero Trust Architecture	3	0	0	3
9	CYE009	Next generation Multimedia and Generative AI Security	3	0	0	3
10	CYE010	Cyber Law and Digital Ethics	3	0	0	3
11	CYE011	Secure and Trustworthy Artificial Intelligence	3	0	0	3
12	CYE012	Quantum Computing and Post Quantum Cryptography	3	0	0	3
13	CYE013	Blockchain and Web3 Security	3	0	0	3
14	CYE014	Lightweight Cryptography	3	0	0	3
15	CYE015	Secure Mobile Computing	3	0	0	3

4. List of Open Electives (OE)

Courses offered by other departments or approved online platforms (e.g. NPTEL), subject to the maximum permitted under the B.Tech. Ordinances and Regulations. PE courses offered by the parent department are also available to its own students under this category.

Sl.	Code	Course Title	L	T	P	Credits
1	CYO001	Financial Cyber Crimes and Fraud Analytics	2	1	0	3
2	IOECXXX	Foreign Language (NPTEL)	2	0	0	2
3	IOECXXX	Design Thinking & Innovation	3	0	0	3
4	CYO002	Malware Analysis and Reverse Engineering	3	0	0	3
5	CYO003	Dark Web Intelligence and Investigation	3	0	0	3
6	CYO004	Web Application Security	3	0	0	3
7	CYO005	Agentic AI for Cyber Security	3	0	0	3

5. Projects (P)

The major project represents the culmination of study towards the B.Tech. degree. It is offered in two parts – Project-1 and Project-2 – and gives students the opportunity to apply and extend the knowledge gained across the programme. The work may be analytical, simulation-based, hardware design, or a combination, in an emerging area of COMPUTER SCIENCE AND ENGINEERING with Specialization in Cyber Security, carried out under the supervision of a department faculty member. Project work may be carried out individually or in groups (size decided by the department) and may build on an industrial problem identified during an internship.

6. Activity Credits (AC)

A minimum of 40 Activity Points must be earned to obtain the 10 credits required in the curriculum; these credits are not counted towards CGPA. The scheme of activity points, as approved by the Senate, is announced at the start of the first semester. A minimum of 20 points must be earned by the end of the 4th semester and 40 points by the end of the 7th semester.

PROGRAMME STRUCTURE

Semester-wise Course Structure (I–VIII)

Semester I						**Semester II**					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
MAC111	Calculus and Linear Algebra	3	1	0	4	MAC121	Discrete Mathematics	3	1	0	4
CSC111	Computer Programming and Problem solving	3	1	0	4	CSC121	Data Structures	3	1	0	4
ECC111	Electronic Circuits and Digital Logic Design	3	1	0	4	CSC122	Computer Organization	3	1	0	4
HMC111	Communication Skills for the Digital Age	3	0	0	3	HMC121	Personality Development and Soft Skills	2	1	0	3
CSC112	Foundations of Computing Systems	2	0	0	2	CAC121	Artificial Intelligence	3	0	0	3
CSL111	Computer Programming Lab	0	0	2	1	CSC123	Object Oriented Programming	2	1	0	3
CSL113	IT Workshop Lab	0	1	2	2	CSL121	Data Structures lab(C,C++)	0	0	2	1
ECL111	Electronic Circuits and Digital Logic Design Lab	0	0	2	1	CSL123	Object Oriented Programming Lab	0	0	2	1
Total		15	3	6	21	**Total**		16	5	4	23
Cumulative Credits at the End of First Year:44											
Semester III						**Semester IV**					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
MAC211	Probability, Statistics, and Random Processes	3	1	0	4	MAC223	Mathematical Primitives for Cyber Security	3	1	0	4
HMC211	Engineering Management, Entrepreneurship Development and start ups	2	0	0	2	HMC221	Engg Economics, Finance and Business Analytics	2	0	0	2
CSC211	Computer Networks	2	1	0	3	CAC221	Machine Learning and Data Analysis	2	1	0	3
CSC212	Design and Analysis of Algorithms	2	1	0	3	CSC222	Language Processor	2	1	0	3
CSC213	Theory of Computation	3	1	0	4	CSC223	Database Management Systems	2	1	0	3
CSC214	Operating Systems	3	0	0	3	CSC224	Advanced Data Structures	2	1	0	3
CSL212	Algorithm Design Lab	0	0	2	1	CSL222	Language Processor Lab	0	0	2	1
CSL211	Computer Networks Lab	0	0	2	1	CSL223	Database Management Systems Lab	0	0	2	1
CSL214	Operating Systems Lab	0	0	2	1	CAL221	Machine Learning and Data Analysis Lab	0	0	2	1
						CYC221	Introduction to Cybersecurity and Cybercrimes	3	0	0	3
Total		15	4	6	22	**Total**		16	5	6	24
Cumulative Credits at the End of Second Year:90											
Semester V						**Semester VI**					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
CYC311	Modern Cryptography	3	1	0	4	CYC321	Ethical Hacking	3	0	0	3
CYC312	Secure Software Engineering and DevSecOps	2	1	0	3	CYEXXX	Deparment Elective-I	3	0	0	3
CYOXXX	Dept Open Elective-I	2	1	0	3	CYEXXX	Deparment Elective-II	3	0	0	3
CYC313	Cloud and Critical Infrastructure Security	3	0	0	3	IOECXXX	Foreign Language(NPTEL)	2	0	0	2
CYC314	IoT and Cyber-Physical Systems Security	3	0	0	3	IOECXXX	Design Thinking and Innovation	3	0	0	3
CYI311	Computer Forensics	0	1	2	2	CYEXXX	Deparment Elective-III	3	0	0	3
CYI312	Artificial Intelligence for Cyber Defense	0	1	2	2	CYI321	Threat Intelligence and Security Operations Centre(SOC)	0	1	2	2
CYL311	Cloud Computing and IoT Systems Lab	0	0	2	1	CYI322	Smart Phone Forensics and Application Security	0	1	2	2
						CYL321	Ethical Hacking Lab	0	0	2	1
							Advanced course-1 for Hon.	3	0	0	3
Total		13	5	6	21	**Total**		17	2	6	22
Cumulative Credits at the End of Third Year:133											

Semester VII						Semester VIII					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
CYOXXX	Dept Open Elective-II	3	0	0	3	CYP421	B.Tech Project Phase-II	0	0	18	9
CYEXXX	Deparment Elective-IV	3	0	0	3		Hons. Project Phase-II	0	0	22	11
CYEXXX	Deparment Elective-V	3	0	0	3						
CYEXXX	Deparment Elective-VI	3	0	0	3						
CYC411	Cyber Warfare,Adversarial Operations and Defensive Architectures	2	1	0	3						
	Advanced course-2 for Hon(NPTEL)	3	0	0	3						
CYP411	B.Tech Project Phase-I	0	0	6	3						
	Hons.Project Phase- 1	0	0	6	3						
Total		15	0	6	18	**Total**		0	0	18	9
Total Credits at the End of Fourth Year:**160** (excluding Activity Credits) 160+20=180 (B.Tech(Hon.));Activity points:40											

Semester XI						Semester X					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
	Research Project-1	0	0	16	8		Research Project-2	0	0	24	12
Total		0	0	16	8	**Total**		0	0	24	12
Total Cumulative Credits at the End of Fifth Year:180+20=200(B.Tech-MS)											

Detailed Semester-wise Structure

Semester I

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC111	Calculus and Linear Algebra	3	1	0	4	IC
2	CSC111	Computer Programming & Problem Solving	3	1	0	4	IC
3	ECC111	Electronic Circuits and Digital Logic Design	3	1	0	4	IC
4	HMC111	Communication Skills for the Digital Age	3	0	0	3	IC
5	CSC112	Foundations of Computing Systems	2	0	0	2	IC
6	CSL111	Computer Programming Lab	0	0	2	1	IC
7	CSL113	IT Workshop Lab	0	1	2	2	IC
8	ECL111	Electronic Circuits and Digital Logic Design Lab	0	0	2	1	IC
		Total	15	3	6	21	

Semester II

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC121	Discrete Mathematics	3	1	0	4	IC
2	CSC121	Data Structures	3	1	0	4	IC
3	CSC122	Computer Organization &Architecture	3	1	0	4	IC
4	HMC121	Personality Development and Soft Skills	2	1	0	3	IC
4	CAC121	Artificial Intelligence	3	0	0	3	IC
6	CSC123	Object Oriented Programming	2	1	0	3	IC
7	CSL121	Data Structures lab(C,C++)	0	0	2	1	IC
8	CSL123	Object Oriented Programming Lab	0	0	2	1	IC
		Total	16	5	4	23	

Semester III

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC211	Probability, Statistics, and Random Processes	3	1	0	4	IC
2	HMC211	Engineering Management, Entrepreneurship Development and start ups	2	0	0	2	IC
3	CSC211	Computer Networks	2	1	0	3	IC
4	CSC212	Design and Analysis of Algorithms	2	1	0	3	IC
5	CSC213	Theory of Computation	3	1	0	4	IC
6	CSC214	Operating Systems	3	0	0	3	IC
7	CSL212	Algorithm Design Lab	0	0	2	1	IC
8	CSL211	Computer Networks Lab	0	0	2	1	IC
9	CSL214	Operating Systems Lab	0	0	2	1	IC
		Total	15	4	6	22	

Semester IV

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC223	Mathematical Primitives for Cyber Security	3	1	9	4	IC
2	HMC221	Engg Economics, Finance and Business Analytics	2	0	0	2	IC
3	CAC221	Machine Learning and Data Analysis	2	1	0	3	IC
4	CSC222	Language Processor	2	1	0	3	IC
5	CSC223	Database Management Systems	2	1	0	3	IC
6	CSC224	Advanced Data Structures	2	1	0	3	IC
7	CSL222	Language Processor Lab	0	0	2	1	IC
8	CSL223	Database Management Systems Lab	0	0	2	1	IC
9	CAL221	Machine Learning and Data Analysis Lab	0	0	2	1	IC
10	CYC221	Introduction to Cybersecurity and Cybercrimes	3	0	0	3	PC
		Total	16	5	6	24	

Semester V

Sl.	Code	Course Title	L	T	P	Credits	Category
1	CYC311	Modern Cryptography	3	0	0	3	PC
2	CYC312	Secure Software Engineering and DevSecOps	2	1	0	3	PC
3	CYOXXX	Dept Open Elective-I	2	1	0	3	OE
4	CYC313	Cloud and Critical Infrastructure Security	3	0	0	3	PC
5	CYC314	IoT and Cyber-Physical Systems Security	3	0	0	3	PC
6	CYI311	Computer Forensics	0	1	2	2	PC
7	CYI312	Artificial Intelligence for Cyber Defense	0	1	2	2	PC
8	CYL311	Cloud Computing and IoT Systems Lab	0	0	2	1	PC
		Total	13	4	6	20	

Semester VI

Sl.	Code	Course Title	L	T	P	Credits	Category
1	CYC321	Ethical Hacking	3	0	0	3	PC
2	CYEXXX	Deparment Elective-I	3	0	0	3	PE
3	CYEXXX	Deparment Elective-II	3	0	0	3	PE
4	IOECXXX	Foreign Language(NPTEL)	2	0	0	2	OE
5	IOECXXX	Design Thinking and Innovation	3	0	0	3	OE
6	CYEXXX	Deparment Elective-III	3	0	0	3	PE
7	CYI321	Threat Intelligence and Security Operations Centre(SOC)	0	1	2	2	PC
8	CYI322	Smart Phone Forensics and Application Security	0	1	2	2	PC
9	CYL321	Ethical Hacking Lab	0	0	2	1	PC
		Advanced course-1 for Hon.					
		Total	17	2	6	22	

Semester VII

Sl.	Code	Course Title	L	T	P	Credits	Category
1	CYOXXX	Dept Open Elective-II	3	0	0	3	PE
2	CYEXXX	Deparment Elective-IV	3	0	0	3	PE
3	CYEXXX	Deparment Elective-V	3	0	0	3	PE
4	CYEXXX	Deparment Elective-VI	3	0	0	3	PE
5	CYC411	Cyber Warfare,Adversarial Operations and Defensive Architectures	2	1	0	3	PC
6		Advanced course-2 for Hon(NPTEL)	3	0	0	3	
7	CYP411	B.Tech Project Phase-I	0	0	6	3	
8		Hons.Project Phase- 1	0	0	6	3	
		Total	15	0	6	18	

Semester VIII

Sl.	Code	Course Title	L	T	P	Credits	Category
1	CYP421	B.Tech Project Phase-II	0	0	18	9	
2		Hons. Project Phase-II	0	0	22	11	
		Total	0	0	18	9	

DETAILED SYLLABUS

SEMESTER I

MAC111 Calculus and Linear Algebra [3-1-0-4]

Course Objectives

- To study the basic topological properties of the real numbers.
- To develop knowledge of sequences of real numbers and their convergence.
- To study the notion of continuous functions and their properties.
- To gain an understanding of systems of linear equations.
- To introduce the fundamental concepts of vector spaces.
- To impart the basics of linear transformations, orthogonalization, basis, dimension, and eigenvalues.
- To provide the knowledge required to apply concepts of linear algebra in engineering applications.

Course Outcomes

- CO1: Understand the concepts of limits, continuity, differentiability, and classical theorems (Rolle's, Mean Value, Fundamental Theorem of Calculus) to model and solve real-world problems in science and engineering.
- CO2: Analyze convergence of sequences and series, and optimize multivariable functions using Hessian matrices to address practical problems in data modelling, optimization and artificial intelligence.
- CO3: Formulate and solve systems of linear equations using matrix methods, and interpret solutions in applied contexts such as network flows, resource allocation, and computational simulations.
- CO4: Apply vector space concepts, orthogonalization techniques, and inner product structures to design efficient representations in signal processing, machine learning, and numerical computation.
- CO5: Compute eigenvalues and eigenvectors, perform matrix diagonalization, and apply these techniques in stability analysis, principal component analysis, engineering design etc.

Syllabus

Calculus: Real Numbers, Properties of Real Numbers, Limit, Continuity, Discontinuity, Uniform continuity, Differentiability, Rolle's theorem, Mean Value theorem, Reimann Integration, fundamental theorem of calculus. Sequences and Series: Convergence and limit laws, Finite and infinite series, Sums of non-negative numbers, Absolute and conditional convergence of an infinite series, tests of convergence; Function of several variables: Limit, Continuity, Partial derivatives, Local maxima and local minima, Saddle point, Hessian Matrix.

Linear Algebra: System of linear equations, Row reduced echelon matrices, Rank of a matrix. Definition of a linear vector space and examples; linear independence of vectors, basis and dimension, Subspaces; Linear transformations, Matrix representation of Linear transformation, Inner product, norms, orthogonal basis, Gram-Schmidt orthogonalization process, Eigen vectors of a matrix and matrix diagonalization, Applications.

Textbooks / References

1. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th Edition, Wiley, 2011.
2. T. M. Apostol, *Calculus*, Volume I, 2nd Edition, Wiley, 2007.
3. G. Strang, *Linear Algebra and Its Applications*, 5th Edition, Wellesley–Cambridge Press/SIAM, 2016.
4. K. Hoffman and R. Kunze, *Linear Algebra*, 2nd Edition, PHI, 2009.
5. E. Kreyszig, *Advanced Engineering Mathematics*, 10th Edition, Wiley, 2015.
6. Thomas and Finney, *Calculus*, 9th Edition, Addison Wesley Publishing, 1998.
7. Anton, Biven, *Calculus*, 11th Edition, Wiley, 2015
8. Gilbert Strang, *Linear Algebra and Learning from Data*, Wellesley-Cambridge Press, 2019.

CSC111 Computer Programming and Problem Solving [3-1-0-4]

Course Objectives

- To introduce problem solving techniques and their representation using algorithms, flowcharts, and pseudocode.
- To teach core concepts of C programming including data types, operators, control flow constructs, and input/output mechanisms.
- To familiarize students with arrays, character arrays, and user-defined functions with parameter passing mechanisms in C.
- To introduce the concepts of recursion and pointer-based programming in C.
- To familiarize students with derived data types such as typedef, enumerations, unions, and structures in C.

Course Outcomes

- CO1: Gain knowledge of problem-solving techniques using algorithmic thinking, flowcharts, and pseudocode.
- CO2: Students learn to apply core C programming concepts including data types, control flow, arrays, functions, recursion, and pointers in problem solving

- CO3: Students learn to use derived data types and structured programming constructs for developing efficient solutions in C

Syllabus

Introduction to Problem Solving and Computational Thinking: Problem decomposition and abstraction, Algorithm design-Pseudocode and flowcharts. Problem-solving strategies (brute force, divide-and-conquer, iterative and recursive thinking), Algorithm verification and testing.

Introduction to C Programming: Introduction to algorithm, flowchart, pseudocode. Introduction to C programming - Character set of C, basic data types, operators, scope of variables, formatted input and output using scanf and printf. Writing and compiling simple C programs.

Control Flow Constructs: Conditional statements, if-else and switch constructs, nested conditions. Entry tested loops, exit tested loops, nested loop programs.

Arrays and Functions: Single-dimensional arrays, multidimensional arrays, character arrays. User-defined functions, function declaration, definition, call, return types, actual and formal parameters, parameter passing by value.

Recursion and Pointers: Recursive problem definitions, base case and recursive case. Example programs: factorial using recursion, Fibonacci series, Printing patterns. Introduction to pointers, address and dereferencing, pass-by-reference using pointers.

Derived Data Types: Use of typedef for user-defined type names, enumeration, unions and structures in C. Defining and accessing union members and structure members.

Textbooks / References

1. V. Anton Spraul, Think Like a Programmer: An Introduction to Creative Problem Solving is a programming, No Starch Press, Inc., 2012
2. Pólya, George. How to Solve It: A New Aspect of Mathematical Method. Princeton University Press, 2004. (Princeton Science Library Edition)
3. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Second Edition, Prentice Hall, 1988.
4. E. Balagurusamy, Programming in ANSI C, Ninth Edition, McGraw Hill, 2024.
5. Paul Deitel and Harvey Deitel, C: How to Program, Ninth Edition, Pearson Education, 2022.
6. V. Rajaraman, Computer Programming in C, Second Edition, Prentice Hall India, 2004.
7. R. G. Dromey, How to Solve It by Computer, Pearson Education India, 2007.
8. Reema Thareja, Computer Fundamentals and Programming in C, Second Edition, Oxford University Press, 2016.

ECC111 Electronic Devices and Digital Logic Design [3-1-0-4]

Course Objectives

- To impart the basic concepts of semiconductor devices.
- To analyse the operation of fundamental electronic devices in various circuits.
- To develop a strong foundation in number systems, Boolean algebra, logic gates, and simplification techniques used in digital electronic systems.
- To analyze and design combinational and sequential logic circuits using standard digital building blocks such as multiplexers, counters, flip-flops, and shift registers.

Course Outcomes

- CO1: Analyse pn-junction behaviour under various operating conditions and design diode circuits.
- CO2:Understand the working of BJT/MOSFETs as switching devices.
- CO3: Apply number system conversions, Boolean algebra, Karnaugh maps, and logic gate principles to analyze and simplify digital circuits.
- CO4: Design and evaluate combinational and sequential circuits including adders, multiplexers, counters, flip-flops, and shift registers with appropriate timing analysis.

Syllabus

Introduction to Semiconductors: Intrinsic and extrinsic semiconductors, pn junction diode, diode equation (derivation not required), diode characteristics, ideal and practical diode (Silicon), diode applications: clipping and clamping circuits. Field Effect Transistors: voltage-current relation in MOSFET (derivation not required), MOSFETs as switches. Number Systems and base conversion, Binary Arithmetic, Representation of Negative Numbers: sign and magnitude, 2's complement, 1's complement forms, shifts, and overflow, Binary Coded Decimal (BCD). Boolean algebra, De Morgan's laws, Basic Theorems, truthtables, SOP/POS. Minterm and Maxterm Expansions, Incompletely Specified Functions, don't–care conditions, Karnaugh map, Karnaugh map SOP and POS minimizations (Two, Three, Four and Five-variable Karnaugh maps), NAND/NOR realization. Combinational Logic Circuits–Half and Full Adders, Parallel Binary Adders, Binary Subtractor, Magnitude Comparators, Decoders, Encoders, Code Converters, Multiplexers, Demultiplexers. Sequential Logic Circuits – Latches and Flip–Flops, SR Latch, Gated D Latch, Edge–Triggered D Flip-Flop, SR Flip–Flop, JK Flip-Flop, T Flip–Flop, Flip-Flops with Asynchronous Clear and Preset inputs, Master-Slave Flip–Flops, Excitation table, Flip-Flop conversions, Counters – Asynchronous Counters, Synchronous Counters, Up/Down Synchronous Counters, Design of Synchronous Counters. Shift Registers – data movements in shift registers, SISO, SIPO, PISO, PIPO shift registers, Design of sequence detector circuits.

Textbooks / References

1. Sedra A. and Smith K. C., Microelectronic Circuits, Oxford University Press, Seventh Edition, 2015.
2. Robert L. Boylestad and Louis Nashelsky, Electronic Devices and Circuit Theory, Pearson Education, Eleventh Edition, 2015.
3. Ben G. Streetman and Sanjay Kumar Banerjee, Solid State Electronic Devices, Seventh Edition, Pearson Education, 2016.
4. M. Morris Mano and Michael D. Ciletti, Digital Design, 6th Edition, Pearson Education, 2018.
5. Thomas L. Floyd, Digital Fundamentals, 11th Edition, Pearson Education, 2015.
6. Charles H. Roth Jr. and Larry L. Kinney, Fundamentals of Logic Design, 7th Edition, Cengage Learning, 2013.

HMC111 Communication Skills for the Digital Age [3-0-0-3]

Course Objectives

- To understand the fundamentals of communication and technical communication.
- To develop effective listening, speaking, reading, and writing skills for academic and professional contexts.
- To apply grammatical structures and vocabulary appropriately in oral and written communication.
- To analyze communication barriers, audience needs, and communication styles in different contexts.
- To create professional documents, presentations, and technical content using appropriate language and format.
- To understand the impact of Artificial Intelligence and Digital Humanities on contemporary communication practices.
- To practice ethical, responsible, and inclusive communication in digital and multilingual environments.

Course Outcomes

- CO1: Apply effective listening, speaking, reading, and writing strategies in academic and professional contexts
- CO2: Analyze communication barriers, audience needs, and communication styles in different contexts
- CO3: Create professional documents, presentations, and technical content using appropriate language and format
- CO4: Evaluate the role of AI and digital technologies in communication with ethical awareness
- CO5: Demonstrate responsible and effective communication in multilingual and digital environments

Syllabus

Fundamentals of Communication: Process and types of communication- Technical communication and its importance - Barriers to communication - Non-verbal communication -Communication in academic and professional contexts -Shifts in communication in the digital age.

Listening and Speaking Skills: Listening process and active listening - Barriers to listening and strategies to overcome them - Fluency and accuracy in speech - Self-expression and tonal variations - Group discussion and interpersonal communication - Public speaking and presentation skills - Interviews and professional interaction - Responsible use of AI in communication.

Reading, Writing and Digital Communication: Reading comprehension and critical reading - Elements of effective writing - Professional emails and formal letters - Technical reports and presentations - Introduction to academic writing - AI in multilingual communication and translation - Ethical and legal considerations in AI-mediated communication - Introduction to Digital Humanities and digital communication practices.

Grammar and Vocabulary for Communication: Parts of speech and sentence structures - Tenses and Subject–Verb Agreement - Reported Speech and Active–Passive Voice - Common errors in English usage - Punctuation and mechanics of writing - Vocabulary enhancement in context.

Practical / Language Laboratory Component (1 Hour per Week): Listening to recorded talks and comprehension activities - Active listening exercises - Group discussion and interpersonal conversation - Pronunciation and tonal variation practice - Extempore speech and presentation activities -Mock interviews and persuasive speaking exercises -Speech assessment and peer feedback-Technology-assisted listening and speaking practice.

Textbooks / References

1. Anderson, Paul V., and Kerry Surman. Technical communication: A reader -centered approach. Thomson/Wadsworth, 2007.
2. Berry, David M. "Introduction: Understanding the digital humanities." Understanding digital humanities. London: Palgrave Macmillan UK, 2012. 1-20.
3. Davis, Marjorie T. "Assessing technical communication within engineering contexts tutorial." IEEE Transactions on professional communication 53.1 (2010): 33-45.
4. Duin, Ann Hill, and Isabel Pedersen. Augmentation technologies and artificial intelligence in technical communication: Designing ethical futures. Routledge, 2023.
5. Durack, K. T. (1997). Gender, Technology, and the History of Technical Communication. Technical Communication Quarterly, 6(3), 249–260.
6. Eggleston, A. G., & Rabb, R. J. (2018, June), Technical Communication for Engineers: Improving Professional and Technical Skills Paper presented at 2018 ASEE Annual Conference & Exposition , Salt Lake City, Utah. 10.18260/1-2—31068
7. Hart-Davidson, Bill, et al. "The history of technical communication and the future of generative AI." Proceedings of the 42nd ACM International Conference on Design of Communication. 2024.
8. Raman, Meenakshi, and Sangeeta Sharma. Technical

communication: Principles and practice. Oxford University Press, 2004.
9. Ranade, Nupoor, and Marly Saravia. "Teaching AI ethics in technical and professional communication: A systematic review." IEEE Transactions on Professional Communication 67.4 (2024): 422-436.
10. Reeves, Carol, and J. J. Sylvia IV. "Generative AI in technical communication: A review of research from 2023 to 2024." Journal of technical writing and communication 54.4 (2024): 439-462.
11. Thomson, Audrey Jean, and Agnes V. Martinet. A practical English grammar. New York: Oxford University Press, 2015.
12. Verma, Shalini. Technical communication for engineers. Vikas Publishing House, 2015.
13. Yule, George. Oxford practice grammar advanced. Oxford University Press, 2015.

CSC112 Foundations of Computing Systems [2-0-0-2]

Course Objectives

- Develop a foundational understanding of computing systems by examining computation as information processing, computational models, and methods of organizing and representing data.
- Establish the mathematical and logical fundamentals governing digital data representation, reliability, and physical hardware execution.
- Provide a unified view of computer systems by bridging the structural hierarchy from physical logic gates up through software layers and networking frameworks.
- Introduce the mechanics of distributed execution, concurrency, and real-world system modeling.
- Evaluate the physical and mathematical limits of classical computing while exploring the architecture of emerging, next-generation paradigms.

Course Outcomes

- CO1: Explain computation as information processing, distinguish between major models of computation, and describe how information is organized using fundamental logical structures and elementary data structures.
- CO2: Apply number system conversions and explain how digital information is represented, encoded, compressed, and protected against transmission errors.
- CO3: Map the abstraction layers of a computing system, linking high-level software execution down to machine code, processor cycles, and physical hardware switches.
- CO4: Analyze data flow and structural topologies by mapping algorithmic design strategies (such as graphs or recursion) onto packet routing mechanisms and network layered frameworks.
- CO5: Construct conceptual models for distributed and cloud-based applications, evaluating how communication topologies and architectural frameworks influence systemic performance.
- CO6: Formulate simulation workflows for complex, real-world processes while appraising basic computational complexity (P vs. NP), the necessity of algorithmic ethics, and adaptability to non-classical hardware.

Syllabus

Foundations of Computational Systems: Computing as information processing, Models of computation, Data Organization and logical structures. Introduction to Linear Data Structures: conceptual overview of storing relationships using Arrays, lists, matrices, and basic graphs.

The mathematical foundations of digital data storage and transmission. Number Systems: positional notation, binary, octal, decimal, and hexadecimal bases; standard conversion techniques. Data Encoding & Multimedia: text representation (ASCII, Unicode), and the digitization of continuous media (image pixels, color depth, audio sampling). Introduction to Compression & Reliability: conceptual difference between lossy and lossless compression; basic error detection (parity bits).

The basic structural hierarchy of physical computer hardware and system software. The Hardware Layer: transistors as binary switches, basic logic gates (AND, OR, NOT, XOR), and simple Boolean expressions. Digital Logic Circuits: introductory overview of combinational logic (adders) and sequential logic components (registers). Processor Architecture: the classic Von Neumann model framework (ALU, Control Unit, Memory) and a conceptual overview of the Fetch-Decode-Execute instruction cycle. Software Layer Abstraction: high-level languages vs. machine code, and the basic role of an Operating System.

Introduction to interconnected processing units and network communication. Networking Foundations: physical topologies (star, mesh), and layered abstraction frameworks (basic concepts of the OSI and TCP/IP models). The Internet Stack: packet routing, logical addressing (IP), and physical addressing (MAC). Distributed Frameworks: conceptual introduction to Distributed Systems and message passing / Remote Procedure Calls (RPC), foundational overview of Cloud Computing models (SaaS, PaaS, IaaS) and client-server architectures.

System Modeling, Complexity Horizons, and Computing Ethics. Computational approaches to mimicking reality, future paradigms, and societal impact. System Modeling & Simulation: introduction to using computers to simulate real-world processes (stochastic vs. deterministic context, queuing systems). Complexity Horizons: Basic notions of computational complexity (time and space trade-offs, scalability), conceptual overview of what makes a problem computationally hard (P vs. NP mapping). Emerging Paradigms: qualitative overview of Quantum Computing (qubits, superposition), neuromorphic architectures, and the Internet of Things (IoT). Responsible Computing and Ethics: introduction to algorithmic bias, data privacy, user confidentiality, and computing sustainability (carbon footprint of large-scale systems).

Textbooks / References

1. Brookshear, J. G., & Brylow, D. Computer Science: An Overview. 13th Edition, Pearson, 2019.
2. Patterson, D. A., & Hennessy, J. L. Computer Organization and Design: The Hardware/Software Interface. 6th Edition, Morgan Kaufmann, 2020.
3. Kurose, J. F., & Ross, K. W. Computer Networking: A Top-Down Approach. 8th Edition, Pearson, 2020.
4. Patt, Yale N., and Sanjay J. Patel. Introduction to Computing Systems: From Bits & Gates to C/C++ & Beyond. 3rd ed., McGraw-Hill Education, 2020.
5. Tanenbaum, A. S., & Austin, T. Structured Computer Organization. 6th Edition, Prentice Hall, 2012.
6. George Coulouris, Jean Dollimore, and Tim Kindberg. (2011). Distributed Systems: Concepts and Design (5th ed.). Addison-Wesley.
7. Buyya, R., Vecchiola, C., & Selvi, S. T. (2013). Mastering Cloud Computing: Foundations and Applications Programming. McGraw-Hill Education.
8. Sipser, Michael. Introduction to the Theory of Computation. 3rd ed., Cengage Learning, 2012.
9. Quinn, M. J. (2020). Ethics for the Information Age (8th/9th Edition). Pearson.
10. Zeigler, Bernard P., et al. Theory of Modeling and Simulation: Discrete Event & Iterative System Computational Foundations. 3rd ed., Academic Press, 2018.

CSL113 IT Workshop Lab [0-1-2-2]

Course Objectives

- To introduce students to modern software development environments, workflows, and collaborative programming practices.
- To develop proficiency in Linux command-line tools, file system management, shell scripting, and developer productivity utilities.
- To enable effective use of Git and GitHub for version control, repository management, collaborative development, and open-source contribution workflows.
- To train students in the use of integrated development environments, debugging techniques, and software documentation standards for professional software development.
- To introduce foundational web programming concepts and modern front-end development using HTML, CSS, JavaScript, and basic React JS for building interactive web applications.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Demonstrate proficiency in Linux environments, command-line tools, shell scripting, and software development utilities.
- CO2: Apply Git and GitHub workflows for repository creation, version control, collaboration, branching, merging, and remote repository management.
- CO3: Develop, debug, and document software projects using modern IDEs, debugging tools, and software documentation practices.
- CO4: Design and implement basic web pages using HTML, CSS, and JavaScript while understanding web application architecture.

Lab Practice

1. **Linux Environment Familiarization and Shell Scripting:** Linux file system, terminal basics, directory navigation, file manipulation commands, package management overview, environment variables, basic shell commands, simple shell scripts, and automation concepts.
2. **Git Fundamentals, Branching, and GitHub Repository Management:** Git installation, repository initialization, staging and committing, viewing history, undoing changes, branch creation, feature branch workflow, merge operations, conflict resolution, introduction to rebase, remote repositories, clone, push, pull, fetch, SSH and HTTPS authentication, repository hosting and management, and forking repositories.
3. **Collaborative Development and Technical Documentation:** Pull requests, issues and discussions, code reviews, open-source contribution workflow, README creation, Markdown syntax, code documentation, and documentation standards.
4. **IDE Setup, Debugging, and Project Management:** Modern IDE features, extensions and plugins, integrated terminal, workspace management, breakpoints, stepping through code, variable inspection, error diagnosis, debugging best practices, and basic project organization.
5. **Introduction to Web Programming:** World Wide Web, client-server architecture, static and dynamic websites, browser developer tools, HTML page structure, forms, tables, lists, semantic elements, CSS selectors, layouts, responsive design, JavaScript variables, functions, events, DOM manipulation, and browser developer tools.
6. React JS Fundamentals: Setting up a React application using Vite, understanding components, JSX fundamentals, conceptual introduction to props and state, event handling in React, and development of a simple interactive component.

Textbooks / References

1. Dean, J., *Web Programming with HTML5, CSS, and JavaScript*, Jones & Bartlett Learning, 2018.
2. Robbins, J., *Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics*, 6th Edition, O'Reilly Media, 2024.
3. Banks, A., and Porcello, E., *Learning React: Modern Patterns for Developing React Apps*, 3rd Edition, O'Reilly Media, 2024.

CSL111 Computer Programming Lab [0-0-2-1]

Course Objectives

- To enable students to write, compile, and execute C programs systematically
- To familiarize students with core C programming concepts including data types, operators, control flow constructs, and input/output mechanisms
- To develop programming skills using arrays, character arrays, and user-defined functions with parameter passing mechanisms.
- To introduce recursive and pointer-based programming through hands-on implementation in C.
- To enable students to implement programs using derived data types such as typedef, enumerations, unions, and structures in C.

Course Outcomes

- CO1: Students learn to translate programming concepts into C programs using structured programming techniques.
- CO2: Students learn to develop recursive and pointer-based solutions for computational problems.
- CO3: Students learn to use C programs for solving real world problems.

Syllabus

Introduction to C Programming: Basic data types, operators, scope of variables, formatted input and output using scanf and printf. Writing and executing simple C programs.

Control Flow Constructs: Implementation of programs using conditional statements and loop constructs.

Arrays and Functions: Implementation of programs using arrays and character arrays. Programs involving user-defined functions with parameter passing by value.

Recursion and Pointers: Implementation of recursive programs. Implementation of programs using pointers and pass-by-reference mechanisms.

Derived Data Types: Implementation of programs using typedef, enumerations, unions, and structures in C. Defining and accessing union members and structure members.

Textbooks / References

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Second Edition, Prentice Hall, 1988.
2. E. Balagurusamy, Programming in ANSI C, Ninth Edition, McGraw Hill, 2024.
3. Paul Deitel and Harvey Deitel, C: How to Program, Ninth Edition, Pearson Education, 2022.
4. Ashok N. Kamthane, Programming in C, Third Edition, Pearson India, 2015

ECL111 Electronic Devices and Digital Logic Design Lab [0-0-2-1]

Course Objectives

- To introduce the fundamental concepts and characteristics of semiconductor devices.
- To analyse the operation and applications of fundamental electronic devices in various circuits.
- To develop a strong foundation in number systems, Boolean algebra, logic gates, and logic simplification techniques used in digital electronic systems.
- To enable the analysis and design of combinational and sequential logic circuits using standard digital building blocks such as multiplexers, counters, flip-flops, and shift registers.

Course Outcomes

- CO1 Identify and operate basic electronic laboratory equipment, and analyse the characteristics and applications of diode-based circuits such as clippers and clampers.
- CO2 Design, implement, and verify combinational logic circuits including adders, multiplexers, and demultiplexers using digital electronic components.
- CO3 Analyse the operation of sequential logic circuits by implementing various flip-flops, latches, and asynchronous counters.
- CO4 Design, implement, and evaluate synchronous counters and shift registers for digital system applications.

Experiments

1. Familiarization of basic electronic lab equipment and components.
2. Diode Clippers.
3. Diode Clampers.
4. Design of Combinational Logic Circuits - Adders, Multiplexers, Demultiplexers
5. Familiarization of Flip-Flops and Latches. SR, D and JK.
6. Design of asynchronous counters.
7. Design of synchronous counters
8. Shift registers.

Textbooks / References

1. Sedra A. and Smith K. C., Microelectronic Circuits, Oxford University Press, Seventh Edition, 2015.
2. Robert L. Boylestad and Louis Nashelsky, Electronic Devices and Circuit Theory, Pearson Education, Eleventh Edition, 2015.
3. M. Morris Mano and Michael D. Ciletti, Digital Design, 6th Edition, Pearson Education, 2018.
4. Thomas L. Floyd, Digital Fundamentals, 11th Edition, Pearson Education, 2015.
5. Charles H. Roth Jr. and Larry L. Kinney, Fundamentals of Logic Design, 7th Edition, Cengage Learning, 2013.

SEMESTER II

IMA111 Discrete Mathematics [3-1-0-4]

Course Objectives

- To extend students' logical and mathematical maturity and their ability to deal with abstraction.
- To introduce the fundamental terminologies and concepts used in electronics, communication engineering, and computer science.
- To explain and apply the basic methods of discrete mathematics in engineering and computing applications.
- To develop the ability to write clear, concise, and correct mathematical proofs.
- To solve counting problems involving permutations, combinations, and the Pigeonhole Principle.
- To understand the fundamental concepts of graph theory and abstract algebra.

Course Outcomes

- CO1: Apply propositional and predicate logic, rules of inference, and proof techniques to analyze mathematical statements and algorithms.
- CO2: Analyze sets, relations, functions, and recursive structures used in discrete mathematical modeling.
- CO3: Apply counting principles, recurrence relations, and generating functions to solve combinatorial problems.
- CO4: Solve problems involving graphs, trees, groups, rings, fields, and lattices in engineering and computing contexts.

Syllabus

Logic: Propositions, negation, conjunction, disjunction, implication and equivalence, truth tables, predicates, quantifiers, rules of inference, methods of proof.

Set Theory: Definitions and simple proofs in set theory, inductive definition of sets and proof by induction, inclusion and exclusion principle, relations, representation of relations by graphs, properties of relations, equivalence relations and partitions, partial orderings.

Functions: Mappings, injections and surjections, composition of functions, inverse functions, special functions, recursive function theory.

Elementary Combinatorics: Counting techniques, Pigeonhole Principle, recurrence relations, generating functions.

Graph Theory: Elements of graph theory, Euler graphs, Hamiltonian paths, trees, tree traversals, spanning trees.

Algebra: Groups, Lagrange's theorem, homomorphism theorem, rings and fields, structure of the ring $\mathbb{Z}_n$ and the unit group $\mathbb{Z}_n$, lattices.

Textbooks / References

1. Kenneth H. Rosen, *Discrete Mathematics and Its Applications*, 7th Edition, McGraw-Hill, 2017.
2. Norman L. Biggs, *Discrete Mathematics*, 2nd Edition, Oxford University Press, 2003.
3. J. P. Tremblay and R. Manohar, *Discrete Mathematical Structures with Applications to Computer Science*, McGraw-Hill, 2017.
4. K. A. Ross and C. R. B. Wright, *Discrete Mathematics*, 5th Edition, Pearson, 2003.
5. P. B. Bhattacharya, S. K. Jain, and S. R. Nagpaul, *Basic Abstract Algebra*, 2nd Edition, Cambridge University Press, 2003.
6. J. A. Gallian, *Contemporary Abstract Algebra*, 9th Edition, Cengage Learning, 2017.

CSC121 Data Structures [3-1-0-4]

Course Objectives

- Define and describe simple data structures like arrays, linked lists, trees and graphs.
- Design and specify algorithms for searching and sorting, and those associated with the above data structures .
- Analyse simple algorithms, like sorting and searching using mathematical tools, like formulation and solving of recurrences.
- Analyse application problems and abstract them to formulate solutions involving data structures and algorithms.

Course Outcomes

- CO1: Students learn to define operations of data structures like arrays, linked lists, trees and graphs.
- CO2: Students learn to design and specify algorithms involving above types of data structures.
- CO3: Students learn to analyze simple algorithms and solve recurrences, asymptotic analysis and probabilistic analysis.
- CO4: Students learn to analyze application problems and abstract them to formulate solutions involving data structures and algorithms

Syllabus

Introduction- Algorithm Analysis, Finding Complexity. Fundamental data structures - List, Sorted Lists, Single and Double Linked Lists, Skip list. Dynamic Arrays, Circular Linked Lists, Applications of Lists in Browser History and Text Editors.

Stack & Queue with applications.

Binary Trees- Insertion and Deletion of nodes, Tree Traversals, Polish Notations, AVL Trees, Heaps, Priority Queues. Optimal binary search tree, Application problems on Optimal binary search Tree.

Sorting -Bubble, Selection, Insertion, Merge Sort, Quick Sort, Radix Sort, Heap sort. Searching. Hashing- Application problems on hashing.

Graphs- Representations, types and properties of graphs, Shortest path algorithms, Minimum Spanning Trees, BFS, DFS.

Textbooks / References

1. Clifford A Shaffer, Data Structures and Algorithm Analysis, Edition 3.2 (Java Version), 2011.
2. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser. Data Structures And Algorithms In Java TM Sixth Edition, Wiley Publishers, 2014.
3. Mark Allen Weiss Data Structures And Algorithm Analysis In Java, Third Edition, 2012.
4. Robert L. Kruse, Data Structures And Program Design In C++, Pearson Education, Second Edition, 2006.
5. Ellis Horowitz, Fundamentals of Data Structures in C++, University Press, 2015.
6. Ajay Agarwal, Data Structure through C, A Complete Reference Guide, Cyber Tech Publications, 2005.

CSC122 Computer Organization [3-1-0-4]

Course Objectives

- This course will introduce to students the fundamental concepts underlying modern computer organization and architecture.
- Students should be able to know the overall working of a computer.
- Students should be able to get a detailed understanding of the design principles involved in developing a computer.
- They should know the representation of data, how programs are represented, executed and how programs manipulate and operate on data.
- They should also be able to appreciate how the memory organization is done and how to organize memory for faster execution of programs.
- Students should also be able to appreciate the concepts in pipelining.

Course Outcomes

- CO1: Understand the basics of computer hardware and how software interacts with computer Hardware.
- CO2:Analyse and evaluate the performance of computers.
- CO3: Understand basics of Instruction Set Architectures and their impact on processor design
- CO4: Understand how computers, represent and manipulate data.
- CO5: Understand how computer, perform arithmetic operations, how they are optimized and made to run faster.
- CO6: Design a pipeline for consistent execution of instructions with minimum hazards.
- CO7: Understand how the memory management takes place in a computer system
- CO8: Design a simple computer with hardware design including data format, instruction format, instruction set, addressing modes, bus structure, input/output, memory, Arithmetic/Logic unit, control unit, and data, instruction and address flow.

Syllabus

Basic principles, hardware components Measuring performance: evaluating, comparing and summarizing performance.

Instructions: operations and operands of the computer hardware, representing instructions, making decision, supporting procedures, character manipulation, styles of addressing, starting a program.

Computer Arithmetic: signed and unsigned numbers, addition and subtraction, logical operations, constructing an ALU, multiplication and division, floating point representation and arithmetic, Parallelism and computer arithmetic.

The processor: building a data path, simple and multicycle implementations, microprogramming, exceptions, Pipelining, pipeline Data path and Control, Hazards in pipelined processors

Memory hierarchy: caches, cache performance, virtual memory, common framework for memory hierarchies

Input/output: I/O performance measures, types and characteristics of I/O devices, buses, interfaces in I/O devices, design of an I/O system, parallelism and I/O. Introduction to multicores and multiprocessors.

Textbooks / References

1. D. A. Pattersen and J. L. Hennesy, Computer Organisation and Design: The Hardware/ Software Interface, Fourth Edition, Morgan Kaufman, 2009.
2. V. P. Heuring and H. F. Jordan, Computer System Design and Architecture, Prentice Hall, 2003.
3. Computer Architecture: A Quantitative Approach, Fifth Edition, Morgan Kaufman, 2011.
4. Carl Hamazher, ZvonkoVranesic and SafwatZaky, Computer Organization, Fifth Edition, McGraw Hill, 2002.

HMC121 Personality Development & Soft Skills [2-1-0-3]

Course Objectives

- To understand the significance of soft skills, professional etiquette, self-awareness, and ethical behaviour in personal and professional development
- To help students understand and explain individual behaviour and personality in interpersonal and professional contexts.
- To apply communication, leadership, teamwork, presentation, and interpersonal skills in academic and workplace contexts
- To create awareness regarding common psychological concerns and the importance of psychological wellbeing in academic and personal life.
- To analyze workplace situations and develop problem-solving, emotional intelligence, adaptability, and stress management skills for professional
- To familiarize students with strategies for emotional regulation, coping, and healthy personality development.

Course Outcomes

- CO1: Apply self-assessment and goal-setting techniques for personal and professional growth
- CO2: Understand one's own personality and that of others, appreciate individual differences, and adapt effectively to people and situations.
- CO3:Analyze workplace situations using communication, leadership, and emotional intelligence skills
- CO4: Develop the ability to cope with personal, academic, and professional challenges through an understanding of human behaviour.
- CO5: Demonstrate effective speaking, presentation, and interpersonal communication skills
- CO6: Evaluate professional challenges related to stress, ethics, teamwork, and work–life balance

Syllabus

Foundations of soft skills and personality development: Introduction to soft skills - Importance and aspects of soft skills - Self-esteem and self-awareness - Professional etiquette and social etiquette - Professional ethics and workplace behavior - Professional body language - SMART goals and personal goal setting - SWOT analysis for personality development.

From Campus to Corporate: Leadership skills and teamwork - Group discussion and meeting management - Adaptability and work ethics - Time management and stress management - Advanced speaking skills - Oral presentations, speeches, and debates - Combating nervousness and stage fear - Planning and preparation of presentations - Making effective presentations - Speeches for various occasions - Interviews and facing job interviews - Emotional intelligence and critical thinking -Work–life balance - Applied grammar for professional communication.

Understanding Personality and Behaviour: Personality: Meaning & Assessment. Psychoanalytic & Neo-Psychoanalytic Approach; Behavioural Approach; Cognitive Approach; Social- Cognitive Approach; Humanistic Approach; The Traits Approach; Models of healthy personality: the notion of the mature person, the self-actualizing personality; Indian perspective on personality.

Mental Health, Institutions, and Social Contexts: Mental health in social, cultural, and institutional contexts; campus and workplace mental health. Institutional violence, bullying, harassment, and discrimination based on caste, religion, gender, sexuality, disability, body image, and other social locations. Structural inequalities, stigma, inclusion, and access to mental health support. Student mental health and suicide prevention, including institutional accountability and recommendations of the Supreme Court's National Task Force on Student Suicides. Critical perspectives on positive psychology and mainstream mental health interventions; indigenous and community-based approaches to wellbeing. Help-seeking, collective care, peer support, and professional mental health services.

Textbooks / References

1. Carnegie, Dale. How to win friends and influence people. John Wiley & Sons, 2026.
2. Covey, Stephen R. "The 7 habits of highly effective people." Personal Finance Magazine 2024.527 (2024): 8-13.
3. Ghai, A. (2015). Rethinking disability in India. Routledge.
4. Guru, G., & Sarukkai, S. (2012). The cracked mirror: An Indian debate on experience and theory. Oxford University Press.
5. Goleman, Daniel. "Emotional Intelligence Bloomsbury Publishing." (2020).
6. Kumar, N. (2023). Mental health and well-being: An Indian psychology perspective. Routledge.
7. Medden, Stephanie, et al. "From Public Speaking to Multimodal Communication: A Reevaluation of the Basic Communication Course." Basic Communication Course Annual 37.1 (2025): 4.
8. Patel, V. (2014). Where there is no psychiatrist: A mental health care manual (2nd ed.). Royal College of Psychiatrists.
9. Patel, V., Saxena, S., Lund, C., Thornicroft, G., Baingana, F., Bolton, P., ... Unützer, J. (2018). The Lancet Commission on global mental health and sustainable development. The Lancet, 392(10157), 1553–1598.
10. Pease, Barbara, and Allan Pease. The definitive book of body language: The hidden meaning behind people's gestures and expressions. Bantam, 2008.
11. Rao, Manchanahalli Satyaranayana. Soft skills-enhancing employability: connecting campus with corporate. IK International Pvt Ltd, 2010.
12. Tracy, Brian. Eat that frog!: 21 great ways to stop procrastinating and get more done in less time. Berrett-Koehler Publishers, 2025.
13. Nevid, J. S., Rathus, S. A., & Greene, B. (2022). Abnormal psychology in a changing world (11th ed.). Pearson.
14. Snyder, C. R., & Lopez, S. J. (Eds.). (2021). Positive psychology: The scientific and practical explorations of human strengths (5th ed.).
15. Carr, A. (2022). Positive psychology: The science of happiness and human strengths (4th ed.).
16. Kabat-Zinn, J. (2018). Mindfulness for beginners: Reclaiming the present moment—and your life.
17. Ryckman, R. M. (2017). Theories of personality (10th ed.).
18. Schultz, D. P., & Schultz, S. E. (2020). Theories of personality (12th ed.).

CAC121 Artificial Intelligence [3-0-0-3]

Course Objectives

- To understand the foundations of Artificial Intelligence and intelligent agents, environments, and rational decision-making processes in AI systems.
- To apply problem-solving techniques in uninformed, in-

formed, adversarial, and constraint search space.
- To understand knowledge representation, reasoning and planning in applications
- To explore advanced AI concepts including natural language processing,expert systems, fuzzy logic, and hybrid intelligent systems.

Course Outcomes

- CO1: Explain the core principles and architectures of Artificial Intelligence systems.
- CO2:Implement search algorithms and reasoning techniques to solve AI problems.
- CO3: Apply knowledge representation and probabilistic reasoning methods in intelligent applications.
- CO4: Analyze the applicability of advanced AI techniques such as NLP, fuzzy systems, expert systems in real-world applications.

Syllabus

Artificial Intelligence & Problem Solving: Introduction to AI: What is AI? Foundation, History, State of the art, Risks and Benefits. Intelligent agent: Agents and environments, concept of rationality, nature of environment, structure of agents. Problem solving: Search space problem, uninformed search strategies, informed search strategies, local search algorithms, constraint satisfaction problems, adversarial search and games.

Knowledge, reasoning and planning: Logical agents: Knowledge-Based Agents, Propositional logic, propositional theorem proving. First order logic: syntax and semantics, knowledge engineering, inference: unification, forward and backward chaining, resolution. Knowledge representation: Ontology, categories, objects, events. Planning: classical planning, heuristics of planning, hierarchical planning.

Uncertain Knowledge and reasoning: Quantifying uncertainty: acting under uncertainty, Bayes' rule, Naïve Bayes model, semantics of Bayes networks, inference in temporal models, combining beliefs and desires under uncertainty, decision networks, sequential decision problems, relational probability models. Fuzzy logic: Crisp logic, Fuzzy logic, Membership function, Membership function, Fuzzy logic Applications.

Expert systems and Advanced AI: Expert Systems: What is Expert system, Conventional systems vs. Expert systems, Basic Concepts, Human Expert Behaviors, Knowledge Types, Inferencing, Rules, Structure of Expert Systems, ES Components, Knowledge Engineer, Expert Systems Working, Problem Areas Addressed by Expert Systems, benefits limitations- Applications of expert systems. Advanced AI: Natural language processing, Hybrid intelligent systems, Introduction to: Ethics of AI, Responsible AI, Generative AI, Explainable AI, Agentic AI.

Textbooks / References

1. Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, Fourth edition, Pearson Education, 2021
2. Elaine rich and kevin knight, introduction to artificial intelligence, mcgraw hill, third edition, 2017.
3. Michael Negnevitsley, artificial intelligence: a guide to intelligent systems, addison wesley, third edition, 2017
4. David L. Poole and Alan K. Mackworth, Artificial Intelligence: Foundations of Computational Agents, 3rd edition,2023. (https://artint.info/3e/html/ArtInt3e.html)
5. Deepak Khemani. A First Course in Artificial Intelligence. McGraw Hill Education (India), 2013.
6. Denis Rothman. Artificial Intelligence by Example, Packt, 2020, 2nd Edition.
7. Explainable AI for Practitioners by Michael Munn, David Pitman, O'Reilly Media 2022, ISBN: 9781098119133
8. Web Resources:
   1.Stanford Encyclopedia of Artificial Intelligence
   2. IBM AI Learning Hub
   3. NPTEL Artificial Intelligence Course

CSC123 Object Oriented Programming [2-1-0-3]

Course Objectives

- Understand the fundamentals of object-oriented programming and the Java programming environment.
- Design and implement programs using classes, objects, constructors, and access control mechanisms.
- Apply inheritance, polymorphism, abstraction, and interfaces to develop reusable and extensible software components.
- Organize programs using packages and manage program robustness through exception handling mechanisms.
- Develop Java applications involving arrays, strings, and object-oriented design principles for real-world problem solving.

Course Outcomes

- CO1: Explain the concepts of object-oriented programming and the architecture of the Java platform, including JVM, JRE, and JDK.
- CO2: Design and implement Java classes using constructors, access modifiers, encapsulation, static members, method overloading, and object management techniques.
- CO3: Develop applications using inheritance, polymorphism, abstract classes, interfaces, and packages to create reusable and modular software systems.
- CO4: Implement exception handling mechanisms, including custom exceptions, to build reliable and fault-tolerant Java applications.
- CO5: Develop Java programs using arrays, strings, string manipulation techniques, and related programming constructs to solve computational problems effectively.

Syllabus

Introduction to OOP and Java Fundamentals: Procedural vs. object-oriented thinking; OOPs Principles, advantages of OOPs, Java ecosystem: Bytecode, JDK, JVM, JRE; compiling and running Java programs, Language Fundamentals

Class and Objects: Introducing Class Fundamentals - Object and Object reference, Defining classes: fields, methods, constructors (default, parameterised, copy), Access modifiers: private, protected, public, package-private, Encapsulation: getters, setters, and data hiding principles, 'this' keyword; static members, method overloading, constructor overloading, Inner Class and Anonymous Class, Object Lifetime and Garbage Collection.

Extending Classes and Inheritance: Use and Benefits of Inheritance in OOP, Types of Inheritance in Java, Role of Constructors in inheritance, Overriding Super Class Methods, use of "super", constructor chaining, Polymorphism in inheritance, dynamic method dispatch, final classes, methods, and variables, Abstract classes and abstract methods; Interfaces: defining contracts, multiple interface implementation, default methods, static methods, functional interfaces, Packages and import statements; Package as access protection.

Exception Handling: Throwable, Error, Exception, RuntimeException, try-catch-finally blocks; multi-catch; Checked vs unchecked exceptions; Creating custom exception classes.

Array & Strings: Defining an Array, Initializing & Accessing Array, Multi –Dimensional Array, Operation on String, Mutable & Immutable String, Using Collection Bases Loop for String, Tokenizing a String, Creating Strings using String Buffer.

Textbooks / References

1. Introduction to Java Programming and Data Structures (12th Ed.), Y. Daniel Liang, Pearson
2. Thinking in Java (4th Ed.), Bruce Eckel, Prentice Hall
3. Core Java Volume I – Fundamentals (12th Ed.), Cay S. Horstmann, Oracle Press / Pearson
4. Head First Java (3rd Ed.), Kathy Sierra & Bert Bates, O'Reilly Media
5. Clean Code: A Handbook of Agile Software Craftsmanship, Robert C. Martin, Prentice Hall
6. C. Thomas Wu, An Introduction to Object Oriented Programming with Java, Fifth Edition, 2009.
7. Ken Arnold, James Gosling and David Holmes, The Java Programming Language, Fourth Edition, 2005.
8. Herbert Schildt, Java: The Complete Reference, McGraw Hill Education(India), 11th Edition 2018. 4. 1. Steven Holzner, PHP: The Complete Reference, McGraw Hill Education, 2017.

CSL121 Data Structures lab (C, C++) [0-0-2-1]

Course Objectives

- Develop proficiency in implementing fundamental data structures and algorithms.
- Analyze the time and space complexity of data structure operations.
- Apply appropriate data structures to solve computational problems efficiently.
- Gain practical experience with tree, graph, sorting, searching, and hashing techniques.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Implement and evaluate linear and non-linear data structures.
- CO2: Analyze the performance of data structure operations and associated algorithms.
- CO3: Apply trees, graphs, hashing, sorting, and searching techniques to solve computational problems.
- CO4: Design efficient solutions using suitable data structures and justify their selection.

Syllabus

Linear Data Structures: Implementation and analysis of Lists, Sorted Lists, Singly Linked Lists, Doubly Linked Lists, Circular Linked Lists, Dynamic Arrays, and Skip Lists. Applications of lists in Browser History Management and Text Editors.

Stacks and Queues: Implementation using arrays and linked lists. Applications including Expression Evaluation, Celebrity Problem, Largest Rectangular Area in a Histogram, and Queue-based Scheduling Problems.

Trees: Binary Tree Operations, Insertion and Deletion of Nodes, Tree Traversals, Polish Notations, AVL Trees, Heaps, Priority Queues, Optimal Binary Search Trees, and Application Problems on Optimal Binary Search Trees.

Sorting and Searching: Implementation and performance analysis of Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort, Radix Sort, Heap Sort, Linear Search, and Binary Search.

Hashing: Hash Functions, Collision Resolution Techniques, and Application Problems on Hashing.

Graphs: Graph Representations, Types and Properties of Graphs, Breadth First Search (BFS), Depth First Search (DFS), Shortest Path Algorithms, and Minimum Spanning Tree Algorithms.

Mini Project / Case Study (optional): Design, implementation, analysis, and presentation of a data structure-based solution to a real-world problem.

Textbooks / References

1. Clifford A Shaffer, Data Structures and Algorithm Analysis, Edition 3.2 (Java Version), 2011.
2. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser. Data Structures And Algorithms In Java TM Sixth Edition, Wiley Publishers, 2014.
3. Mark Allen Weiss Data Structures And Algorithm Analysis In Java, Third Edition, 2012.

4. Robert L. Kruse, Data Structures And Program Design In C++, Pearson Education, Second Edition, 2006.
5. Ellis Horowitz, Fundamentals of Data Structures in C++, University Press, 2015.
6. Ajay Agarwal, Data Structure through C, A Complete Reference Guide, Cyber Tech Publications, 2005.

CSL123 Object Oriented Programming Lab [0-0-2-1]

Course Description

This laboratory course provides hands-on experience in developing Java applications using object-oriented programming principles. Students gain practical knowledge of the Java programming environment and learn to design and implement programs using classes, objects, constructors, and access control mechanisms. The course emphasizes software development through inheritance, polymorphism, abstraction, interfaces, packages, and exception handling. Students also develop applications involving arrays, strings, and object-oriented design principles to solve real-world problems.

Course Objectives

- To enable students to write, compile, and execute Java programs using the Java programming environment.
- To familiarize students with object-oriented programming concepts including classes, objects, constructors, and access control mechanisms.
- To develop programming skills using inheritance, polymorphism, abstraction, interfaces, and packages for modular software development.
- To introduce exception handling techniques for developing robust Java applications.
- To enable students to implement object-oriented applications using arrays, strings, and Java class libraries.

Course Outcomes

- CO1:Students learn to design and implement Java programs using classes, objects, constructors, and object-oriented programming principles.
- CO2: Students learn to develop reusable and extensible applications using inheritance, polymorphism, abstraction, interfaces, and packages.
- CO3: Students learn to develop robust Java applications using exception handling, arrays, strings, and object-oriented design principles for solving real-world problems.

Syllabus

Introduction to Java Programming: Writing, compiling, and executing Java programs. Basic language constructs, data types, operators, input/output, and program structure. Classes and Objects: Implementation of programs using classes, objects, constructors, access modifiers, encapsulation, static members, method overloading, constructor overloading, inner classes, and anonymous classes.
Inheritance and Polymorphism: Implementation of programs using inheritance, constructor chaining, method overriding, dynamic method dispatch, abstract classes, interfaces, and packages.
Exception Handling: Implementation of programs using exception handling mechanisms including try-catch-finally blocks, multiple catch blocks, checked and unchecked exceptions, and user-defined exceptions.
Arrays and Strings: Implementation of programs using one-dimensional and multidimensional arrays, string manipulation, String, StringBuffer, StringBuilder, tokenizing strings.
Object-Oriented Applications: Development of Java applications integrating object-oriented concepts, exception handling, arrays, strings, and packages to solve real-world problems.

Textbooks / References

1. Introduction to Java Programming and Data Structures (12th Ed.), Y. Daniel Liang, Pearson
2. Thinking in Java (4th Ed.), Bruce Eckel, Prentice Hall
3. David A. Bell,
4. Core Java Volume I – Fundamentals (12th Ed.), Cay S. Horstmann, Oracle Press / Pearson
5. C. Thomas Wu, An Introduction to Object Oriented Programming with Java, Fifth Edition, 2009.
6. Ken Arnold, James Gosling and David Holmes, The Java Programming Language, Fourth Edition, 2005.

SEMESTER III

MAC211 Probability, Statistics and Random Processes [3-1-0-4]

Course Objectives

- To expose the students to the modern theory of probability, concept of random variables and their expectations.
- To introduce various discrete and continuous distributions and concept of estimation theory, confidence interval.
- To develop the ability to perform statistical inference and data analysis using estimation, confidence intervals, and hypothesis testing techniques.
- To illustrate the concept of hypothesis testing, tests for means and variances, Goodness-of-fit tests.
- To introduce the concept of random processes, Markov chains, and Brownian Motion.
- To introduce stochastic modeling techniques and their applications in data-driven systems.

Course Outcomes

- CO1: Define and apply the concepts of probability and conditional probability.

- CO2: Define and illustrate discrete and continuous random variables, their probability mass functions and probability density functions.
- CO3: Understand the concepts and significance of estimation theory and hypothesis testing.
- CO4: Apply estimation and hypothesis testing techniques to construct confidence intervals for means and variances and conduct goodness-of-fit tests for statistical inference and data-driven decision making.
- CO5: Analyze random processes, Markov chains, and Brownian motion and explain their applications in stochastic modelling and computing systems.

Syllabus

Basics of Probability: Axiomatic construction of the theory of probability, independence, conditional probability, and basic formulas.

Random Variables and Distributions: Univariate, Bivariate and multivariate random variables, cumulative and marginal distribution functions, conditional and multivariate distributions, functions of random variables: sum, product, ratio, change of variables, mathematical expectations, moments, moment generating function, characteristic functions

Special Probability Distributions: discrete and continuous distributions, Limit theorems, Binomial distribution, Geometric distribution, Poisson distribution, Normal distribution, Exponential distribution, Gamma distribution, Beta distribution, Central Limit Theorem, Tchebeychev's inequality, Law of Large Numbers.

Estimation Theory: Bias of estimates, confidence intervals, minimum variance unbiased estimation, Bayes' estimators, moment estimators, maximum likelihood estimators, Chi-square distribution, confidence intervals for parameters of normal distribution.

Hypothesis Testing: Tests for means and variances, hypothesis testing and confidence intervals, Bayes' decision rules, power of tests, Goodness-of-fit tests, Kolmogorov-Smirnov Goodness-of-fit test.

Random Processes: Definition and classification of random processes, discrete-time Markov chains, Poisson process, continuous-time Markov chains, stationary processes, Gaussian process, Brownian motion.

Textbooks / References

1. S. Ross, *Introduction to Probability and Statistics for Engineers and Scientists*, 3rd ed., Elsevier, 2004.
2. P. G. Hoel, S. C. Port, and C. J. Stone, *Introduction to Probability Theory*, Universal Book Stall, 2000.
3. S. M. Ross, *Introductory Statistics*, 2nd ed., Academic Press, 2009.
4. J. Medhi, *Stochastic Processes*, 3rd ed., New Age International, 2009.
5. V. K. Rohatgi and A. K. Saleh, *An Introduction to Probability and Statistics*, 3rd ed., Wiley Student Edition, 2006.
6. G. R. Grimmett and D. R. Stirzaker, *Probability and Random Processes*, Oxford Univ. Press, 2001.
7. W. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1, 3rd ed., Wiley, 1968.
8. S. M. Ross, *Stochastic Processes*, 2nd ed., Wiley, 1996.
9. C. M. Grinstead and J. L. Snell, *Introduction to Probability*, 2nd ed., Universities Press India, 2009.
10. S. Ross, *A First Course in Probability*, 10th ed., Pearson Education, Delhi, 2018.
11. K. P. Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022.
12. G. Casella and R. L. Berger, *Statistical Inference*, Chapman and Hall/CRC, 2024.
13. S. Ross, *Introduction to Probability Models*, 13th ed., Elsevier, 2023.
14. R. V. Hogg, J. W. McKean, and A. T. Craig, *Introduction to Mathematical Statistics*, 8th ed., Pearson, 2019.

HMC211 Engineering Management, Entrepreneurship Development & start ups [2-0-0-2]

Course Objectives

- To introduce the principles and practices of engineering management and enable students to understand managerial functions in technology-oriented organizations.
- To provide students with tools and techniques for planning, organizing, leading, and controlling engineering projects and resources.
- To develop an entrepreneurial mindset by exposing students to innovation, opportunity identification, startup ecosystems, and venture creation.
- To equip students with the knowledge required to evaluate business opportunities and prepare sustainable technology-based business models and ventures.

Course Outcomes

- CO1: Understand the functions, responsibilities, and practices of managers in engineering and technology organizations.
- CO2: Apply basic management, project management, and decision-making tools for effective management of engineering projects and teams.
- CO3: Analyze entrepreneurial opportunities, innovation processes, and startup ecosystems for technology-driven ventures.
- CO4: Develop a preliminary business plan incorporating market, financial, legal, and sustainability considerations.

Syllabus

Engineering Management: Meaning, definition, scope and significance of management in engineering organizations; Evolution of management thought; Contributions of F.W. Taylor, Henry Fayol, Elton Mayo and Behavioral School; Functions of management; Levels of management; Managerial roles and skills; Strategic thinking; Management lessons from Indian ethos and philosophy; Teamwork, productivity, effectiveness and efficiency.

Planning and Organizing: Planning process, objectives, policies, strategies and decision making; Management by Objectives (MBO); Organizing principles; Organizational structures and design; Departmentation; Centralization and decentralization; Delegation of authority; Span of control; Staffing and human resource management; Team building; Communication and coordination in engineering organizations.

Leadership, Operations and Project Management: Leadership concepts, leadership styles and motivation theories; Directing and controlling functions; Operations management fundamentals; Productivity and quality management; Lean concepts and continuous improvement; Engineering project management; Project life cycle; Work Breakdown Structure (WBS); PERT and CPM techniques; Resource allocation; Cost estimation; Risk management; Project monitoring and control.

Entrepreneurship and Venture Creation: Entrepreneurship concepts and characteristics; Entrepreneurial competencies; Creativity, innovation and design thinking; Opportunity identification and feasibility analysis; Startup ecosystem in India; Technology entrepreneurship; Incubation and acceleration; Business Model Canvas; Customer discovery and value proposition; Marketing fundamentals for startups; Sources of finance including bootstrapping, angel investment, venture capital and crowdfunding; Intellectual Property Rights (IPR); Business planning; Sustainable and social entrepreneurship; Case studies of successful technology ventures.

Textbooks / References

1. Engineering Management, Addison-Wesley, 1998.
2. Essentials of Management, Tata McGraw-Hill, 1998.
3. Management, Pearson Education, 6th Edition, 2004.
4. Management, Pearson Education, 15th Edition, 2022.
5. Principles of Management, Tata McGraw-Hill, 2017.
6. Fundamentals of Business Organisation and Management, Sultan Chand & Co., 2019.
7. Entrepreneurship, McGraw-Hill Education, 2023.
8. Entrepreneurship Development, S. Chand Publishing, 2022.
9. Business Model Generation, Wiley, 2010.
10. Technology Ventures: From Idea to Enterprise, McGraw-Hill Education, 2020.
11. Innovation and Entrepreneurship, Harper Business, 2006.
12. The Lean Startup, Crown Business, 2011.

CSC211 Computer Networks [2-1-0-3]

Course Objectives

- To understand the principles, architecture, and operation of modern computer networks.
- To study networking protocols and services across different layers of the TCP/IP model.
- To understand routing, switching, and transport mechanisms in computer networks.
- To explore wireless networking technologies and emerging networking paradigms.
- To analyze network performance, reliability, and security mechanisms.
- To provide the foundation required for network design, administration, and advanced networking studies.

Course Outcomes

- CO1: Explain networking architectures, communication models, and protocol layering principles.
- CO2: Analyze data transmission, error control, flow control, and medium access mechanisms in communication networks.
- CO3: Apply Internet addressing, routing, and transport protocols for network communication and management.
- CO4: Explain the operation of common application-layer protocols and network services.
- CO5: Assess wireless networking technologies, network security threats, and emerging networking paradigms.

Syllabus

Introduction to Computer Networks: Evolution of computer networks, network components and network types, network topologies, layered network architecture, OSI reference model, TCP/IP protocol suite, Internet standards and RFCs.

Data Communication and Data Link Layer: Data transmission fundamentals, transmission media, encoding techniques, framing, error detection and correction, flow control, sliding window protocols, HDLC and PPP protocols, medium access control protocols, switching concepts.

Network Layer and Routing: Internet Protocol (IP), addressing and subnetting, ARP, ICMP, DHCP, CIDR and NAT, routing principles, distance vector and link state routing, RIP, OSPF and BGP, introduction to Software Defined Networking (SDN).

Transport and Application Layers: UDP and TCP, reliable data transfer, flow control, congestion control, Quality of Service (QoS), DNS, HTTP and HTTPS, electronic mail protocols, multimedia networking basics.

Wireless, Network Security and Emerging Networks: Wireless LAN and Wi-Fi standards, network security fundamentals, firewalls and intrusion detection concepts, Internet of Things (IoT) networking, edge computing, and future Internet architectures.

Textbooks / References

1. Kurose, J. F. and Ross, K. W., *Computer Networking: A Top-Down Approach*, 8th Edition, Pearson, 2021.
2. Tanenbaum, A. S. and Wetherall, D. J., *Computer Networks*, 6th Edition, Pearson, 2021.

3. Peterson, L. L. and Davie, B. S., *Computer Networks: A Systems Approach*, 6th Edition, Morgan Kaufmann, 2021.
4. Forouzan, B. A., *Data Communications and Networking*, 5th Edition, McGraw-Hill Education, 2013.
5. Comer, D. E., *Internetworking with TCP/IP: Principles, Protocols, and Architecture*, 6th Edition, Pearson, 2013.

CSC212 Design and Analysis of Algorithms [2-1-0-3]

Course Objectives

- Understand the principles of algorithm design and analysis.
- Analyze the correctness and efficiency of algorithms using appropriate mathematical techniques.
- Understand fundamental algorithmic paradigms such as divide-and-conquer, greedy methods, and dynamic programming to solve computational problems.
- Develop solutions using graph algorithms.
- Understand computational complexity, NP-completeness, and approaches for solving intractable problems.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Analyze the time and space complexity of algorithms using asymptotic and amortized analysis techniques.
- CO2: Design and implement algorithms using divide-and-conquer, greedy, and dynamic programming paradigms.
- CO3: Apply graph algorithms to solve real-world computational problems.
- CO4: Classify computational problems based on complexity classes and explain concepts of NP-completeness and approximation algorithms.

Syllabus

Introduction to algorithm design paradigms: Branch and Bound, Backtracking.
Divide and Conquer: Analysis by Recurrence Relations Methods for solving recurrence: Recursion Tree, Substitution Method, Master Theorem. Applications of the Master Theorem.
Greedy Algorithms: Optimization problems, optimal substructures, greedy choice, Applications of advanced data structures - Union-Find, Fibonacci heap.
Dynamic Programming: Reusing work across sub computations, optimal substructures, overlapping subproblems, ordering of subproblems and memoization. **Asymptotic Analysis of algorithms**: Asymptotic notations, Insertion Sort- Proof of Correctness and Complexity.
Amortized Analysis of algorithms: Aggregate Method, Accounting Method, Potential Method, Dynamic Tables– Balanced Trees.
Graph Algorithms: Topological Sorting, Strongly Connected Components, Minimum Spanning Tree, Shortest Path. **Intractable Problems**: Polynomial Time– class P, Polynomial Time Verifiable Algorithms– class NP, NP completeness and reducibility, NP Hard Problems, NP completeness proofs.
Introduction to Approximation Algorithms.

Textbooks / References

1. Introduction to Algorithms, T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, 3rd ed. Cambridge, MA: MIT Press, 2009.
2. Algorithms, 1st Edition, Sanjoy Dasgupta, Christos Papadimitriou, Umesh Vazirani, 2024.
3. The Design and Analysis of Algorithms, Dexter C. Kozen, Monographs in Computer Science, Springer New York, NY, 2011.
4. Algorithm Design, Kleinberg, J., & Tardos, E., Pearson/ Addison-Wesley, 2006.

CSC213 Theory of Computation [3-1-0-4]

Course Objectives

The primary objectives of this course are to enable students to:

- Understand the abstract mathematical models of computation and how they relate to modern computer hardware and software design (such as compilers).
- Analyze and Design various types of automata, including Deterministic Finite Automata (DFA), Nondeterministic Finite Automata (NFA), Pushdown Automata (PDA), and Turing Machines (TM) for specified languages.
- Master the Relationship between formal languages, grammars, and abstract machines across the Chomsky hierarchy.
- Apply Mathematical Tools, such as the Pumping Lemma, to rigorously prove the limitations of specific computational models.
- Distinguish between decidable and undecidable problems, and gain a foundational understanding of computational complexity classes.

Course Outcomes

By the end of this course, learners will be able to:

- CO1: Design, minimize, and interconvert Finite State Machines, regular expressions, and regular grammars.
- CO2: Apply formal mathematical tools, such as the Pumping Lemma, to prove the limitations of regular and context-free languages.
- CO3: Construct, simplify, and transform Context-Free Grammars and design equivalent Pushdown Automata models.
- CO4: Synthesize standard and variant Turing Machines to compute functions and classify complex languages within the Chomsky Hierarchy.

- CO5: Evaluate the decidability of computational problems using reductions and analyze foundational principles of Complexity Theory.

Syllabus

Introduction: Notion of formal language- Strings, Alphabet, Language, Operations, Finite State Machine, definitions, finite automaton model, acceptance of strings and languages.

Finite Automata – deterministic and nondeterministic, Regular operations, Regular Expression, equivalence between DFA, NFA and REs, Interconversions of DFA, NFA and REs, minimization of FSM, Non regular languages and pumping lemma, Moore and Mealy machines. Regular grammars, right linear and left linear grammars equivalence between regular linear grammar and FA, inter conversion between RE and RG.

Context free grammars: Derivation, parse trees. Language generated by a CFG. Eliminating useless symbols, productions, and unit productions, Chomsky Normal Form, Non CFLs and pumping lemma for CFLs.

Pushdown automata: Definition, instantaneous description as a snapshot of PDA computation, notion of acceptance for PDAs, Equivalence of PDA and CFG, Properties of CFLs, DCFLs.

Turing machine: definition, model, design of TM, Computable Functions, recursive enumerable language, Church's Hypothesis, Counter machine, types of TM's, RAM machine, Configuration graph, closure properties of decidable languages, decidability properties of regular languages and CFLs.

Undecidability and classes of problems: Reductions, Chomsky hierarchy of languages, Rice's Theorem, Universal Turing Machine, un-decidability of post's correspondence problem, introduction to complexity theory.

Textbooks / References

1. MSipser, Introduction to the Theory of Computation, Second Edition, Thomson, 2005.
2. Lewis H.P. and Papadimition C.H. Elements of Theory of Computation, Prentice Hall of India, Fourth Edition, 2007.
3. S. Arora and B. Barak, Computational Complexity: A Modern Approach, Cambridge University Press, 2009.
4. C. H. Papadimitriou, Computational Complexity, Addison-Wesley Publishing Company, 1994.
5. D. C. Kozen, Theory of Computation, Springer, 2006.
6. D. S. Garey and G. Johnson, Computers and Intractability: A Guide to the Theory of NP- Completeness, Freeman, New York, 1979.
7. J Hopcroft, JD Ullman and R Motwani, Introduction to Automata Theory, Languages and Computation, Third Edition, Pearson, 2008.

CSC214 Operating Systems [3-0-0-3]

Course Objectives

- To understand the architecture, structures, and services of operating systems.
- To study process management, thread management, CPU scheduling, synchronization, and deadlock handling techniques.
- To understand memory management and virtual memory mechanisms.
- To analyze file systems, disk management, and I/O management techniques.
- To introduce Linux-based systems, multicore systems, and virtualization concepts.
- To develop analytical and problem-solving skills related to operating system resource management and performance.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Explain the structure, services, and operating modes of modern operating systems.
- CO2: Analyze process management, thread management, CPU scheduling, and inter-process communication mechanisms.
- CO3: Apply synchronization and deadlock handling techniques in concurrent systems.
- CO4: Evaluate memory management and virtual memory management strategies used in operating systems.
- CO5: Analyze file systems, disk management, I/O management, and resource management techniques.
- CO6: Understand Linux-based operating system environments, virtualization concepts, containers, and contemporary operating system technologies.

Syllabus

Introduction and Operating System Structures: Evolution and functions of operating systems, Operating system services, system calls, and command interpreter (shell). Modes: User mode and kernel mode. Operating system structures: Monolithic, Layered, Microkernel, and Hybrid architectures. Brief introduction to Linux/xv6 operating systems.

Processes, Threads, and CPU Scheduling: Processes and Process Control Block (PCB), Process states and context switching, Threads and multithreading models, and Inter-process communication (IPC). CPU Scheduling: Scheduling Criteria, Scheduling Algorithms, Multiple-Processor Scheduling, Algorithm Evaluation, and Process Scheduling Models. Introduction to Linux scheduling.

Process Synchronization and Deadlocks: Race conditions and critical section problem, Hardware support for synchronization. Mutexes, semaphores, monitors, condition variables, and classical synchronization problems. **Deadlocks:** System Model, Characterization, Prevention. Avoidance, Detection, and Recovery.

Memory Management: Memory Management Background, Swapping, Contiguous and non-contiguous memory allocation, Paging and segmentation, Virtual memory concepts, and Demand paging. Page Replacement Algorithms, Allocation of Frames, Thrashing, and Overview of Memory management in Linux/Windows systems.

File Systems and I/O Management: File concepts and access methods, directory structures, file sharing, and protection mechanisms. File-system structure and implementation, file allocation methods, and free-space management.

Disk structure and disk scheduling concepts: buffering and caching; RAID concepts; application I/O interface; kernel I/O subsystem; and an overview of journaling file systems.

Modern Operating System Concepts Introduction to virtualization, Basics of containers, Protection and security mechanisms, and Overview of mobile and real-time operating systems.

Textbooks / References

1. Abraham Silberschatz, Peter B. Galvin, Greg Gagne, Operating System Concepts, Wiley.
2. Andrew S. Tanenbaum and Herbert Bos, Modern Operating Systems, Pearson.
3. Robert Love, Linux System Programming, O'Reilly.
4. W. Richard Stevens and Stephen A. Rago, Advanced Programming in the UNIX Environment, Addison-Wesley.
5. Thomas Anderson and Michael Dahlin, Operating Systems: Principles and Practice.

CSL212 Algorithm Design Lab [0-0-2-1]

Course Objectives

- Analyze and compare the performance of algorithms through experimentation.
- Apply algorithm design paradigms to solve computational problems.
- Gain experience with graph algorithms.
- Strengthen problem-solving and programming skills through algorithmic projects and case studies.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Implement and evaluate the performance of fundamental algorithms and data structures.
- CO2: Apply divide-and-conquer, greedy, dynamic programming, and graph-based techniques to solve computational problems.
- CO3: Design efficient algorithmic solutions and analyze their correctness and complexity.

Syllabus

Implementation and Analysis of Basic Algorithms: Insertion Sort, Merge Sort, Binary Search,

Asymptotic and Recurrence Analysis: Experimental verification of asymptotic growth, Solving recurrence relations and complexity comparison

Graph Algorithms like Topological Sorting, Strongly Connected Components, Minimum Spanning Tree (Prim's and Kruskal's Algorithms), Shortest Path Algorithms

Greedy Algorithms like Interval Scheduling, Fractional Knapsack, Huffman Coding, Minimum Spanning Tree Applications

Dynamic Programming algorithms like Rod Cutting Problem, Matrix Chain Multiplication, Longest Common Subsequence, Machine Scheduling Problems, Bellman-Ford and Floyd-Warshall Algorithms

Complexity and Computational Problems: Problem classification (P, NP, NP-Complete, NP-Hard), Simple reductions and complexity analysis

Approximation: Implementation of basic approximation algorithms Mini Project / Case Study (optional): Design, implementation, analysis, and presentation of an algorithmic solution to a real-world problem.

Textbooks / References

1. The algorithm design manual, Skiena, S. S. (2nd ed.). Springer (2008).
2. Algorithms, 1st Edition, Sanjoy Dasgupta, Christos Papadimitriou, Umesh Vazirani, 2024.
3. Algorithm Design, Kleinberg, J., & Tardos, E., Pearson/ Addison-Wesley, 2006.

CSL211 Computer Networks Lab [0-0-2-1]

Course Objectives

- To provide practical exposure to networking protocols, services, and communication mechanisms.
- To develop skills in network programming, protocol analysis, and network simulation.
- To enable students to analyze, configure, and troubleshoot networked systems using modern networking tools.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Use networking tools to analyze network traffic and protocol behavior.
- CO2: Develop client-server applications using socket programming.
- CO3: Simulate and evaluate networking protocols using network simulators.
- CO4: Configure, troubleshoot, and evaluate network communication scenarios.

Syllabus

Network Programming

1. Introduction to Linux Networking Commands
2. Socket Programming – TCP Client/Server
3. Socket Programming – UDP Client/Server
4. Multi-client Communication Applications
5. Remote Procedure Call (RPC) Concepts
6. Implementation of Routing Algorithms

Network Analysis

7. Packet Capture and Analysis using Wireshark
8. Measurement of Network Performance Metrics

Network Simulation

10. Introduction to NS3 Simulator
11. Point-to-Point Network Simulation
12. Routing Protocol Simulation
13. Mini Project on Network Design and Performance Evaluation

Textbooks / References

1. Panwar, S., Mao, S., Ryoo, J.-D., and Li, Y., *Computer Networking Fundamentals*, Cambridge University Press, 2024.
2. Stevens, W. R., *UNIX Network Programming.*
3. Wireshark Documentation.
4. NS-3 Documentation.

CSL214 Operating Systems Lab [0-0-2-1]

Course Objectives

- To provide practical understanding of operating system concepts.
- To develop shell scripting and Linux system programming skills.
- To implement process management, thread synchronization, and inter-process communication techniques.
- To simulate CPU scheduling, deadlock handling, and memory management algorithms.
- To familiarize students with Linux utilities for file handling, disk management, and process/resource monitoring.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Use Linux commands and shell scripting for system operations.
- CO2:Develop programs using process creation and Inter Process Communication mechanisms.
- CO3:Implement synchronization techniques using mutexes and semaphores.
- CO4: Simulate CPU scheduling, deadlock handling, and memory management algorithms.
- CO5: Perform file handling, directory operations, and disk scheduling in Linux environments.
- CO6: Analyze process behavior and operating system resource utilization using Linux monitoring tools.

Syllabus

1. Familiarization with Linux environment and shell commands.
2. Shell scripting: File handling, Pattern matching, and Process automation.
3. Process creation using fork (), exec (), and wait ().
4. Inter-process communication using Pipes, Shared memory, and Message queues.
5. Simulation of CPU scheduling algorithms, including FCFS, SJF, Priority Scheduling, and Round Robin.
6. Thread creation using POSIX threads
7. Synchronization using Mutexes and Semaphores
8. Implementation of Producer–Consumer and Reader–Writer problems using synchronization primitives.
9. Deadlock avoidance using Banker's Algorithm.
10. Simulation of contiguous and non-contiguous memory allocation, and Page replacement algorithms: FIFO, LRU, and Optimal.
11. File handling, process monitoring, and directory operations in Linux,
12. Disk scheduling simulations.

Textbooks / References

1. Abraham Silberschatz, Peter B. Galvin, and Greg Gagne, Operating System Concepts, Wiley.
2. Robert Love, Linux System Programming, O'Reilly.
3. Andrew S. Tanenbaum and Herbert Bos, Modern Operating Systems, Pearson.
4. W. Richard Stevens and Stephen A. Rago, Advanced Programming in the UNIX Environment, Addison-Wesley.
5. Michael Kerrisk, The Linux Programming Interface, No Starch Press.

SEMESTER IV

MAC223 Mathematical Primitives for Cyber Security [3-1-0-4]

Course Prerequisites

Student should have a passing grade in Calculus and Linear Algebra (MAC111) or the instructor's approval.

Course Description

This course introduces the mathematical foundations underlying modern cryptography, post-quantum cryptography, and quantum computing. It covers number theory, abstract algebra, elliptic curve algebra, advanced linear algebra, complex vector spaces, Hilbert spaces, and Fourier analysis. The course equips students with the mathematical tools required to understand cryptographic constructions, computational hardness assumptions, quantum state representations, and the mathematical principles behind quantum algorithms.

Course Objectives

- Develop a strong foundation in number theory and computational mathematics required for classical and post-quantum cryptographic systems.
- Understand and apply algebraic structures and elliptic curves in the design and analysis of cryptographic protocols.
- Acquire advanced linear algebra concepts that form the mathematical basis of quantum computing.

- Analyse complex vector spaces and Hilbert spaces for representing quantum states and transformations.

Course Outcomes

- CO1: Apply number-theoretic techniques to solve cryptographic problems.
- CO2: Analyse computational hardness assumptions and explain their significance in modern and post- quantum cryptography.
- CO3: Utilize algebraic structures to model and evaluate cryptographic constructions.
- CO4: Perform computations involving advanced linear algebra concepts used in quantum computation.
- CO5: Represent and manipulate quantum states using complex vector spaces, Hilbert spaces
- CO6: Analyse the role of DFT and QFT in cryptographic and quantum algorithms and explain their applications in algorithms such as Shor&39 algorithm

Syllabus

Number Theory: Divisibility, Primes, GCD, Modular Arithmetic, Euler's Theorem, Fermat's Little Theorem, Chinese Remainder Theorem, Quadratic Residues, Legendre symbol, Jacobi symbol, Primality Testing (Miller-Rabin), Integer Factorization, Discrete Logarithm Problem, Lattice Basics, SVP, CVP.
Abstract Algebra ,Elliptic Curve Algebra and Advanced Linear Algebra: Groups, Rings, Fields, Finite Fields ($GF(2^n)$), Homomorphisms, Polynomial Rings, Elliptic Curves over finite fields, Point Addition, Scalar multiplication,Tensor products, Kronecker Product, Unitary Matrices, Hermitian Matrices, Spectral Decomposition.
Complex Numbers & Hilbert Spaces: Complex Plane, Polar Form, Euler's Formula, Bra-Ket Notation, Operators, Projectors.
Fourier Analysis: Discrete Fourier Transform, Quantum Fourier Transform (QFT).

Textbooks

1. David M.Burton, Elementary Number Theory, McGraw Hill, 7th Edition, 2011.
2. John B. Fraleigh, A first Course in Abstract Algebra, Pearson, 7th Edition, 2013.
3. Sheldon Axler, Linear Algebra Done Right, Springer, 4th Edition, 2024.

References

1. David S. Dummit and Richard M. Foote, Abstract Algebra, Wiley, 3rd Edition, 2004
2. Steven Roman, Advanced Linear Algebra, Springer, 3rd Edition, 2007
3. Roger A. Horn, Charles R. Johnson, Matrix Analysis, Cambridge University Press, 2nd Edition, 2013
4. Victor Shoup,A Computational Introduction to Number Theory and Algebra, Cambridge University Press, 2nd Edition, 2015
5. Michael A. Nielsen and Issac L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press, 10th Edition, 2011.

HMC221 Engineering Economics, Finance & Business Analytics [2-0-0-2]

Course Objectives

- To introduce the fundamental concepts of engineering economics and their application in engineering decision-making.
- To provide students with knowledge of financial management principles, financial markets, and investment evaluation techniques.
- To familiarize students with business analytics concepts, data-driven decision making, and analytical tools used in modern organizations.
- To develop the ability to evaluate economic, financial, and business alternatives for effective managerial and entrepreneurial decision making.

Course Outcomes

- CO1: Understand the principles of engineering economics, cost analysis, and economic decision-making relevant to engineering and technology sectors.
- CO2: Apply financial concepts and techniques for budgeting, investment appraisal, and financial planning.
- CO3: Analyze business data using descriptive, predictive, and prescriptive analytics approaches to support managerial decisions.
- CO4: Evaluate business opportunities and organizational performance using economic, financial, and analytical tools.

Syllabus

Engineering Economics: Nature and scope of engineering economics; Economic decision making in engineering; Demand, supply and market equilibrium; Cost concepts and cost analysis; Fixed and variable costs; Break-even analysis; Time value of money; Simple and compound interest; Cash flow diagrams; Economic comparison of alternatives; Present worth, annual worth and future worth methods; Depreciation concepts and methods; Inflation and its impact on economic decisions; Replacement and maintenance analysis.

Financial Management and Financial Markets: Introduction to finance and financial management; Objectives and functions of financial management; Financial statements and financial analysis; Ratio analysis; Working capital management; Capital budgeting; Sources of finance; Cost of capital; Risk and return concepts; Financial planning and budgeting; Overview of financial markets and institutions; Equity, debt and mutual funds; Basics of investment management and portfolio concepts; Venture capital and startup financing.

Business Analytics Fundamentals: Introduction to business analytics; Role of analytics in business decision making; Types of analytics—descriptive, diagnostic, predictive and prescriptive analytics; Data collection and data visualization; Measures of central tendency and dispersion; Probability concepts; Hypothesis Testing; Correlation and regression analysis; Forecasting techniques; Data-driven decision making; Introduction to spreadsheet-based analytics and business intelligence tools.

Applications of Analytics in Business and Engineering: Analytics for operations, marketing, finance and human resources; Key Performance Indicators (KPIs); Dashboard reporting; Decision trees and risk analysis; Data ethics and governance; Artificial Intelligence and analytics in business; Case studies in engineering economics, financial decision making and business analytics; Applications in startups, technology ventures and digital enterprises.

Textbooks / References

1. Engineering Economy, McGraw-Hill Education, 2018.
2. Engineering Economy, Pearson Education, 2020.
3. Principles of Engineering Economic Analysis, Wiley, 2012.
4. Fundamentals of Financial Management, Cengage Learning, 2021.
5. Financial Management, Vikas Publishing House, 2021.
6. Financial Management: Theory and Practice, McGraw-Hill Education, 2023.
7. Business Analytics, Pearson Education, 2020.
8. Business Analytics: Data Analysis and Decision Making, Cengage Learning, 2019.
9. Data Science for Business, O'Reilly Media, 2013.
10. Business Intelligence, Analytics, and Data Science, Pearson Education, 2018.
11. Business Analytics: The Science of Data-Driven Decision Making, Wiley India, 2017.
12. Managerial Economics, Oxford University Press, 2020.
13. Managerial Economics, McGraw-Hill Education, 2022.
14. Statistics for Business and Economics, Pearson Education, 2020.
15. Competing on Analytics, Harvard Business Review Press, 2017.

CAC221 Machine Learning and Data Analysis [2-1-0-3]

Course Objectives

- To introduce the mathematical and statistical foundations required for machine learning and data analysis.
- To develop skills in exploratory data analysis, preprocessing, and visualization techniques.
- To provide knowledge of supervised, unsupervised, and reinforcement learning methods.
- To enable students to design, implement, and evaluate machine learning models.
- To develop analytical and problem-solving skills for intelligent data-driven applications.

Course Outcomes

- CO1: Understand the mathematical and statistical foundations of machine learning.
- CO2: Perform exploratory data analysis and preprocessing on datasets.
- CO3: Apply supervised and unsupervised learning algorithms for predictive analytics.
- CO4: Evaluate and optimize machine learning models using appropriate metrics.
- CO5:Develop intelligent data-driven solutions for real-world applications.

Syllabus

Mathematical Foundations and Exploratory Data Analysis Overview of Linear Algebra: Scalars, vectors, and matrices, Matrix operations, Systems of linear equations, Eigenvalues and eigenvectors, Matrix decomposition. Overview of Probability and Statistics: Random variables, Probability distributions, Descriptive statistics, Measures of central tendency, Measures of dispersion, Sampling techniques. Overview of Optimization Techniques: Optimization fundamentals, Objective functions and constraints, Gradient and Hessian matrix, Least squares optimization, Gradient descent methods. Exploratory Data Analysis (EDA): Data collection and preprocessing, Handling missing values, Data transformation, Outlier detection, Correlation analysis, Data visualization techniques.

Statistical Learning and Model Selection Introduction to Machine Learning: Definitions and goals of machine learning, History and applications of machine learning, Types of machine learning. Statistical Learning: Linear classification, Classification errors, Regression techniques, Predictive analytics concepts. Model Selection Techniques: Empirical Risk Minimization (ERM), Structural Risk Minimization (SRM), Cross-validation techniques, Regularization methods, Bias-variance tradeoff, Performance evaluation metrics.

Mining Frequent Patterns, Associations, and Correlations Association Rule Mining: Frequent itemset mining, Apriori algorithm, FP-growth concepts, Association rules, Support and confidence measures, Market basket analysis applications.

Supervised Learning Fundamentals: Supervised learning concepts, Generative and discriminative learning, Parametric and non-parametric learning. Regression Techniques: Linear regression, Model fitting and prediction. Classification Techniques: Decision trees, k-Nearest Neighbors (KNN), Naïve Bayes classifier with Laplace smoothing, Logistic regression with Maximum Likelihood Estimation Model Evaluation:Regression Metrics and Confusion matrix, Precision, Recall, F1-score, ROC curve.

Unsupervised Learning: Unsupervised learning concepts, Clustering techniques, K-Means clustering, Hierarchical clustering. Dimensionality Reduction: Principal Component Analysis (PCA), Feature selection and feature extraction techniques.

Textbooks / References

1. Christopher M. Bishop, Pattern Recognition and Ma-

chine Learning, Springer.
2. Tom M. Mitchell, Machine Learning, McGraw-Hill.
3. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow, O'Reilly Media.
4. Gareth James et al., An Introduction to Statistical Learning, Springer.
5. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning, Springer.
6. Gilbert Strang, Introduction to Linear Algebra, Wellesley-Cambridge Press.
7. 3. Stephen Boyd and Lieven Vandenberghe, Convex Optimization, Cambridge University Press.

CSC222 Language Processors [2-1-0-3]

Course Objectives

- To introduce the major concept areas in language processors and know the various types of Language processors
- To analyze lexical analysis techniques and parsing methods used in compiler construction.
- To understand the various parsing algorithms and comparison of the same.
- To gain knowledge about the various code generation principles.
- To understand the necessity for code optimization

Course Outcomes

- CO1:Apply the knowledge of LEX, YACC, ANTLR tools to develop a scanner and parser.
- CO2: Suggest the necessity for appropriate code optimization techniques.
- CO3: Design a Translator for any programming language
- CO4: Design compiler components and recommend suitable code optimization techniques for efficient program execution.
- CO5: Design a machine learning translators for real-time applications

Syllabus

Introduction to Language Processing: Translators, Assemblers, Compilers, and Interpreters. Phases of a Compiler, System Software, Linkers, Loaders, and macro processors. Scanners: Tokens, lexemes, and patterns, Input Buffering Strategies, Scanner generations tools like Lex or Flex or Antlr.

Parser: CFG, Derivations, parse trees, and ambiguity, Top-Down Parsing: Recursive Descent Parsing, Predictive Parsing, and LL(1) grammars, Bottom-Up Parsing: Shift-Reduce Parsing, Operator Precedence Parsing, Different LR Parsers construction, Parser Generators: Using tools Yacc or Bison.4, or Antlr.

Syntax-Directed Translation (SDT) Syntax-Directed Definitions: Inherited and synthesized attributes, S-Attributed and L-Attributed Definitions. Translation Schemes: Construction of syntax trees. Intermediate Code Generation: Intermediate Representations (IR), Declarations and Assignment Statements, Boolean Expressions and Case Statements.

Code Optimization: Sources of optimization, Optimization Techniques, and data-flow analysis. Code Generation and Runtime Environment Runtime Storage Management: Activation trees, stack allocation, and heap management. Target Machine Code: Instruction selection, register allocation, and basic block generation.

Understanding AI specific compiler Architectures: Machine learning compilers, Multi-Level Intermediate Representations, Quantum Compilers.

Textbooks / References

1. Mössenböck, Hanspeter. Compiler Construction: Fundamentals and Applications. Springer Nature, 2025.
2. Cooper, K. D., &; Torczon, L. (2022). Engineering a compiler (3rd ed.). Morgan Kaufmann.
3. Thain, Douglas. Introduction to compilers and language design. Lulu. com, 2016.
4. Aho A.V., Lam M. S., Sethi R., and Ullman J. D., Compilers: Principles, Techniques and Tools, Pearson Education, 2007.

Web References

1. Machine Learning Compilation, https://book.mlc.ai/
2. Bettina Heim, Mathias Soeken, Sarah Marshall, Chris Granade, Martin Roetteler, Alan Geller, Matthias Troyer & Krysta Svore. Quantum programming languages. Nat Rev Phys 2, 709–722 (2020). https://doi.org/10.1038/s42254-020-00245-7

CSC223 Database Management Systems [2-1-0-3]

Course Objectives

- To understand the fundamentals of database systems and data modeling techniques.
- To design and implement relational databases using SQL and advanced SQL features.
- To apply normalization techniques for efficient database design.
- To understand indexing, query processing, transaction management, and database security mechanisms.
- To introduce modern database technologies including NoSQL, cloud databases, and vector databases.

Course Outcomes

- CO1: Explain database concepts, architectures, and data modeling techniques.
- CO2: Design relational database schemas and write SQL queries for data management and retrieval.
- CO3: Apply normalization techniques to develop efficient database designs.
- CO4: Analyze indexing, query processing, transaction management, and security mechanisms in database systems.

- CO5: Identify and apply modern database technologies such as NoSQL, cloud databases, and vector databases for emerging applications.

Syllabus

Database Modelling: Database system concepts and architecture, Data models, Schemas and instances, Database system environment, Data modelling using Entity Relationship (ER) model, Enhanced ER model, Specialization, Generalization.

Relational Databases: Relational Model, Relational database design using ER to relational mapping, Constraints, Relational algebra, Relational calculus, SQL, Advanced SQL (Views, Stored Procedures, Triggers, CTEs, Window Functions), PL/SQL.

Database Design: Database design theory and methodology, Functional dependencies and normalization of relations, Normal Forms, Properties of relational decomposition, Algorithms for relational database schema design, Denormalization and scalable schema design.

Indexing and Query Processing in DBMS: Single level and multi-level indexing, Dynamic Multi-level Indexing using B Trees and B+ Trees, Hash-Based Indexing (Static and Dynamic Hashing), Query processing, Query evaluation plans, Query optimization.

Database Transactions and Security: Transaction processing concepts, ACID properties, Schedules and serializability, Concurrency control, Two Phase Locking Techniques, Deadlock, Database recovery concepts and techniques. Database Security: Introduction to database security, Authentication and authorization, Access control mechanisms, Privileges using GRANT and REVOKE.

Advanced Database Systems and Emerging Database Technologies: Big Data overview and NoSQL database concepts, NewSQL Databases, Cloud Databases, Vector Databases for AI Applications, Emerging trends in database technologies.

Textbooks / References

1. R. Elmasri and S. B. Navathe, *Fundamentals of Database Systems*, 7th Edition, Pearson, 2022.
2. R. Ramakrishnan and J. Gehrke, *Database Management Systems*, 3rd Edition, McGraw-Hill Higher Education, 2003.
3. M. Kleppmann, *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*, 1st Edition, O'Reilly Media, 2017.

CSC224 Advanced Data Structures [2-1-0-3]

Course Objectives

The course aims to:

- Master Advanced Data Structures.
- Optimize Information Retrieval.
- Solve Network Connectivity.
- Design Probabilistic Solutions.

Course Outcomes

After successful completion of the course, students will be able to:

- CO1: Analyze and Implement Self-Balancing Search Structures: Demonstrate the ability to implement and evaluate balanced search trees to optimize disk access and dynamic data retrieval.
- CO2: Evaluate Advanced Heap Architectures: Compare the structural and amortized performance of Binomial and Fibonacci Heaps to optimize priority queue operations in network and graph algorithms.
- CO3: Design Efficient Pattern Matching and String Processing Systems: Apply advanced string algorithms (KMP, Rabin-Karp) and prefix/suffix structures (Tries, Suffix Trees/Arrays) to solve complex text-processing and biological data problems.
- CO4: Formulate Graph Connectivity Solutions: Deploy advanced graph algorithms to identify strongly connected components, detect cycles, and map topology in complex networks.
- CO5: Synthesize and Apply Probabilistic Algorithmic Strategies.

Syllabus

Self-Balancing Trees & Extensions but not limited to: Red-Black Trees, B-Trees & B+ Trees.

Advanced Heaps & Priority Queues but not limited to: Fibonacci Heaps , Binomial Heaps.

String and Prefix Processing but not limited to:Tries (Prefix Trees), Suffix Trees & Suffix Arrays, KMP Algorithm, Rabin Karp algorithm.

Algorithms on Graph connectivity but not limited to: Tarjan's algorithm and Kosaraju's algorithm for detection of strongly connected components, Detect cycles in an undirected graph.

Network Flow: Max Flow min cut theorem, Ford Fulkerson algorithm, Bipartite matching using flows.

Introduction to Randomized Algorithms but not limited to: Las Vegas Algorithms -Quick Sort, Monte Carlo Algorithms- Miller-Rabin primality test.

Textbooks / References

1. Introduction to Algorithms (4th Edition) by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein.
2. Algorithms (4th Edition) by Robert Sedgewick and Kevin Wayne.
3. Advanced Data Structures by Peter Brass.
4. Randomized Algorithms by Rajeev Motwani and Prabhakar Raghavan.

CSL222 Language Processors Lab [0-0-2-1]

Course Objectives

- To provide practical exposure to the implementation of various language processing components.

- To develop lexical, syntax, semantic analysis, and intermediate code generation modules using compiler construction tools.
- To familiarize students with runtime support mechanisms, code optimization techniques, and interpreter design.
- To introduce modern language processing frameworks including parser generators, intermediate representations, and AI-oriented compiler infrastructures.
- To enable students to design simple domain-specific language (DSL) translators and machine learning-assisted language processing applications.

Course Outcomes

- CO1: Develop lexical analyzers, parsers, and syntax-directed translation modules using compiler construction tools.
- CO2: Implement semantic analysis, symbol table management, intermediate code generation, and runtime support mechanisms.
- CO3: Design translators, interpreters, and simple domain-specific language processors.
- CO4: Apply code optimization techniques and experiment with modern compiler infrastructures for AI and machine learning applications.

Syllabus

- Design and implementation of lexical analyzers using LEX/FLEX or ANTLR.
- Construction of LL(1) and LR parsers using YACC/Bison or ANTLR.
- Development of Recursive Descent Parser for arithmetic expressions.
- Construction of Abstract Syntax Trees (ASTs) and Parse Trees.
- Implementation of Symbol Table Management and Semantic Analysis.
- Syntax Directed Translation for arithmetic expressions and assignment statements.
- Generation of Intermediate Code (Three Address Code, Quadruples and Triples).
- Implementation of Runtime Environment concepts including stack-based activation records and memory allocation simulation.
- Development of a simple Interpreter for arithmetic or expression-based languages.
- Design of a simple Domain Specific Language (DSL) using ANTLR.

Textbooks / References

1. Aho, A. V., Lam, M. S., Sethi, R., &; Ullman, J. D. (2007). Compilers: Principles, techniques, and tools (2nd ed.). Pearson Education.
2. Cooper, K. D., &; Torczon, L. (2022). Engineering a compiler (3rd ed.). Morgan Kaufmann.
3. Muchnick, S. S. (1997). Advanced compiler design and implementation. Morgan Kaufmann.
4. Louden, K. C., &; Lambert, K. A. (2016). Compiler construction: Principles and practice. Cengage Learning.
5. Nystrom, R. (2021). Crafting interpreters. Genever Benning.

CSL223 Database Management Systems Lab [0-0-2-1]

Course Objectives

- To understand the fundamentals of database systems and data modeling techniques.
- To design and implement relational databases using SQL and advanced database programming constructs.
- To apply normalization and schema design principles for efficient database development.
- To understand query processing, indexing, transaction management, recovery, and database security.
- To explore modern database technologies such as NoSQL, cloud databases, and vector databases for contemporary applications.

Course Outcomes

- CO1: Design conceptual and logical database schemas using ER and Enhanced ER models.
- CO2: Implement relational databases and develop SQL/PL-SQL programs for data management.
- CO3: Apply normalization techniques and evaluate database schema quality.
- CO4: Analyze indexing methods, query execution plans, and optimization strategies.
- CO5: Implement transaction management, concurrency control, security mechanisms, and modern database applications using NoSQL and emerging database technologies.

Syllabus

1. Database Design using ER/EER Models: Design ER and Enhanced ER diagrams for real-world applications.
2. Relational Database Design: Convert ER models into relational schemas and implement constraints.
3. SQL Fundamentals: Practice DDL, DML, DCL, and TCL commands.
4. SQL Queries and Joins: Write simple, nested, and join queries for data retrieval.
5. Advanced SQL: Implement views, set operators, CTEs, and window functions.
6. PL/SQL Programming: Develop programs using control structures, cursors, and exception handling.
7. Stored Procedures and Functions: Create and execute procedures and user-defined functions.
8. Database Triggers: Implement triggers for data validation and auditing.
9. Normalization and Schema Design: Apply functional dependencies and normalization techniques.
10. Indexing and Query Optimization: Create indexes and analyze query execution plans.

11. Transaction Management and Security: Perform transaction control, concurrency control, and privilege management.
12. NoSQL Database Operations: Implement CRUD operations and aggregation using MongoDB.
13. Modern Database Technologies: Explore Cloud Databases, NewSQL Databases, and Vector Databases.
14. Mini Project: Design and implement a complete database application for a real-world problem.

Textbooks / References

1. Silberschatz, A., Korth, H. F., & Sudarshan, S., *Database System Concepts*, 7th ed., McGraw-Hill Education, 2019.
2. Elmasri, R., & Navathe, S. B., *Fundamentals of Database Systems*, 7th ed., Pearson Education, 2016.
3. Viescas, J. L., & Hernandez, M. J., *SQL Queries for Mere Mortals: A Hands-On Guide to Data Manipulation in SQL*, 5th ed., Addison-Wesley Professional, 2023.
4. Beaulieu, A., *Learning SQL: Generate, Manipulate, and Retrieve Data*, 3rd ed., O'Reilly Media, 2020.
5. Kleppmann, M., *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*, 1st ed., O'Reilly Media, 2017.
6. Bradshaw, S., Brazil, E., & Chodorow, K., *MongoDB: The Definitive Guide: Powerful and Scalable Data Storage*, 3rd ed., O'Reilly Media, 2019.
7. Thusoo, A., & Sen Sarma, J., *Vector Search: Building Intelligent Search Systems with Embeddings*, 1st ed., O'Reilly Media, 2025.

CAL221 Machine Learning and Data Analysis Lab [0-0-2-1]

Course Overview

This laboratory course provides hands-on experience in Python programming, data preprocessing, exploratory data analysis, visualization, and machine learning techniques. Students implement regression, classification, clustering, association rule mining, and model evaluation methods using Python libraries to analyze real-world datasets and develop practical data-driven solutions.

Course Objectives

- Recall the core terms, concepts, and mathematical foundations used in data analysis and machine learning.
- Interpret and understand statistical methods, learning paradigms, and visualization principles.
- Apply and analyse data preprocessing, statistical, mining, and machine learning techniques on datasets.
- Synthesize and create a complete data analysis and machine learning workflow for a real problem.

Course Outcomes

- CO1: Identify the essential concepts, tools, and mathematical ideas used in data analysis and machine learning.
- CO2: Explain statistical procedures, learning methods, and visualization principles in practical contexts.
- CO3: Implement and evaluate preprocessing, prediction, clustering, mining, and visualization techniques on datasets.
- CO4: Design and develop an end-to-end analytical solution using machine learning and visualization tools.

Syllabus

1. **Introduction to Python Programming:** Installation, development environments, Python packages, and scientific computing libraries for machine learning and data analysis.
2. **Mathematical Foundations using Python:** Implementation of matrix operations, eigenvalues and eigenvectors, gradients, and probability distributions.
3. **Data Preprocessing and Feature Engineering:** Data cleaning, transformation, normalization, feature selection, and dimensionality reduction techniques.
4. **Exploratory Data Analysis (EDA):** Computation of measures of central tendency and dispersion, and exploratory analysis using Python.
5. **Regression Analysis:** Implementation of simple and multiple linear regression models with performance evaluation and error analysis.
6. **Model Selection and Generalization:** Application of cross-validation, regularization techniques, and analysis of the bias–variance trade-off.
7. **Supervised Learning:** Implementation and evaluation of parametric and non-parametric classification algorithms such as Decision Trees, Naïve Bayes, and k-Nearest Neighbors (k-NN).
8. **Unsupervised Learning:** Implementation of clustering, dimensionality reduction, and outlier detection techniques.
9. **Association Rule Mining:** Implementation of frequent pattern mining algorithms such as Apriori and FP-Growth for association rule discovery.
10. **Data Visualization:** Generation of line plots, bar charts, scatter plots, histograms, box plots, heat maps, and treemaps using Matplotlib, Seaborn, and Plotly.

Textbooks / References

1. Hastie, T., Tibshirani, R., and Friedman, J. H. The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd Edition, Springer Series in Statistics, Springer, New York, USA, 2009. ISBN: 978-0-387-84857-0 (Hardcover), ISBN: 978-0-387-84858-7 (eBook), 745 pages.
2. Bishop, C. M. Pattern Recognition and Machine Learning, Information Science and Statistics Series, Springer, New York, USA, 2006. ISBN: 978-0-387-31073-2 (Hardcover), ISBN: 978-0-387-45528-0 (eBook), 738 pages.
3. Han, J., Kamber, M., and Pei, J. Data Mining: Concepts and Techniques, 3rd Edition, Morgan Kaufmann (Elsevier), Waltham, MA, USA, 2011. ISBN: 978-0-12-381479-1, 744 pages.

4. Murphy, K. P. Machine Learning: A Probabilistic Perspective, Adaptive Computation and Machine Learning Series, The MIT Press, Cambridge, MA, USA, 2012. ISBN: 978-0-262-01802-9 (Hardcover), ISBN: 978-0-262-30432-0 (eBook), 1104 pages.
5. Géron, A. Hands-On Machine Learning with Scikit-Learn and PyTorch: Concepts, Tools, and Techniques to Build Intelligent Systems, 1st Edition, O'Reilly Media, Sebastopol, CA, USA, 2025. ISBN: 979-8341607989.

CYC221 Introduction to Cybersecurity and Cybercrimes [3-0-0-3]

Course Prerequisites

None

Course Description

This course introduces fundamental concepts of cybersecurity and cybercrimes, focusing on threats, vulnerabilities, attack mechanisms, and defensive strategies.

Course Objectives

- To understand the fundamentals of cybersecurity, threats, and vulnerabilities.
- To analyze different types of cybercrimes and their impact.
- To introduce basic security mechanisms, policies, and legal frameworks.

Course Outcomes

At the end of the course, students should be able to:

- CO1: Identify common cybersecurity threats, vulnerabilities, and attack vectors
- CO2:Explain various cyber attack techniques and malware types .
- CO3: Describe fundamental security mechanisms such as cryptography, authentication, and network security controls
- CO4: Apply basic security practices to improve the protection of digital systems and data.

Syllabus

Fundamentals of Cybersecurity: Introduction to information security, security goals (CIA triad), Principles of Cybersecurity, Threat, Vulnerability, Attacks, Threat landscape, Categories of threat actors.

Cyber Threats and Attack Mechanisms: Malware (viruses, worms, trojans, ransomware), Social engineering, Password attacks, Network-based attacks, Web application attacks,Relevant Case studies of recent cyber incidents

Security Mechanisms and Best Practices: Cyber Technologies-Encryption, Authentication, Access control, Hashing.

Network security Technologies: Firewalls, Intrusion detection systems, Endpoint protection technologies

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	2	1			1	
CO2	2		1		2	
CO3		2			1	
CO4		2		1	1	

Textbooks

1. Pfleeger, C. P., Pfleeger, S. L., and Margulies, J. Security in Computing. 5th ed., Noida: Pearson, 2018. .
2. Goutham, R. K. Cybersecurity Fundamentals. New Delhi: BPB Publications, 2025. ancis Group, 2022.

References

1. Moore, R. Cybercrime: Investigating High-Technology Computer Crime. Routledge, 2010.
2. Moallem, A. Understanding Cybersecurity Technologies: A Guide to Selecting the Right Cybersecurity Tools. 1st ed., Boca Raton: CRC Press, Taylor and Francis Group, 2022.

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CYC311 Modern Cryptography [3-0-0-3]

Course Prerequisites

Student should have a passing grade in Mathematical Primitives of Cyber Security(CYC221) or the instructor's approval.

Course Description

This course introduces the foundations of modern cryptography and its role in securing digital communication and computing systems. It covers the design and security principles of symmetric and public-key cryptography, cryptographic hash functions, authentication mechanisms, and secure communication protocols. The course also explores implementation security challenges, secure messaging concepts, and emerging privacy-preserving cryptographic techniques, providing students with both theoretical foundations and practical insights into modern cryptographic systems..

Course Objectives

- To understand the foundations and security principles of modern cryptographic systems.
- To study symmetric and asymmetric cryptographic algorithms and authentication mechanisms
- To analyse authentication mechanisms and cryptographic protocols used in secure communication.
- To understand implementation security issues and secure deployment of cryptographic systems
- To introduce emerging applications and privacy-preserving cryptographic techniques.

Course Outcomes

At the end of the course, students should be able to:

- CO1:Understand the principles, attack models, and security goals of modern cryptographic systems.
- CO2:Analyse symmetric and asymmetric cryptographic techniques and authentication mechanisms
- CO3:Evaluate cryptographic protocols used in secure communication and web security systems..
- CO4:Identify implementation-level vulnerabilities and secure cryptographic engineering practices.
- CO5:Explain modern applications of cryptography, including secure messaging and privacy-preserving technologies.

Syllabus

Introduction to cryptography, cryptanalysis, and security services. Classical cryptographic techniques and their limitations. Security goals and attack models.Introduction to security definitions and proof-based cryptography. Concepts of chosen-plaintext (CPA) and chosen-ciphertext (CCA) security.

Symmetric encryption and security notions: Introduction to pseudorandom generators (PRGs) and pseudorandom functions (PRFs). Block cipher principles. DES, AES. Modes of operation. Stream ciphers. Message Authentication Codes (MACs) and HMAC. Authenticated encryption.

Cryptographic hash functions and security properties: preimage resistance, second-preimage resistance, and collision resistance. Merkle–Damgård transformation and Sponge construction. Public-key cryptography: RSA, Diffie–Hellman key exchange, ElGamal encryption, Elliptic Curve Cryptography (ECC). Computational hardness assumptions underlying public-key cryptography. Digital signatures. Certificates and Public Key Infrastructure (PKI). Key management concepts.

Authentication protocols, TLS and its evolution, IPSec, web authentication mechanisms, secure session management, token-based authentication mechanisms, and vulnerabilities in cryptographic protocols and applications.

Implementation-level security challenges in cryptographic systems. Side-channel attacks and countermeasure concepts. Secure key management. Trusted hardware and secure boot concepts. End-to-end encryption and forward secrecy concepts in secure messaging systems. Introduction to privacy-preserving cryptographic primitives — homomorphic encryption, secure multi-party computation, and zero-knowledge proofs at a conceptual level.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	3	2				
CO2	3	2				1
CO3		3				2
CO4		3	1			1
CO5	2	2				3

Textbooks

1. Katz, Jonathan, and Yehuda Lindell. Introduction to Modern Cryptography. 3rd ed., CRC Press, 2021.

References

1. Boneh, Dan, and Victor Shoup. A Graduate Course in Applied Cryptography. 2023.
2. Rosulek, Mike. The Joy of Cryptography. 2021.
3. Ferguson, Niels, Bruce Schneier, and Tadayoshi Kohno. Cryptography Engineering: Design Principles and Practical Applications. Wiley, 2010
4. Aumasson, Jean-Philippe. Serious cryptography: a practical introduction to modern encryption. No Starch Press, Inc, 2024.
5. Paar, Christof, Jan Pelzl, and Tim Güneysu. Understanding cryptography: From established symmetric and asymmetric ciphers to post-quantum algorithms. Springer Nature, 2024
6. Stallings, William. Cryptography and Network Security: Principles and Practice. 7th ed., Pearson, 2017.

CYC312 Secure Software Engineering and DevSecOps [2-1-0-3]

Course Prerequisites

None

Course Description

This course introduces the principles and practices of secure software engineering and DevSecOps for developing resilient and trustworthy software systems. The course covers software development life cycle models, secure software development methodologies, threat modelling, secure coding practices, software security testing, and vulnerability management. Students are exposed to common software security risks, secure development frameworks, cloud-native and container security, and secure deployment practices. The course also explores DevSecOps principles, security automation, governance, risk and compliance, and the integration of security controls throughout the software development and delivery lifecycle.

Course Objectives

- Understand the fundamental concepts of software engineering, software development life cycles, and process models, and analyse their suitability for developing secure software systems.
- Apply Secure Software Development Lifecycle (Secure SDLC) practices, threat modelling techniques, and security frameworks to identify and mitigate security risks during software development.
- Understand secure coding principles, common software vulnerabilities, and software testing techniques to improve software quality and security.
- Evaluate security risks in web applications, APIs, cloud-native applications, and containerized environments using industry-recognized standards and best practices.
- Integrate security controls, automation, and compliance requirements into DevOps workflows through DevSecOps practices and secure CI/CD pipelines.

Course Outcomes

At the end of the course, students should be able to:

- CO1: Analyse software development life cycle models and select appropriate software engineering approaches for developing secure and reliable software systems.
- CO2: Apply Secure SDLC concepts, SQUARE methodology, STRIDE, and DREAD threat modelling techniques to identify security requirements and assess software security risks.
- CO3: Evaluate and compare secure software development frameworks such as Microsoft SDL, OWASP practices, and security approaches in Agile and DevOps environments.
- CO4: Identify, analyze, and mitigate common software vulnerabilities using secure coding practices, OWASP Top 10 Web and API Security Risks, CWE classifications, and CVE information.
- CO5: Implement secure deployment and maintenance practices for cloud-native applications, Docker containers, and Kubernetes environments using industry best practices.
- CO6: Integrate security controls into CI/CD pipelines and apply DevSecOps principles to automate security throughout the software delivery process.

Syllabus

Introduction to Software Engineering, Software Development Life Cycle (SDLC) and Software life cycle models: Waterfall model, Iterative waterfall model, Prototyping model, Spiral model, Agile development methodologies.
Secure Software Development Lifecycle: SQUARE process model, Threat modelling- STRIDE, DREAD, Secure SDLC frameworks: Microsoft SDL, OWASP, Security in Agile and DevOps environments.
Basic concepts in user interface design: Secure Coding practices, Common Vulnerabilities: OWASP Top 10 Web and API Security Risks, CWE and; CVE, Secure Coding guidelines; Basics of Software Testing.
Secure Deployment and Maintenance: Secure deployment in cloud- native environments, Container security: Docker, Kubernetes best practices.
DevSecOps: Integrating security into CI/CD pipelines, Governance, risk, and compliance (GRC) in software engineering.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1		3				
CO2		3	1			
CO3		3		1	1	
CO4		3	2		2	
CO5		3				3
CO6		3	2			2

Textbooks

1. Gary McGraw, Software Security: Building Security In, Addison-Wesley Professional,1 st Edition, 2006.
2. Robert C. Seacord, Secure Coding in C and C++, Addison-Wesley Professional, 2 nd Edition, 2013.
3. Adam Shostack, Threat Modeling: Designing for Security.Wiley, 1 st Edition, 2014.

References

1. Dafydd Stuttard, Marcus Pinto, The Web Application Hacker's Handbook, Wiley, 2 nd Edition, 2011.
2. Tony Hsiang-Chih Hsu, Deepak Gupta, Practical DevSecOps, Packt Publishing, 1 st Edition, 2023.
3. Liz Rice, Container Security, OReilly Media, 1 st Edition, 2020.
4. Nicole Forsgren, Jez Humble, Gene Kim, Accelerate, IT Revolution Press,1 st Edition, 2018.

CYC313 Cloud and Critical Infrastructure Security [3-0-0-3]

Course Prerequisites

Student should have a passing grade in Computer Networks (CSC211), Operating Systems (CSC214), Introduction to Cybersecurity and Cybercrimes (CYC222) and IT Workshop Lab (CSL112) or the instructor's approval. .

Course Description

This course provides an introduction to the security principles, challenges, and protection mechanisms associated with cloud computing, edge computing, and critical infrastructure systems. The course covers cloud security fundamentals, identity and access management, data protection, virtualization and container security, edge and IoT security, and industrial control system security. Students are introduced to security challenges in distributed and cyber-physical environments, including threats targeting cloud platforms, edge devices, operational technology, and critical infrastructure. The course also explores emerging approaches such as Zero Trust architectures, AI-assisted threat detection, and secure cloud-edge ecosystems through real-world case studies and contemporary security practices.

Course Objectives

- To introduce the fundamentals of cloud, edge, and critical infrastructure security
- To understand security challenges in distributed and cyber-physical environments.
- To study security mechanisms for cloud platforms, edge devices, and industrial systems.
- To analyze threats and protection strategies for critical infrastructure and Operational Technology environments
- To familiarise students with emerging trends and real-world cyberattacks on critical systems.

Course Outcomes

At the end of the course, students should be able to:

- CO1:Analyse cloud, edge, and critical infrastructure architectures and security concepts.

- CO2:Identify and analyze threats and vulnerabilities in cloud and edge systems.
- CO3:Apply security techniques for cloud services, containers and edge devices
- CO4:Evaluate security challenges in SCADA, ICS, and critical infrastructure systems.
- CO5:Design secure and resilient cloud-edge infrastructure solutions.

Syllabus

Cloud Computing and Security Fundamentals: Introduction to Cloud Computing - Characteristics and Service Models (IaaS, PaaS, SaaS) - Deployment Models: Public, Private, Hybrid Cloud - Virtualisation and Containers - Cloud Threats and Attack Surface - Shared Responsibility Model - Basics of Zero Trust Security.

Cloud Infrastructure and Data Security : Identity and Access Management (IAM) - Authentication and Multi-Factor Authentication - Cloud Network Security and Firewalls - Data Security: Encryption at Rest and in Transit - Key Management Basics - Logging, Monitoring, and Security Auditing - eBPF-based Security: kernel observability, runtime monitoring, network visibility, and threat detection - DevSecOps and Compliance

Edge Computing Security: Introduction to Edge and Fog Computing - Edge Computing Architecture - IoT and Edge Device Security - Secure Communication Protocols (MQTT, CoAP) - Lightweight Cryptography for IoT - Firmware and Device Security - Privacy Issues in Edge Computing.

Critical Infrastructure and Industrial Security : Introduction to Critical Infrastructure Protection - Industrial Control Systems (ICS) and SCADA Basics - Operational Technology (OT) Security - Common Attacks on Critical Infrastructure - Smart Grid and Industrial IoT Security - Risk Assessment and Incident Response - Case Study: Colonial Pipeline Attack.

Emerging Trends and Case Studies: Secure Multi-Cloud and Hybrid Cloud - Edge AI Security Concepts - Zero Trust for Cloud and Edge - AI-based Threat Detection - MITRE ATT&CK for ICS (Introduction) - Cloud and Infrastructure Breach Case Studies - Future Trends in Cloud and Critical Infrastructure Security.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	2	2				3
CO2		2	1			2
CO3		3				3
CO4		2	2			2
CO5		3				3

Textbooks

1. Tim Mather,Subra Kumaraswamy, Shahed Latif, Cloud Security and Privacy – O'Reilly..
2. Rajkumar Buyya, Satish Narayana Srirama, Fog and Edge Computing: Principles and Paradigms, Wiley Publications,2019
3. Eric D. Knapp, Industrial Network Security , 2nd edition, Syngress ,2014

References

1. Brian Russell, Drew Van Duren, Practical Internet of Things Security: Beat IoT security threats by strengthening your security strategy and posture against IoT vulnerabilities, Packt Open Source
2. Ronald Krutz and Russell Vines, Cloud Security: A Comprehensive Guide to Secure Cloud Computing, Wiley Publications.
3. Tyson Macaulay and Bryan Singer, Cybersecurity for Industrial Control Systems – CRC Taylor and Francis Group.
4. NIST Guide to Industrial Control Systems Security.

CYC314 IoT and Cyber-Physical Systems Security [3-0-0-3]

Course Prerequisites

SStudent should have a passing grade in Computer Networks (CSC211), Mathematical Primitives of Cyber Security(CYC221) or the instructor's approval.

Course Description

This course provides an in-depth understanding of security, trust, and authentication mechanisms for Internet of Things (IoT) and Cyber-Physical Systems (CPS). It explores the unique security challenges arising from the integration of physical devices, communication networks, cloud platforms, and intelligent applications in domains such as smart cities, industrial automation, healthcare, transportation, and critical infrastructure.

Course Objectives

- introduce the architecture, communication frameworks, and operational principles of Internet of Things (IoT) and Cyber-Physical Systems (CPS).
- To understand the security challenges, threat landscape, vulnerabilities, and attack surfaces in IoT and CPS environments.
- To study secure communication protocols, cryptographic mechanisms, and security solutions for IoT and industrial systems.
- To develop knowledge of device identity management, authentication, authorization, and trust establishment mechanisms for resource-constrained devices.
- To analyze security issues and best practices through real-world IoT and CPS case studies.

Course Outcomes

At the end of the course, students should be able to:

- CO1:Explain the architecture, communication models, protocols, and operational characteristics of IoT and Cyber-Physical Systems.
- CO2:Analyze security requirements, threats, vulnerabilities, and attack vectors in IoT, CPS, industrial control systems, and critical infrastructure environments.

- CO3:Apply secure communication protocols, cryptographic techniques, and key management mechanisms to protect IoT and CPS networks.
- CO4:Design and evaluate device identity management, authentication, authorization, and trust management mechanisms for secure IoT deployments.
- CO5:Assess security incidents, emerging threats, and real-world case studies to recommend appropriate security controls and best practices for IoT and CPS environments.

Syllabus

Fundamentals of IoT and CPS: Evolution of IoT, IIoT, Smart Cities, Industry 4.0, Introduction to IoT architecture, IoT devices and data sensing (Data acquisition, sensing and actuation), IoT connectivity (Device to device, device to cloud, device to gateway, cloud to Gateway), Wireless Sensor Networks (WSN), Machine to Machine (M2M) architecture, Introduction to Cyber-Physical Systems architecture, Industrial Control Systems (ICS)

Introduction to IoT and CPS Security: Non-repudiation in IoT, Data Classification and criteria (Public, Private, Sensitive, Confidential, Proprietary), Privacy Issues in IoT, IoT Security life cycle, Different IoT attack surfaces, Common attack methods in IoT (Botnet recruitment, Lateral movement, Data exfiltration), SCADA Security, PLC security challenges, Digital Twin Security, Supply Chain Security, Secure Firmware Updates (OTA)

IoT and CPS secure communication protocols: Secure IoT communication protocols - Application Layer Protocols (MQTT, CoAP, AMQP), Transport Layer Protocols (TLS, DTLS), Network Layer Protocols (RPL, 6LowPAN, LoRa, Bluetooth, Zigbee, Z-wave), Secure Industrial communication protocols – Modbus, DNP3, OPC-UA, PROFINET.

IoT Device Identity, Authentication and Trust Management: Device identity lifecycle management, Lightweight authentication protocols, ECC-based authentication, PUF-based authentication, Biometric based authentication, Mutual Authentication mechanisms, Multi-factor authentication for IoT, Threat assessment & Trust management in IoT communications, Secure Elements and TPMs, Trusted Execution Environments (TEE), Zero Trust Architecture for IoT.

Industry Case Studies: Smart Grid attacks, Connected vehicle security incidents, Smart healthcare device vulnerabilities, Relevant case studies.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	2	2				3
CO2			2	1	2	
CO3		2	2			2
CO4		3				3
CO5	2	3	2		2	3

Textbooks

1. Mahmoud Daneshmand, Amir H. Jahangir, Kamesh Namuduri, "Security and Privacy in Internet of Things (IoTs): Models, Algorithms, and Implementations", Wiley
2. Brian Russell, Drew Van Duren, "Practical Internet of Things Security: Design a security framework for an Internet connected ecosystem", 2nd Edition,Packt Publication 2018
3. Houbing Song, Danda B. Rawat, Sabina Jeschke, Christian Brecher, "Cyber-Physical Systems: Foundations, Principles and Applications", Elsevier

References

1. Pascal Ackerman, "Industrial Cybersecurity", Packt Industrial Cybersecurity: Efficiently monitor the cybersecurity posture of your ICS environment , Second Edition,Packt Publication,2021
2. Madhusanka Liyanage, An Braeken, Pardeep Kumar, Mika Ylianttila, "IoT Security: Advances in Authentication,John Wiley & Sons ,2020, ISBN:9781119527978
3. David Etter, " IoT Security: Practical guide book " Create Space, 1st Edition,CreateSpace Independent Publishing Platform, 2016.
4. Sean Smith, "The Internet of Risky Things", O'Reilly Media, 1st Edition, 2017

CYI311 Computer Forensics [0-1-2-2]

Course Prerequisites

Student should have a passing grade in Operating Systems (CSC214) or the instructor's approval.

Course Description

This course provides practical training in digital forensic investigation, covering the forensic acquisition, preservation, analysis, and reporting of digital evidence from computer systems. Students will gain hands-on experience in acquiring and analysing memory and disk images using industry-standard forensic tools and hardware write blockers. Through case studies and practical investigations, students will develop the skills required to conduct systematic and legally defensible digital forensic examinations.

Course Objectives

- Understand the principles and stages of digital forensic investigation.
- Understand how to acquire digital data forensically following best practices
- Perform analysis over evidence extracted from computers
- Prepare reports by analysing digital forensic evidence

Course Outcomes

At the end of the course, students should be able to:

- CO1:Acquire data forensically from memory and hard disk utilizing forensic tools .
- CO2:Analyse memory to detect presence of malicious executables

- CO3:Analyse artifacts from RAM and Hard disk utilizing forensic software's.
- CO4:Prepare forensic reports utilizing forensic tools

Syllabus

Digital Forensic Investigation and Data Acquisition Process, Comparison of Tools to Acquire Memory and Disk Images in a Legally Acceptable Manner, Familiarization of Hardware to Acquire Disk Images Using Write Blockers. Analyze Disk Images and Prepare Forensic Reports, Anti-Forensics Techniques and Mitigation, Tools to Analyze Memory Images for Malicious Executables. Analysis of Artifacts from Registry, Operating System, Web, Email, Media and Cloud. Case Studies and Mini Project Involving Forensic Images.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1			3	2	3	
CO2			3		3	
CO3			3		3	
CO4			2	3	2	

Textbooks

1. Bill Nelson, Amelia Phillips, Christopher Steuart, and Frank Enfinger, Guide to Computer Forensics and Investigations, 7th Edition, Cengage Learning, 2022.
2. Easttom, C., Digital Forensics, Investigation, and Response, 4th Edition, Jones & Bartlett Learning, 2022.
3. Ayman Shaaban and Konstantin Sapronov,Practical Windows Forensics:definitive reference for analyzing Windows-based systems, Packt, Publishing Limited, ISBN: - 139781783554096, 2016

References

1. Michael Hale Ligh, Andrew Case, Jamie Levy, and AAron Walters, The Art of Memory Forensics, Wiley, 2014.
2. Lee, R., & Webb, T., Hands-On Digital Forensics with Open-Source Tools, Packt Publishing, 2019.
3. Easttom, C., Digital Forensics, Investigation, and Response, 4th Edition, Jones & Bartlett Learning, 2022.
4. National Institute of Standards and Technology (NIST), Computer Forensic Reference Data Sets (CFReDS) for sample case studies

CYI312 Artificial Intelligence for Cyber Defense [0-1-2-2]

Course Prerequisites

Student should have a passing grade in Machine Learning and Data Analysis(CAC221) and knowledge of Python programming or the instructor's approval.

Course Description

This lab provides hands-on experience in applying artificial intelligence techniques to detect, prevent, and respond to cyber threats. Students will work with real-world datasets and tools to build intelligent systems for intrusion detection, malware analysis, anomaly detection, and threat intelligence. The lab emphasizes practical implementation of machine learning and deep learning models in cybersecurity scenarios.

Course Objectives

- To develop practical skills in applying artificial intelligence techniques for cyber defense applications.
- To provide hands-on experience in analyzing network, system, and security data for identifying cyber threats and malicious activities.
- To enable students to implement and evaluate machine learning and deep learning models for intrusion detection, anomaly detection, malware analysis, and phishing detection.
- To familiarize students with adversarial machine learning concepts and defense mechanisms in cybersecurity.
- To provide exposure to cybersecurity datasets, tools, and real-world cyber defense scenarios through practical experimentation and projects.

Course Outcomes

At the end of the course, students should be able to:

- CO1:Design and develop AI-based solutions for cyber defense applications.
- CO2:Apply AI and machine learning techniques for intrusion detection, anomaly detection, and threat analysis.
- CO3:Analyze real-world cybersecurity datasets to extract actionable security insights.
- CO4:Evaluate adversarial machine learning techniques and their implications for cybersecurity.
- CO5:Implement and assess AI-driven approaches for solving practical cybersecurity problems.

Syllabus

Intrusion detection and anomaly detection: Introduction to Cybersecurity Datasets: Explore datasets like KDD Cup 99, NSL-KDD, CICIDS, Data preprocessing and feature analysis, Intrusion Detection using Machine Learning: Implement classification models (Logistic Regression, Decision Tree), Evaluate performance using accuracy, precision, recall, Anomaly Detection in Network Traffic: Use clustering techniques (K-Means, DBSCAN), Identify unusual patterns in network data.

Malware and Phishing Detection using AI: Explore the various malware dataset, Feature extraction from malware datasets, Train classifiers to distinguish benign vs malicious files, Phishing Detection using NLP: Text preprocessing of URLs/emails, Apply NLP models for classification.

Threat Intelligence and AI-based Log Analysis: Use open-source threat data, Visualize attack patterns and trends, Analyze system logs using ML techniques, Detect suspicious activities.

Adversarial Attacks on ML Models: Generate adversarial examples, Analyze model vulnerability, Defense Against

Adversarial Attacks Implement defensive techniques (adversarial training, input filtering).

Deep Learning for Intrusion Detection : Build ANN/CNN models for attack detection, Compare with traditional ML models. Case Study & Final Project: Solve a real-world cyber defense problem, Submit report and presentation.

Tools and Techniques used: Python, Jupyter Notebook, Scikit-learn, TensorFlow / PyTorch, Wireshark (network analysis)

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1		2	3			3
CO2			3		3	
CO3	2		3		2	
CO4	2	2				3
CO5	2	2	3	2	2	2

Textbooks

1. Aurélien Géron 1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow , O'Reilly Media,2022
2. Sebastian Raschka, Vahid Mirjalilis,Python Machine Learning Packt, Publishing Limited,2019

References

1. Saxe, J., x& Sanders, H., Malware Data Science: Attack Detection and Attribution, No Starch Press, 2018.
2. Joseph, A. D., Nelson, B., Rubinstein, B. I. P., & Tygar, J. D., Adversarial Machine Learning, Cambridge University Press, 2024.
3. Shokri, R. (Ed.), Trustworthy Machine Learning, Springer, 2025.
4. Goodfellow, I., Bengio, Y., & Courville, A., Deep Learning, MIT Press, 2016.
5. Apruzzese, G., Colajanni, M., Ferretti, L., Guido, A., & Marchetti, M., Machine Learning and Deep Learning for Cybersecurity, Springer, 2024.
6. Alpaydin, E., Introduction to Machine Learning (4th Edition), MIT Press, 2020.
7. Buczak, A. L., & Guven, E., A Survey of Data Mining and Machine Learning Methods for Cyber Security Intrusion Detection, IEEE Communications Surveys & Tutorials, 2016.
8. Kucharavy, A., et al. (Eds.), Large Language Models in Cybersecurity: Threats, Exposure and Mitigation, Springer, 2024.

CYL311 Cloud Computing and IoT Systems Lab [0-0-2-1]

Course Prerequisites

Student should have a passing grade in Computer Networks (CSC211), Operating Systems (CSC214), or the instructor's approval.

Course Description

This laboratory course provides hands-on experience in developing, deploying, and managing Internet of Things (IoT) applications integrated with cloud computing platforms. Students learn to connect sensors and embedded devices to cloud services, implement data acquisition and analytics pipelines, develop IoT dashboards, and deploy scalable cloud-based solutions. The course emphasizes real-world IoT architectures, cloud services, edge computing, security, and industry-standard platforms used in smart homes, smart cities, healthcare, agriculture, and industrial automation.

Course Objectives

- To interface sensors and actuators with IoT hardware platforms.
- To develop cloud-connected IoT applications using industry-standard protocols.
- To store, process, and visualize IoT data using cloud services.
- To implement edge and cloud analytics for IoT systems.
- To deploy secure and scalable IoT solutions using cloud platforms.
- To design end-to-end IoT applications for real-world use cases.

Course Outcomes

- CO1:Configure IoT devices, sensors, and communication modules for data acquisition and transmission.
- CO2:Develop cloud-connected IoT applications using MQTT, HTTP, and REST APIs.
- CO3:Implement cloud-based data storage, processing, and visualization solutions for IoT applications.
- CO4:Deploy and manage IoT services using cloud platforms and edge computing frameworks.
- CO5:Apply security mechanisms for device authentication, secure communication, and access control in IoT systems.
- CO6:Design and develop complete IoT-cloud solutions for real-world industrial and smart-system applications.

Syllabus

IoT Device Connectivity: Data sensing and serial communication, Wireless communication using Wi-Fi, Bluetooth, Zigbee, and LoRa.

Cloud Integration: Real-time data transmission from IoT devices to cloud services, Creating cloud databases for IoT data storage, Implementing serverless functions for IoT event processing, Building IoT dashboards and data visualization.

Data Analytics and Edge Computing: Real-time sensor data analytics using cloud services, Edge processing with Raspberry Pi or Edge Gateway, Rule-based automation and alert generation, Predictive analytics using cloud AI services. IoT Communication Protocols: Publishing and subscribing sensor data using MQTT, Configuring an MQTT broker and topic management, Secure MQTT Communication using TLS, CoAP server client communication, Secure CoAP communication using DTLS, RPL Network Security Analysis, Vulnerability Assessment

IoT Security and Device Management: Secure device authentication – Cryptography based authentication, PUF based authentication, Biometric based authentication, Device monitoring and remote management.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1		2	2			3
CO2		2	2			3
CO3			2			2
CO4		2	2			3
CO5		3	2			3
CO6		3	2			3

Textbooks

1. Bahga, A., & Madisetti, V., Internet of Things: A Hands-On Approach, Universities Press, First Edition, 2015.
2. Dennis, A., Practical Internet of Things with Python, Packt Publishing, First Edition, 2018.
3. Erl, T., Puttini, R., & Mahmood, Z., Cloud Computing: Concepts, Technology & Architecture, Pearson Education, First Edition, 2013.

References

1. Minoli, D., Building the Internet of Things: Implement New Business Models, Disrupt Competitors, Transform Your Industry, Wiley, First Edition, 2016.
2. Mahmood, Z., Mastering Cloud Computing, McGraw Hill Education, 2013.
3. Vermesan, O., & Friess, P. (Eds.), Internet of Things: Principles and Paradigms, Morgan Kaufmann, 2011.
4. Buyya, R., Broberg, J., & Goscinski, A. (Eds.), Cloud Computing: Principles and Paradigms, Wiley, 2011.
5. Wilmshurst, T., Hands-On Internet of Things with MQTT, Packt Publishing, 2017.

SEMESTER VI

CYC321 Ethical Hacking [3-0-0-3]

Course Prerequisites

Student should have a passing grade in Introduction to Cybersecurity and Cybercrimes(CYC222) or the instructor's approval.

Course Description

This course provides a comprehensive exploration of offensive security methodologies, transitioning from theoretical threat modelling to hands-on exploitation and remediation. Students will master the penetration testing lifecycle, including advanced reconnaissance, network exploitation, Active Directory attacks, and web application security. By integrating modern AI-driven tools and industry-standard frameworks like OWASP and MITRE ATT&CK, the course prepares students to identify vulnerabilities, execute ethical hacks, and generate professional security reports.

Course Objectives

- To impart a deep understanding of the legal, ethical, and professional frameworks governing offensive security operations
- To develop technical proficiency in reconnaissance, vulnerability research, and exploitation techniques across diverse network environments.
- To equip students with the skills to perform post-exploitation and lateral movement within complex enterprise infrastructures like Active Directory.
- To foster the ability to critically analyze web application vulnerabilities and document findings in a professional, industry-aligned reporting format.

Course Outcomes

At the end of the course, students should be able to:

- CO1: Outline the penetration testing lifecycle and apply ethical and legal constraints to define the scope and rules of engagement for a security assessment.
- CO2:Analyze a target's attack surface using OSINT and automated scanning tools to prioritize vulnerabilities based on CVSS scores and NIST standards.
- CO3:Execute network-level exploits using frameworks like Metasploit and evaluate the effectiveness of bind vs. reverse shells in bypassing basic security controls.
- CO4:Formulate multi-stage attack paths involving privilege escalation and lateral movement to demonstrate the risk of total domain compromise in Active Directory environments
- CO5:Validate web application vulnerabilities using the OWASP Top 10 framework and produce a comprehensive technical report with actionable remediation strategies using AI-assisted tools.

Syllabus

Foundations of Offensive Security: CIA Triad - Cyber Kill chain - MITRE ATT&CK - Penetration Testing Lifecycle - Legal and Ethical Constraints - Social Engineering

Information Gathering and Vulnerability Research: Foot Printing Techniques - OSINT - Service Enumeration - Automated vs Manual Vulnerability Assessment Techniques and Tools - Overview of CVE and CVSS Scores - NIST Standards for Information Security

Network Attacks and Exploitation: Understanding Payloads - Bind and Reverse Shell - Metasploit Framework - Payload Generation - Exploiting Common Protocols (SSH, FTP, RDP) - Man in the Middle Attacks

Post Exploitation and Lateral Movement: Privilege Escalation Techniques (Windows & Linux) - Active Directory (AD) Attacks - Pivoting and Tunneling - Creating Backdoors

Web Application Security and Reporting: Introduction to OWASP Top 10 - Server-side vulnerabilities - Client-Side Vulnerabilities - API Security - AI in Offensive Security - Documenting Penetration Test Report - Ethics and Legalities (ROE, Scope definition) and Compliance

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1			2	3	3	
CO2			3		3	
CO3			3		3	
CO4			3		3	
CO5			3	3	3	

Textbooks

1. Samir Kumar Rakshit,Ethical Hacker's Penetration Testing Guide,2022, ISBN 978-93-55512-154
2. Mariya Ouaissa and Mariyam Ouaissa,2. Offensive and Defensive Cyber Security Strategies - Fundamentals, Theory and Practice,, CRC Press, 2025
3. Dr. Catherine J. Ullman,The Active Defender - Immersion in the Offensive Security Mindset ,Wiley, 2023

References

1. Weidman, G., Penetration Testing: A Hands-On Introduction to Hacking, No Starch Press, 2014.
2. Kim, P., The Hacker Playbook 3: Practical Guide to Penetration Testing, Secure Planet LLC, 2018.
3. Stuttard, D., & Pinto, M., The Web Application Hacker's Handbook, Wiley, 2011.
4. Engebretson, P., The Basics of Hacking and Penetration Testing, Syngress, 2021.

IOEC XXX Design Thinking and Innovation [2-1-0-3]

Course Prerequisites

None

Course Description

TThe Design Thinking and Innovation course introduces students to human-centered approaches for solving complex real-world problems through creativity, collaboration, and innovation. The course covers the fundamentals of design thinking, user empathy, ideation techniques, prototyping,interaction design, usability evaluation, innovation strategies, and social impact design. Students learn through case studies, collaborative activities, hands-on exercises, and project-based learning. The course emphasizes interdisciplinary teamwork, communication, and iterative problem-solving methodologies widely used in engineering, product development, entrepreneurship, digital design,and social innovation sectors.

Course Objectives

- To introduce students to the principles, processes, and applications of design thinking and innovation.
- To develop skills in user-centered problem identification, empathy building, and user research methods.
- To foster creativity and innovation through brainstorming, ideation, prototyping, and testing activities.
- To equip students with practical skills in interaction design, usability evaluation, and design communication.
- To encourage interdisciplinary collaboration, ethical thinking, and socially responsible innovation through project-based learning.

Course Outcomes

At the end of the course, students should be able to:

- Explain the concepts, stages, and significance of the design thinking process in solving real-world problems.
- Apply user research methods, empathy techniques, and problem-definition strategies to understand user needs effectively.
- Generate innovative ideas using brainstorming and ideation techniques and develop prototypes for testing and evaluation.
- Design user-centered solutions using interaction design principles, usability evaluation methods, and iterative improvements.
- Collaborate effectively in interdisciplinary teams and communicate design solutions through presentations, reports, and project demonstrations.

Syllabus

Introduction to Design Thinking - Introduction to design - Characteristics of good and bad design - Importance of design thinking in engineering and problem solving - Human-centered design approach - The five stages of design thinking: Empathize, Define, Ideate, Prototype, and Test - Design thinking versus traditional problem-solving approaches - Individual and team-based problem solving.

User Empathy and Problem Identification - Understanding user needs and stakeholder analysis - Empathy development techniques - User research methods including interviews, observations, surveys, and focus groups - Problem definition and reframing techniques - User-centered design principles - Ergonomics and accessibility considerations.

Ideation and Concept Development - Brainstorming methods and innovation heuristics - Creativity enhancement techniques - Affinity diagrams, empathy maps, mind maps, and customer journey maps - Concept generation and evaluation - Scenario building and storytelling techniques - Collaborative ideation activities.

Prototyping and Usability Testing - Low-fidelity and high-fidelity prototyping methods - Prototype development tools and techniques - Wireframes - Usability testing methods - Gathering and analyzing user feedback - Iterative design improvement - Heuristic evaluation techniques - Basics of data analysis using SPSS for design evaluation.

Interaction Design and User Experience - Fundamentals of interaction design - History and evolution of interaction design - User Interface (UI) and User Experience (UX) principles - Information architecture - Personas and task flows - Laws and principles of interaction design - Qualitative and quantitative research methods - Usability evaluation techniques.

Innovation, Ethics, and Social Impact - Principles and processes of successful innovation - Modes of design innovation - Social entrepreneurship and innovation for social good - Ethics in design and responsible innovation - Inclusive and sustainable design practices - Industry case studies on innovation and design impact - Familiarization with modern design tools - Design-based mini project development and presentation.

Textbooks References

1. Kumar, K., & Kurni, M. (Eds.), Design Thinking: A Forefront Insight, CRC Press, 2022.
2. Vijay Kumar, 101 Design Methods: A Structured Approach for Driving Innovation in Your Organization, John Wiley & Sons, 2013.
3. Dan Saffer, Designing for Interaction: Creating Innovative Applications and Devices, New Riders Publishing, 2010.
4. Lewrick, M., Link, P., & Leifer, L., The Design Thinking Toolbox: A Guide to Mastering the Most Popular and Valuable Innovation Methods, Wiley, 2020.
5. Metts, M. J., & Welfle, A., Writing is Designing: Words and the User Experience, Rosenfeld Media, 2020.
6. Uebernickel, F., Jiang, L., Brenner, W., Pukall, B., Naef, T., & Schindlholzer, B., Design Thinking: The Handbook, World Scientific, 2020.
7. Gallagher, A., & Thordarson, K., Design Thinking in Play: An Action Guide for Educators, ASCD, 2020.
8. Regine Gilbert, Inclusive Design for a Digital World: Designing with Accessibility in Mind, Apress, 2019.
9. Michael G. Luchs, Design Thinking: New Product Development Essentials from the PDMA, Wiley, 2015.
10. D'Source: Digital Online Environment for Design Learning – https://www.dsource.in

CYI321 Threat Intelligence and Security Operations Centre(SOC) [0-1-2-2]

Course Prerequisites

Students should have completed Introduction to Cybersecurity and Cybercrimes(CYC222) and should possess basic knowledge of Operating Systems(CSC214), Computer Networks(CSC211) or the instructor's approval

Course Description

This laboratory course introduces students to the operational processes, tools, and techniques used in Security Operations Centres (SOCs). The course covers security monitoring, log analysis, threat intelligence, threat hunting, incident investigation, and incident response. Through hands-on exercises using industry-relevant tools and frameworks, students develop practical skills in detecting, analyzing, and responding to cybersecurity threats within a SOC environment

Course Objectives

- To introduce students to the architecture, functions, and workflows of a Security Operations Centre (SOC).
- To develop practical skills in security monitoring, log analysis, and threat detection.
- To familiarize students with cyber threat intelligence collection, analysis, and dissemination techniques.
- To provide hands-on experience in threat hunting, incident triage, and security investigations.
- To expose students to industry-standard SOC tools, frameworks, and operational procedures.

Course Outcomes

At the end of the course, students should be able to:

- CO1: Configure and utilize security monitoring and log analysis tools.
- CO2:Collect, analyze, and correlate threat intelligence from multiple sources.
- CO3: Identify indicators of compromise and investigate security incidents.
- CO4: Perform threat hunting using security telemetry and threat intelligence data.
- CO5: Generate incident reports and communicate findings in a professional SOC environment.

Syllabus

Security Operations Centre (SOC): SOC architecture, roles and responsibilities, SOC workflows, security monitoring lifecycle, incident management process.

Security Information and Event Management (SIEM): Log collection, normalization, correlation, alert generation, dashboard creation, event analysis and prioritization.

Cyber Threat Intelligence: Threat intelligence lifecycle, strategic, operational and tactical intelligence, Indicators of Compromise (IOCs), Indicators of Attack (IOAs), Open-Source Intelligence (OSINT), threat actor profiling, MITRE ATT&CK framework.

Threat Hunting and Incident Investigation: Threat hunting methodologies, hypothesis-driven hunting, IOC-based hunting, behavior-based detection, malware and phishing investigations, evidence collection and analysis.

Incident Response and Reporting: Alert triage, incident classification, containment and recovery planning, incident documentation, SOC metrics, reporting and communication

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1			3		3	
CO2			3		3	
CO3			3		3	
CO4			3		3	
CO5			2	3	2	

Textbooks

1. Joseph Muniz and Gary McIntyre, Security Operations Center: Building, Operating, and Maintaining Your SOC, Cisco Press.
2. Chris Sanders and Jason Smith, Applied Network Security Monitoring: Collection, Detection, and Analysis, Syngress.

References

1. Richard Bejtlich, The Practice of Network Security Monitoring: Understanding Incident Detection and Response, No Starch Press.
2. Michael Collins, Network Security Through Data Analysis: From Data to Action, O'Reilly Media.

CYI322 Smart Phone Forensics and Application Security [0-1-2-2]

Course Prerequisites

Student should have a passing grade in Digital Forensics and Investigation Lab (CYI311) or the instructor's approval.

Course Description

This course provides hands-on training in mobile device forensics and Android application security analysis. Topics include mobile forensic investigation procedures, Android architecture, data acquisition and analysis, forensic image examination, Android application reverse engineering, malware investigation, and static and dynamic vulnerability analysis. Practical exercises, case studies, and a mini-project enable students to apply forensic and security analysis techniques to real-world mobile environments.

Course Objectives

- Understand the mobile forensic investigation process, mobile operating system architectures, and data acquisition techniques for Android and iOS devices.
- Apply forensic tools and methodologies to acquire, preserve, and analyze digital evidence from mobile devices and forensic images
- Perform reverse engineering and security analysis of Android applications to identify malicious behaviors and vulnerabilities.
- Conduct forensic investigations and security assessments through practical case studies involving forensic images and vulnerable mobile applications.

Course Outcomes

At the end of the course, students should be able to:

- CO1:Examine Android architecture, file systems, and data storage structures to identify potential sources of digital evidence.
- CO2:Acquire and extract forensic data from mobile devices and forensic images using ADB, imaging tools, and established forensic procedures.
- CO3:Analyze Android and iOS forensic images to recover, interpret, and correlate digital evidence relevant to an investigation
- CO4:Reverse engineer Android applications and perform static and dynamic analysis to identify malicious behaviour and security vulnerabilities.

Syllabus

Mobile Forensic Investigation Process, Android Architecture and File Hierarchy from a Security Perspective, Android File Structure Analysis and Data Retrieval using Android Debug Bridge (ADB).

Data Acquisition using Tableau Imaging Kit, Difference between Quick, File System and Full Images, and Challenges in Acquiring Them, Analysis of iOS and Android Forensic Images.

Reverse Engineering Android Applications and Investigation of Malicious Code, Modifying and Recompiling Android Applications, Analysis of Vulnerabilities in Android Applications through Static and Dynamic Analysis.

Case Studies and Mini Project Involving Forensic Images and Vulnerable APKs.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	2		2		3	
CO2			3	1	3	
CO3			3		3	
CO4			3		3	

Textbooks

1. S. Bommisetty, R. Tamma, and H. Mahalik, Practical Mobile Forensics, 4th ed. Birmingham, UK: Packt Publishing, 2021.
2. R. Tamma, O. Skulkin, H. Mahalik, and S. Bommisetty, Learning Android Forensics, 2nd ed. Birmingham, UK: Packt Publishing, 2018.
3. Hoog, Android Forensics: Investigation, Analysis and Mobile Security for Google Android, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2014.

References

1. N. Elenkov, Android Security Internals: An In-Depth Guide to Android's Security Architecture, San Francisco, CA, USA: No Starch Press, 2014.
2. Gupta, A.. Learning Pentesting for Android Devices: A Practical Guide to Learning Penetration Testing for Android Devices and Applications (1st ed.). Packt Publishing. ISBN: 978-1783288984.
3. J. J. Drake, Z. Lanier, C. Mulliner, P. Oliva Fora, S. A. Ridley, and G. Wicherski, Android Hacker's Handbook, 1st ed. Indianapolis, IN, USA: John Wiley & Sons, 2014. ISBN: 978-1-118-60864-7

CYL321 Ethical Hacking Lab [0-0-2-1]

sblockCourse Prerequisites Student should have a passing grade in Introduction to Cybersecurity and Cybercrimes(CYC222) or the instructor's approval.

Course Description

This laboratory course provides hands-on experience in offensive security and penetration testing methodologies across networks, systems, web applications, and enterprise environments. Students will learn reconnaissance, vulnerability assessment, exploitation, privilege escalation, and post-exploitation techniques using industry-standard tools and frameworks. The course emphasizes ethical security testing practices within controlled environments while developing practical skills for identifying and assessing cybersecurity risks.

Course Objectives

- To develop practical skills in reconnaissance, vulnerability assessment, and penetration testing using industry-standard security tools and frameworks.
- To understand and apply offensive security techniques for identifying and exploiting vulnerabilities in networks, operating systems, and web applications.
- To perform security assessments of enterprise environments, including Active Directory infrastructures and modern web technologies.
- To analyze advanced attack methodologies, threat actor techniques, and emerging security challenges such as AI/LLM-related vulnerabilities.
- To promote ethical hacking practices and responsible disclosure while strengthening cybersecurity risk assessment and mitigation capabilities.

Course Outcomes

At the end of the course, students should be able to:

- CO1: Perform reconnaissance and information gathering using passive scanning and OSINT techniques.
- CO2: Conduct vulnerability assessments and penetration testing of networks, systems, and web applications using industry-standard tools.
- CO3: Identify, exploit, and analyze common security vulnerabilities in Windows, Linux, web, and enterprise environments.
- CO4: Apply post-exploitation, privilege escalation, and Active Directory assessment techniques in controlled laboratory settings.
- CO5: Evaluate emerging cybersecurity threats and recommend appropriate security controls and mitigation strategies.

Syllabus

- APT Tactics, Techniques, and Procedures (TTPs) – Analysis of Advanced Persistent Threats using the MITRE ATT&CK Framework.
- Phishing Campaigns and Social Engineering – Setup and execution of phishing simulations using GoPhish, SET, and Kali Linux tools.
- Passive Reconnaissance – Information gathering using Nmap, Whois, DNS Dumpster, and Hunter.io.
- Open-Source Intelligence (OSINT) – Metadata analysis, image geolocation, and social media investigations.
- Vulnerability Assessment – Security assessment using Burp Suite, OWASP ZAP, Nessus, and Nuclei on controlled targets.
- Shell Establishment and Post-Exploitation – Reverse shells, web shells, persistence mechanisms, and Metasploit Framework usage.
- Man-in-the-Middle Attacks – ARP spoofing and DNS spoofing techniques using Ettercap.
- Windows Privilege Escalation – Identification and exploitation of common privilege escalation vectors in Windows environments.
- Linux Privilege Escalation – Enumeration and exploitation of privilege escalation opportunities in Linux systems.
- Active Directory Security Assessment – Reconnaissance, enumeration, and exploitation of Active Directory environments.
- Web Application Enumeration and Exploitation – Automated vs Manual Scanning, Enumeration Techniques, Nmap NSE Scripts for Vulnerability Assessment
- LLM Security Testing – Understanding and assessment of security risks in LargeLanguage Model applications.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1			3		3	
CO2			3		3	
CO3			3		3	
CO4			3		3	
CO5		2	2		2	3

Textbooks

1. Weidman, G., Penetration Testing: A Hands-On Introduction to Hacking, No Starch Press, 2014.
2. Stuttard, D., and Pinto, M., The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws (2nd Edition), Wiley Publishing, 2011.
3. Kim, P., The Hacker Playbook 3: Practical Guide to Penetration Testing, Secure Planet LLC, 2018.

References

1. Bazzell, M., OSINT Techniques: Resources for Uncovering Online Information (9th Edition), Intel Techniques, 2020.
2. Johnson, J., Jackson, R., and Lodovici, C., Attacking and Defending Active Directory: Modern Authentication and Attack Techniques for Security Professionals, O'Reilly Media, 2022.
3. Murdoch, D., Blue Team Handbook: SOC, SIEM and Threat Hunting (2nd Edition), CreateSpace Independent Publishing Platform, 2020.
4. O'Gorman, J., and McDonald, D., The Art of Memory Forensics: Detecting Malware and Threats in Windows, Linux, and Mac Memory, Wiley, 2012.
5. Kucharavy, A., Bécue, A., Mé, L., and Totel, E. (Eds.), Large Language Models in Cybersecurity: Threats, Exposure and Mitigation, Springer Nature, 2024.
6. Huang, K., Wang, Y., Goertzel, B., and Li, J. (Eds.), Generative AI Security: Theories and Practices, Springer Nature, 2024.

SEMESTER VII

CYC411 Cyber Warfare,Adversarial Tactics and Resilience [2-1-0-3]

Course Prerequisites

Student should have a passing grade in Offensive Security and Penetration Testing(CYC321), Secure Software Engineering and DevSecOps(CYC312), or the instructor's approval.

Course Description

This capstone course provides a comprehensive exploration of full-spectrum cyber operations, bridging the strategic gap between offensive tactics and defensive architectures. Students will analyse architectural vulnerabilities in modern applications and study the theoretical lifecycles of Red Team adversarial simulations. The curriculum strongly emphasizes architecting robust Blue Team defences, including the design of threat hunting frameworks, SIEM configurations, and isolated malware analysis sandboxes. By evaluating collaborative Purple Team dynamics, students will learn to synthesize simulated attack-and-defence outcomes to validate enterprise security posture, assess organizational resilience, and formulate actionable remediation strategies.

Course Objectives

- To analyse architectural vulnerabilities in modern applications and understand the frameworks and methodologies required for systematic security assessments.
- To explore the strategic planning and lifecycles of adversarial simulations, including advanced exploitation and post-exploitation operations in enterprise environments.
- To design robust defensive architectures, encompassing threat hunting frameworks, SIEM configurations, and isolated environments for malware analysis.
- To evaluate organizational resilience by synthesizing offensive and defensive operations (Purple Teaming) and formulating effective remediation strategies.

Course Outcomes

At the end of the course, students should be able to:

- Apply industry-standard methodologies and threat modeling tools to identify and analyze architectural flaws in web, mobile, and API environments.
- Analyse the theoretical lifecycles of adversarial operations—from reconnaissance to post-exploitation—to plan comprehensive red team simulations.
- Architect defensive frameworks, including SIEM configurations, threat hunting playbooks, and incident response strategies, to correlate and mitigate cyber attacks.
- Design isolated sandbox environments for malware analysis and formulate automation scripts to streamline security validation and SOC workflows.
- Evaluate purple team dynamics to assess enterprise security posture and synthesize simulated attack-and-defense outcomes into actionable remediation strategies.

Syllabus

Architecture of Vulnerable Systems & Cyber Ranges: Study of the architectural flaws in modern web, mobile, and API environments. Principles of designing cyber range topologies, Capture the Flag (CTF) frameworks, and theme-based challenge rooms for security validation.

Methodologies for Application Security Assessment: Theoretical frameworks and industry-standard methodologies for evaluating software systems. Focus on vulnerability discovery processes, threat modelling tools, and standardizing reporting practices.

Adversarial Simulation & Red Team Operations: Planning and designing adversarial simulations. Examining the theoretical lifecycles of reconnaissance, exploitation, privilege escalation, persistence, and post-exploitation within enterprise environments.

Defensive Architectures: Blue Team & Threat Hunting: Architecting detection engineering frameworks, threat hunting playbooks, and attack correlation strategies. Analysing SIEM platform configurations, log analysis techniques, and incident response lifecycles.

Malware Analysis Frameworks & Automation: Design principles of isolated sandbox environments for static and dynamic analysis. Formulating scripts and frameworks for automating vulnerability assessment, security validation, and SOC workflows . Purple Team Dynamics & Security Validation: Analysing the collaborative frameworks between offensive and defensive operations. Assessing organizational resilience and formulating remediation strategies based on simulated attack and defence outcomes.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	2	3	1		1	2
CO2	1	3	2		2	2
CO3	1	3	3		2	3
CO4	1	2	3		3	2
CO4		2	2	2	3	3

Textbooks

1. Stuttard, D., & Pinto, M. The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws (2nd Edition). Wiley Publishing,2011. ISBN: 978-1118026472..
2. Weidman, G. (2014). Penetration Testing: A Hands-On Introduction to Hacking. No Starch Press, 2014. ISBN: 978-1593275648.
3. Sikorski, M., & Honig, A. Practical Malware Analysis: The Hands-On Guide to Dissecting Malicious Software. No Starch Press,2012. ISBN: 978-1593272906.

References

1. Murdoch, D. Blue Team Handbook: SOC, SIEM and Threat Hunting (2nd Edition). CreateSpace Independent Publishing Platform,2020 ISBN: 978-1091493896.

2. Murdoch, D. Blue Team Handbook: Incident Response (2nd Edition). CreateSpace Independent Publishing Platform,2020.ISBN: 978-1091493919.

Program Electives

CYE001 Cybersecurity Governance, Risk, and Compliance [2-1-0-3]

Course Prerequisites

Student should have a passing grade in Introduction to Cybersecurity and Cybercrimes(CYC222) or the instructor's approval.

Course Description

This course provides a comprehensive understanding of governance, risk management, and compliance (GRC) in cybersecurity. It focuses on aligning security strategies with organizational objectives, identifying and managing risks, and ensuring adherence to legal, regulatory, and industry standards.

Course Objectives

- To understand principles of cybersecurity governance and organizational policies
- To analyze risk management processes and assessment techniques
- To study compliance requirements, standards, and regulatory frameworks

Course Outcomes

At the end of the course, students should be able to:

- CO1:Demonstrate a clear understanding of cybersecurity governance structures, organizational roles, and security policy frameworks.
- CO2:Identify and evaluate cybersecurity risks using standard risk assessment methods and tools.
- CO3:Apply appropriate risk treatment strategies and security controls to mitigate identified risks.
- CO4:Explain compliance processes and audit mechanisms practices in organizational environments

Syllabus

Overview of cybersecurity governance : Organizational roles and responsibilities, Security policies and procedures, Governance frameworks overview (e.g., COBIT, NIST Cybersecurity framework, ISO/IEC 27001).

Risk Management: Phases of cyber risk management, Asset management, Risk identification, Risk analysis, Risk assessment methodologies.

Risk control and treatment strategies: NIST Risk management Framework.Basics of incident response planning.

Concept of compliance in organizations: Internal audits, Need for IT Audit- IT Governance, Audit lifecycle, Documentation and reporting practices, IT auditing basics.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1				3		
CO2		1	2	3		
CO3		3		2		
CO4				3		

Textbooks

1. Michael E. Whitman, Herbert J. Mattord, "Principles of In formation Security," 6th Edition, Cengage Learning, 2018.
2. Darril Gibson, Andy Igonor. Managing Risk in Information Systems. Indianapolis: Pearson IT Certification, 2015.
3. Ravi Das. Assessing and Insuring Cybersecurity Risk. Boca Raton: CRC Press, Taylor & Francis Group, 2019.

References

1. Angel R. Otero. Information Technology Control and Audit. Boca Raton: CRC Press, Taylor & Francis Group, 2018.

CYE002 Criminal Psychology [3-0-0-3]

Course Prerequisites

Student should have a passing grade in Introduction to Cybersecurity and Cybercrimes(CYC222) or the instructor's approval.

Course Description

This course explores the psychological and behavioural mechanisms driving cybercrime. It equips students with the foundational frameworks needed to profile various digital threat actors, understand the exploitation of cognitive biases in social engineering, and analyse the psychological motives behind insider threats. By examining victimology and the human impact of digital extortion, students will learn to bridge the critical gap between human behaviour and technical cybersecurity defence strategies.

Course Objectives

- To introduce the foundational psychological mechanisms and behavioural frameworks that drive criminal activity in digital environments.
- To categorize varying cyber threat actors and understand how they weaponize cognitive biases and psychological principles to breach security.
- To analyse the psychological drivers behind insider threats and examine the profound impact of digital extortion and cyberstalking on victims.

Course Outcomes

At the end of the course, students should be able to:

- CO1: Explain the foundational psychological concepts, behavioural principles, and profiling methodologies relevant to cyber criminology.

- CO2: Analyse the distinct motivations of various threat actors and deconstruct how cognitive biases and persuasion principles are exploited in digital attacks.
- CO3: Evaluate the psychological indicators of insider threats and assess the behavioural impacts of online extortion and cyberstalking on victims.

Syllabus

Introduction to psychological principles in criminology, Motive, Means, and Opportunity framework in digital spaces, Behavioral psychology basics, Conditioning and Reinforcement.

Traditional criminal profiling methodologies and Digital Behavioral Profiling, Script Kiddies, Hacktivists, Cyber Mercenaries/Organized Crime, State-Sponsored Actors, Cognitive biases, Confirmation bias, Authority bias, Scarcity principle, Robert Cialdini's 6 Principles of Persuasion.

Psychology of betrayal and corporate espionage, Profiling the Insider Threat, Victimology principles, Psychology of Cyberstalking, Cyberbullying, and Online Extortion, Stockholm Syndrome.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	1			2	1	
CO2			1	1	2	
CO3				2	2	

Textbooks

1. K. Jaishankar, Cyber Criminology: Exploring Internet Crimes and Criminal Behavior, 1st Edition, 2011, Taylor & Francis Group.
2. Gráinne Kirwan, The Psychology of Cyber Crime: Concepts and Principles, 1st Edition, 2011, IGI Global.
3. Christopher Hadnagy, Social Engineering: The Science of Human Hacking, 2nd Edition, 2018, Wiley.

References

1. Robert B. Cialdini, Influence: The Psychology of Persuasion, Revised Edition, 2009, HarperCollins
2. Kent L. Norman, Cyberpsychology: An Introduction to Human-Computer Interaction, 2nd Edition, 2017, Cambridge University Press.

CYE003 Secure and Privacy-Preserving Machine Learning [3-0-0-3]

Course Prerequisites

tudent should have a passing grade in Machine Learning and Data Analysis(CAC221), Modern Cryptography (CYC311) or the instructor's approval.

Course Description

This course introduces the fundamental concepts, techniques, and practical approaches for building machine learning systems that protect sensitive data. Students will explore key privacy-preserving techniques such as differential privacy, homomorphic encryption, and secure multi-party computation. The course also covers various attack models and defense mechanisms to mitigate these risks.

Course Objectives

- Understand privacy risks in machine learning
- Analyze different types of privacy attacks
- Learn core privacy-preserving techniques
- Apply cryptographic methods in ML systems
- Design privacy-aware machine learning models

Course Outcomes

At the end of the course, students should be able to:

- CO1:Understand privacy risks in machine learning systems
- CO2:Learn techniques to protect sensitive data during training and inference
- CO3:Explore cryptographic and statistical privacy methods
- CO4:Apply privacy preserving machine learning techniques in real-world applications (healthcare, finance, etc.)

Syllabus

Introduction to Privacy-Preserving ML : Motivations and risks in sensitive data environments , data anonymity and data privacy, Data anonymization techniques; Deanonymization attacks; Linkage and Reconstruction attacks. Privacy threats in machine learning (data leakage, re-identification), Types of attacks (Membership inference attacks, Model inversion attacks). Overview of privacy-preserving ML techniques.

Differential Privacy: Definition and intuition of differential privacy, Mathematics of differential privacy; Privacy budget (ϵ - epsilon), Laplace and Gaussian mechanisms, Local vs Global differential privacy, Applying Differential privacy in data publishing and Machine learning model training, Differentially Private Stochastic Gradient Descent.

Foundations of Cryptography for ML: Secure Multi-Party Computation protocols and use cases, Encryption-based vs secret-sharing approaches, Homomorphic Encryption

Federated Learning: Concept and architecture of federated learning, Client-server synchronization and aggregation, Applications and Industry Case Studies: Privacy in healthcare: federated learning for medical AI, Secure collaboration in finance: risk models and compliance, Defense and government use cases.

Secure Machine Learning Techniques: Privacy-preserving model training, Encrypted inference, Model compression & privacy, Trusted Execution Environments (TEE)

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	2	2				3
CO2		3				3
CO3	3	2				3
CO4		3		2		3

Textbooks

1. Dwork, C., and Roth, A., The Algorithmic Foundations of Differential Privacy, Foundations and Trends in Theoretical Computer Science, Now Publishers, 2014
2. Bellet, A., Habrard, A., Sebban, M., and Tommasi, M., A Primer on Privacy-Preserving Machine Learning, Inria Research Report, 2021
3. Shokri, R. (Ed.), Trustworthy Machine Learning, Springer, Cham, Switzerland, 2025.

References

1. Nissim, K., et al., Differential Privacy: A Primer for a Non-Technical Audience, Journal of Entertainment and Technology Law, 2018.
2. Vhang, J. M., Zhuang, D., and Samaraweera, G. D., Privacy-Preserving Machine Learning, CRC Press, Boca Raton, FL, USA, 2024.
3. Kairouz, P., McMahan, H. B., Avent, B., et al., Advances and Open Problems in Federated Learning, Foundations and Trends in Machine Learning, Now Publishers, 2021.
4. Acar, A., Aksu, H., Uluagac, A. S., and Conti, M., A Survey on Homomorphic Encryption Schemes: Theory and Implementation, ACM Computing Surveys, Vol. 51, No. 4, 2018.
5. Goldreich, O., Foundations of Cryptography, Volume 2: Basic Applications, Cambridge University Press, 2004. (For Secure Multi-Party Computation

CYE004 Large Language Models Security and Prompt Engineering [3-0-0-3]

Course Prerequisites

student should have a passing grade in Machine Learning and Data Analysis(CAC221) or the instructor's approval.

Course Description

This course provides an introduction to Large Language Models (LLMs), prompt engineering techniques, and security challenges associated with LLM-based applications. It covers LLM architectures, prompt design strategies, prompt injection and jailbreaking attacks, guardrails, alignment techniques, and secure deployment practices. The course also introduces machine learning-based approaches for detecting and mitigating security threats in LLM systems, enabling students to develop secure and trustworthy AI-driven applications.

Course Objectives

- To provide basic understanding of core concepts of Large Language Models (LLMs) and how they work.
- To provide an introduction to prompt engineering and its application to security problems.
- To design and implement simple LLM-based solutions for security applications

Course Outcomes

At the end of the course, students should be able to:

- CO1:Understand LLM basics and their role in security problems.
- CO2: Learn basic prompt engineering techniques such as zero-shot, few-shot, and chain-of-thought.
- CO3: Learn how LLMs can be attacked such as prompt injection, jailbreaking and defended.
- CO4:Develop an ability to choose appropriate prompts and guardrails for simple security tasks

Syllabus

Large Language Models Overview: Introduction to LLMs, Transformer basics, Tokenization, Pre-training and fine-tuning, Working with LLMs via APIs and local models. Prompt Engineering: Zero-shot prompting, Few-shot prompting, Chain-of-Thought reasoning, Role-prompting, Basic prompt design best practices.

LLM Security Vulnerabilities: Prompt Injection, Jailbreaking techniques, Data leakage, Hallucinations and over-reliance, Basic overview of OWASP for LLMs.

Defensive Strategies: Input and output filtering, Guardrails, Red-teaming basics, Alignment concepts such as RLHF, Constitutional AI, Secure deployment practices.

Machine Learning for LLM Security: Using ML to detect prompt injections and malicious outputs, Anomaly detection for unusual queries.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	2					3
CO2			2			3
CO3		2	2		1	3
CO4		3				3

Textbooks

1. Jurafsky, D., and Martin, J. H., Speech and Language Processing (3rd edition), PEARSON, 2023.
2. Kucharavy, A., et al. (Eds.), Large Language Models in Cybersecurity: Threats, Exposure and Mitigation, Springer, 2024.
3. Huang, K., Wang, Y., Goertzel, B., et al. (Eds.), Generative AI Security: Theories and Practices, Springer, 2024.

References

1. OWASP Foundation, OWASP Top 10 for Large Language Model Applications, 2025.
2. Vaswani, A., et al., Attention Is All You Need, NeurIPS, 2017.
3. Wei, J., et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS, 2022.

CYE005 Distributed and Federated Learning Security [3-0-0-3]

Course Prerequisites

Student should have a passing grade in Machine Learning and Data Analysis(CAC221) or the instructor's approval.

Course Description

This course provides an introduction to the principles, techniques, and challenges of embedded systems security. It covers hardware security foundations, cryptographic implementations, secure hardware design, side-channel attacks, hardware Trojans, FPGA-based security solutions, and hardware trust mechanisms. The course enables students to understand security threats in embedded platforms and apply appropriate protection techniques for designing secure and trustworthy embedded systems.

Course Objectives

- To provide basic understanding of core concepts of distributed and federated learning and how they work.
- To provide an introduction to security threats in federated learning and their impact on privacy and model integrity.
- To design and implement simple defense strategies for secure federated learning applications

Course Outcomes

At the end of the course, students should be able to:

- CO1: Understand federated learning basics and their role in privacy-preserving applications.
- CO2:Learn common attack techniques such as poisoning attacks and inference attacks.
- CO3: Learn how federated learning systems can be attacked and defended.
- CO4: Develop an ability to choose appropriate defense mechanisms for simple security tasks in federated learning.

Syllabus

Distributed and Federated Learning Security: Introduction to Distributed and Federated Learning- Centralized vs. Decentralized learning, Horizontal FL and Vertical FL, Federated averaging, Non-IID data challenges.

IIntroduction to Federated Learning Security Issues: Security challenges in decentralized learning, Common vulnerabilities such as Client-side attacks, and Server-side attacks, Insider attacks, Model poisoning attacks, Data poisoning attacks, Free-rider attacks, Byzantine failures.

Federated Learning Security Methods and Techniques: Robust aggregation methods such as Krum, Median, and Trimmed Mean, Anomaly detection for malicious updates, Secure aggregation basics, Privacy-preserving techniques.

Inference attacks in Federated Learning: Introduction to inference attacks, Types of inference attacks, Membership inference, Data reconstruction, Gradient leakage, Model inversion attacks, Mitigation using differential privacy and gradient clipping.

Federated Learning Auditing and Monitoring: Client behavior auditing and logging, Vulnerability assessment for federated systems, Penetration testing using adversarial client simulation, Basic trust management for clients.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	2					3
CO2		2	2			3
CO3		3	2			3
CO4		3				3

Textbooks

1. Buyya, R., Mukherjee, A., and Das, S. K. (Eds.), Federated Learning: Foundations and Applications, Elsevier, 2026. ISBN: 978-0-443-28979-8.
2. Tripathy, S., Kasyap, H., and Fang, M., Federated Learning: Security and Privacy, CRC Press/Taylor & Francis, 2025. ISBN: 978-1-032-89304-2.
3. Yu, S., and Cui, L., Security and Privacy in Federated Learning, Springer, 2023. DOI: 10.1007/978-981-19-8692-5. ISBN: 978-981-198-692-5.

References

1. Yang, Q., Liu, Y., Chen, T., and Tong, Y., Federated Learning, Morgan & Claypool, 2019. DOI: 10.2200/S00960ED2V01Y201910AIM043. ISBN: 978-1-68173-697-4.
2. Thai, M. T., Phan, H. N., and Thuraisingham, B. (Eds.), Handbook of Trustworthy Federated Learning, Springer, 2025. DOI: 10.1007/978-3-031-58923-2. ISBN: 978-3-031-58922-5

CYE006 Secure and Resilient UAV Systems [3-0-0-3]

Course Prerequistes

The Student should have a passing grade in Modern Cryptography (CYC311), IoT and Cyber-Physical Systems Security and Authentication (CYC314) or the instructor's approval.

Course Description

This course provides a comprehensive framework for engineering secure and resilient Unmanned Aerial Vehicle (UAV) systems by combining communication stack hardening, lightweight cryptography, and hardware-rooted swarm authentication schemes. It equips students to design robust, GPS-denied fallback navigation architectures integrated with machine learning-driven intrusion detection systems to mitigate real-time cyber threats. Furthermore, it addresses the synchronization of ultra-low-latency digital twins to enable predictive failure analysis and dynamic hardware reconfiguration during active cyber-attacks.

Course Objectives

- Analyze the UAV technology stack to provide a comprehensive understanding of the UAV hardware and controller layers
- Examine the mathematical foundations of lightweight algorithms for energy and memory constraints of aerial platforms.

- Evaluate protocol security to investigate vulnerabilities in wireless and transport layer protocols and the implementation physically unclonable functions (PUFs) for drone authentication.
- Mitigate cyber-physical attacks by exploring defensive strategies against false data injection (FDI), jamming, and spoofing
- To achieve resilient UAV networks through mesh topologies, and the use of digital twins for proactive system reconfiguration where the synchronized virtual models predict, simulate, and mitigate cyber-physical threats in real-time.

Course Outcomes

At the end of the course, students should be able to:

- CO1: Secure the Communication Stack by hardening MAVLink telemetry and command-and-control (C2) links to prevent unauthorized command injection and interception.
- CO2: Implement Lightweight Ciphers and deploy efficient cryptographic solutions for real-time processing needs of a UAV flight controller.
- CO3: Demonstrate authentication handshakes for robust authentication protocols using PUF-based digital fingerprints and group authentication schemes to secure UAV swarms in cross-domain environments..
- CO4 Design resilient navigation as a fall-back navigation systems that utilize SLAM and other techniques to maintain stable flight when GPS signals are compromised or denied as well detect any possible breach using Intrusion Detection Systems (IDS).
- CO5: Synchronize digital twins by utilizing low-latency communication and to predict failures and reconfigure physical UAV hardware during active cyber-attacks

Syllabus

Building Blocks: technology stack- hardware layer (sensors & actuators), the controller layer- trusted execution environments (TEEs), communication protocols- secure telemetry & command and control (c2) links, MAVLink protocol, cryptographic primitives for UAV systems

Security Protocols: attacks on wireless protocol handshake, attacks in transport layer, Authentication- Physically Unclonable Functions (PUF), Group Authentication, Cross-Domain Authentication, Authentication Protocols, Applications, Attacks, Random Number Generator: design principles, NIST and AIS Test.

Security requirements: False data injection and jamming attacks, replay and communication spoofing detection mechanisms and mitigation, GPS-denied navigation using SLAM (Simultaneous Localization and Mapping) and fall-back systems. ML-based Intrusion Detection Systems (IDS) for UAV security

Resilience: network topologies, mesh networking, frequency hopping spread spectrum (FHSS), Digital Twins: reconfigurable UAVs, synchronization- low-latency communication between the UAV and its twin

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1		3				3
CO2	3	2				2
CO3		3				2
CO4		3	2		1	2
CO5		3				3

Textbooks

1. Doug Stinson, Cryptography Theory and Practice, CRC Press, 2005.
2. William Stallings, "Cryptography and Network Security", Pearson, 2013.
3. Lee, E. A., & Seshia, S. A., "Introduction to Embedded Systems: A Cyber-Physical Systems Approach", MIT Press, 2017

References

1. Bhunia, S., & Tehranipoor, M., "Hardware Security: A Hands-on Learning Approach", Morgan Kaufmann, 2018
2. Maes, R, "Physically Unclonable Functions: Constructions, Properties and Applications", Springer, 2013
3. NIST Special Publication 800-22, "A Statistical Test Suite for Random and Pseudorandom Number Generators for Cryptographic Applications", NIST , 2010, doi: https://doi.org/10.6028/NIST.SP.800-22r1a
4. Poschmann, A. S., "Lightweight Cryptography: Cryptographic Engineering for a Pervasive World", Cryptology ePrint Archive, 2009

CYE007 Hardware and Embedded System Security [3-0-0-3]

Course Prerequisites

None

Course Description

This course provides an introduction to the principles, techniques, and challenges of embedded systems security. It covers hardware security foundations, cryptographic implementations, secure hardware design, side-channel attacks, hardware Trojans, FPGA-based security solutions, and hardware trust mechanisms. The course enables students to understand security threats in embedded platforms and apply appropriate protection techniques for designing secure and trustworthy embedded systems.

Course Objectives

- Understand concepts, issues, principles, and mechanisms in embedded systems security such as embedded security trends, software vulnerabilities, physical attacks and security policies;
- Obtain hands-on skills in securing practical embedded systems;
- Learn recent research advances in embedded systems security and prepare for graduate research in embedded systems security

Course Outcomes

At the end of the course, students should be able to:

- CO1:Demonstrate proficiencies in hardware implementations of a popular crypto primitive.
- CO2:Demonstrate proficiencies in understanding hardware security issues.
- CO3:Demonstrate proficiencies in understanding hardware security primitives.
- CO4:Demonstrate proficiencies in applying cryptography and security primitives to address hardware security issues.
- CO5:Demonstrate critical thinking and analytical skills through a summary and evaluation of an open hardware security problem

Syllabus

Introduction:Introduction to embedded systems, Embedded system trends Hardware Security & Trust, Security & Protection Objectives, Why choose hardware? - Performance, Protection Environment, Scalability,Security by design:Physical or implementation attacks,Side-channel attacks Hardware reverse engineering attacks, Hardware Trojans.

FPGAs: FPGA Versus Software Programming: Why, When, and How?, High-Level Synthesis, High-Level Synthesis Solutions for FPGAs.

Embedded Cryptography:Secret key cryptography, public key cryptography, hash functions, authentication techniques, etc. - Key management for embedded systems Useful Hardware Security Primitives: Cryptographic Hardware and their Implementation, Optimization of Cryptographic Hardware on FPGA, Physically Unclonable Functions (PUFs), PUF Implementations.

Side-channel Attacks on Cryptographic Hardware: Basic Idea, Current-measurement based Side-channel Attacks (Case Study: Kocher's Attack on DES), Design Techniques to Prevent Side-channel Attacks, Cache Attacks.

Hardware Trojans: Hardware Trojan Nomenclature and Operating Modes, Countermeasures Such as Design and Manufacturing Techniques to Prevent/Detect Hardware Trojans, Logic Testing and Side-channel Analysis based Techniques for Trojan Detection, Impact of Hardware Security Compromise on Public Infrastructure, Defense Techniques.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1		3	2			
CO2	2	2				
CO3	2	3				
CO4	3	3				2
CO5			2	1		

Textbooks

1. David Kleidermacher and Mike Kleidermacher, Embedded Systems Security: Practical Methods
2. Debdeep Mukhopadhyay and Rajat Subhra Chakraborty, "Hardware Security: Design, Threats, and Safeguards", CRC Press
3. item M. Tehranipoor and C. Wang, Introduction to Hardware Security and Trust, Springer, 2012.

References

1. for Safe and Secure Software and Systems Development, 1st Edition, Newnes, 2012.
2. Ahmad-Reza Sadeghi and David Naccache (eds.): Towards Hardware-intrinsic Security: Theory and Practice, Springer.
3. M. Tehranipoor and C. Wang, Introduction to Hardware Security and Trust, Springer, 2012.
4. Koch, D., Hannig, F., & Ziener, D. (Eds.). (2016). FPGAs for software programmers. Berlin, Germany: Springer.
5. Ted Huffmire et al: Handbook of FPGA Design Security, Springer.
6. Stefan Mangard, Elisabeth Oswald, Thomas Popp: Power analysis attacks - revealing the secrets of smart cards. Springer 2007.
7. Doug Stinson, Cryptography Theory and Practice, CRC Press.
8. Wagner, M. (2016). The hard truth about hardware in cyber-security: it's more important. Network Security, 2016(12), 16-19.

CYE008 Advanced Network Security and Zero Trust Architecture [3-0-0-3]

Course Prerequisites

Student should have a passing grade in computer Networks(CSC211), Modern Cryptography (CYC311) or the instructor's approval.

Course Description

This course introduces the various attacks possible on a network. The course also covers the principles of various network protection systems like Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS). It also explores the design concepts of various types of Firewalls that operate in the Application, Transport, and Network layers of the networking paradigm. Further, the course offers insights on to the concepts and protocols for wireless security. Finally, the course provides details on the operating principles and implementation of Zero Trust Architecture for modern cyber security.

Course Objectives

- To study and analyze different network attacks.
- To understand the operating principles of intrusion detection and prevention.
- To familiarize the design concepts of different types of Firewalls.
- To explore the cryptographic techniques for wireless security.
- To understand the principles of Zero Trust Architecture for modern cyber security

Course Outcomes

At the end of the course, students should be able to:

- CO1:Analyze the impacts of various network attacks.
- CO2:Understand and differentiate between the principles of intrusion detection and prevention.
- CO3: Understand the working of different types of Firewalls.
- CO4:Evaluate the different cryptographic algorithms used for wireless security.
- CO5:Identify the benefits of Zero Trust Architecture for modern cyber security.

Syllabus

Network Attacks: Overview of network attacks, Port scanning techniques, IP spoofing, ARP poisoning, SYN flood attack, TCP spoofing.

Network Protection: Intrusion Detection Systems – Signature-based and anomaly-based IDS, HIDS and NIDS, Issues in intrusion detection; Intrusion Prevention Systems – Difference from IDS, Working; Firewalls – Packet filtering firewalls, Stateful packet filters, Application-level gateways, Next generation firewalls.

Wireless Security: Overview; Wireless security protocols – WEP, WEP encryption, Limitations of WEP, WPA, WPA Encryption.

Zero Trust: Overview of zero trust principles, Benefits of ZTA in modern cyber security, Components of ZTA, Zero trust implementation in identity and access management, Zero Trust Network Access, Micro-segmentation.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	3	2			1	
CO2	2	3	1		1	
CO3	2	3	1		1	1
CO4	3	2				2
CO5	1	3	1			3

Textbooks

1. William Stallings, "Cryptography and Network Security", 6th Edition, Pearson Education.
2. J. F. Kurose and K. W. Ross, "Computer Networking: A Top Down Approach" Seventh Edition, Pearson India, 2017.
3. Charlie Kaufman, Radia Perlman, Mike Speciner, and Ray Perlner, "Network Security: PRIVATE Communication in a PUBLIC world", Second Edition, Prentice Hall, 2022.

References

1. Jon Ericson, "Hacking: The Art of Exploitation", Second Edition, No Starch Press, 2008
2. Scott Rose, Oliver Borchert, Stu Mitchell, and Sean Connelly, "Zero Trust Architecture", NIST Special Publication 800-207.
3. Razi Rais, Christina Morillo, Evan Gilman, Doug Barth, "Zero Trust Networks: Building Secure Systems in Untrusted Networks", O'Reilly Media, Incorporated, 2024.

CYE009 Next Generation Multimedia and Generative AI Security [3-0-0-3]

Course Prerequisites

None

Course Description

This course introduces the fundamentals of multimedia security, multimedia forensics, and generative AI in multimedia systems. It covers multimedia content protection, information hiding, digital watermarking, and rights management, along with forensic techniques for content authentication, tampering detection, and source attribution. The course also examines security and privacy challenges associated with AI-generated multimedia content, including deepfakes and synthetic media, and explores approaches for establishing trust through authentication, provenance, and watermarking mechanisms.

Course Objectives

- To introduce the fundamental concepts of multimedia systems, multimedia content representation, and generative approaches for multimedia content creation.
- To provide an understanding of security requirements, content protection mechanisms, and information hiding techniques in multimedia systems.
- To develop knowledge of multimedia forensic principles and techniques for content authentication, tampering detection, and source attribution.
- To examine security, privacy, and forensic challenges associated with AI-generated multimedia content and synthetic media.
- To familiarize students with emerging approaches for establishing trust, authenticity, and provenance in multimedia and generative AI ecosystems.

Course Outcomes

At the end of the course, students should be able to:

- CO1:Explain the fundamental concepts of multimedia systems, multimedia content representation, and generative approaches for multimedia content creation.
- CO2: Analyze multimedia security requirements and apply appropriate techniques for multimedia content protection, digital watermarking, and information hiding.
- CO3: Investigate the authenticity and integrity of multimedia content using forensic analysis, tampering detection, source attribution, and content authentication techniques.
- CO4:Evaluate security, privacy, and forensic challenges associated with AI-generated multimedia content, deepfakes, synthetic identities, and related threats.
- CO5:Assess mechanisms for establishing trust in multimedia ecosystems, including provenance tracking, authentication, watermarking, and verification of AI-generated content.

Syllabus

Multimedia and Generative AI Foundations : Introduction to Multimedia, Multimedia Data Types and Characteristics, Multimedia Representation, Storage, Transmission and Compression, Emerging Multimedia Applications, Generative Approaches for Multimedia Content Creation

Multimedia Security and Information Hiding: Security Requirements in Multimedia Systems, Multimedia Encryption and Content Protection, Digital Watermarking and Fingerprinting, Information Hiding Techniques and Analysis, Digital Rights Management, Privacy and Security Challenges in Multimedia Systems.

Multimedia Forensics and Content Authentication : Principles of Multimedia Forensics and digital Evidence Handling, Forensic Analysis of Multimedia Content, Multimedia Tampering Detection, Source Attribution and Device Identification, Multimedia Authentication, Provenance and Traceability, Forensic Tools and Case Studies.

Security, Forensics and Trust in Generative AI : Generative Techniques for Multimedia Synthesis, AI-generated Multimedia Content, Synthetic Media and Associated Threats, Deepfakes, Synthetic Identity Attacks, Detection and Verification of AI-generated Multimedia Content, Security and Privacy Challenges in Generative AI Systems, Authentication of AI-generated Content, Watermarking and Provenance Mechanisms for Synthetic Media

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	2					3
CO2		3				2
CO3			3		3	
CO4		2		1		3
CO5		3		2		3

Textbooks

1. Frank Y. Shih, Multimedia Security: Watermarking, Steganography, and Forensics, CRC Press.
2. Borko Furht and Darko Kirovski (Eds.), Multimedia Security Handbook, CRC Press.
3. William Stallings and Dejan Milojicic, Artificial Intelligence for Cybersecurity, Pearson.

References

1. Husrev Taha Sencar and Nasir Memon, Digital Image Forensics, Springer.
2. Eoghan Casey, Handbook of Digital Forensics and Investigation, Academic Press.
3. Anthony T. S. Ho and Shujun Li (Eds.), Multimedia Forensics, Springer.
4. Kush R. Varshney, Trustworthy Machine Learning, Cambridge University Press.
5. John MacIntyre et al., Deepfakes: Creation, Detection and Impact, Springer.

CYE010 Cyber Law and Digital Ethics [3-0-0-3]

Course Prerequisites

Student should have a passing grade in Introduction to Cybersecurity and Cybercrimes(CYC222) or the instructor's approval.

Course Description

This course examines the legal and ethical frameworks governing the digital ecosystem. Students will explore cybercrime challenges, the evolution of the IT Act, jurisdictional hurdles, and legal liabilities such as 'mens rea' and 'actus reus'. The curriculum also covers consumer protection in e-commerce, electronic signatures, and online transaction security, equipping students with the legal knowledge and ethical principles required to navigate digital environments responsibly.

Course Objectives

- tO introduce the legal frameworks governing cyberspace, focusing on the evolution of the IT Act, components of cyber law, and the classification of penalties and offences.
- To examine the intersection of consumer protection and digital business, focusing on product liability, restrictive trade practices, and foundational legal principles.
- To explore the complexities of cyberspace jurisdiction, the legal recognition of electronic signatures, and the ethical and security challenges inherent in modern E-Commerce.

Course Outcomes

At the end of the course, students should be able to:

- CO1: dentify various forms of cybercrimes and explain the relevant provisions, components, and penalties under the IT Act.
- CO2:Analyze consumer protection legalities, trade practices, and liability principles within e-commerce scenarios.
- CO3:Evaluate cyberspace jurisdictional conflicts, the application of electronic signatures, and ethical dilemmas in modern E-Commerce using relevant case studies.

Syllabus

Introduction and Challenges associated with Cyber Crimes,Purpose of Law, Legal Rights, Evolution of the IT Act, IT Act and Components of Cyber law, Penalties & Offences, amendments. Consumer Protection Act and E-commerce, Product liability, Unfair Trade Practice, Restrictive Trade Practices, False advertisement and penalties, Employment bonds, Actus reus, Mens rea, Strict liability. Case Laws on Cyber Space Jurisdiction and Jurisdiction issues under IT Act, Electronic Signature in Indian Laws, E-Commerce Ethics and Case Studies. Online payment and Security issues.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1				3		
CO2				3		
CO3				3		

Textbooks

1. Sushma Arora, Raman Arora, Cyber Crimes & Laws, 4th Edition 2021, Publisher: Taxmann, ISBN-10: 9390712491
2. N S Nappinai, Technology Laws Decoded, 1st Edition, Publisher: Lexis Nexis, ISBN:9789350359723
3. Suresh T. Vishwanathan, The Indian Cyber Law, Bharat Law House New Delhi

References

1. P.M. Bukshi and R.K. Suri, Guide to Cyber and E–Commerce Laws, Bharat Law House, New Delhi.
2. Rodney D. Ryder, Guide to Cyber Laws; Wadhwa and Company, Nagpur
3. The Information Technology Act, 2000; Bare Act– Professional Book Publishers, New Delhi.

CYE011 Secure and Trustworthy Artificial Intelligence [3-0-0-3]

Course Prerequisites

Student should have a passing grade in Machine Learning and Data Analysis(CAC221) or the instructor's approval.

Course Description

This course focuses on the design, development, and deployment of artificial intelligence systems that are secure, reliable, fair, and explainable. As AI systems are increasingly used in critical domains such as healthcare, finance, and cybersecurity, ensuring their trustworthiness has become essential. The course explores vulnerabilities in AI models, including adversarial attacks, data poisoning, and model theft, and presents methods to defend against these threats. Students will also study key aspects of trustworthy AI such as robustness, fairness, accountability, transparency, and privacy. Techniques like explainable AI (XAI), differential privacy, federated learning, and secure model deployment are covered. Ethical considerations and global AI governance frameworks are also discussed.

Course Objectives

- Understand the principles of secure and trustworthy AI systems
- Identify vulnerabilities and threats in AI/ML models
- Learn techniques to improve robustness against adversarial attacks
- Apply privacy-preserving methods in AI systems
- Analyze fairness, bias, and ethical issues in AI

Course Outcomes

At the end of the course, students should be able to:

- CO1:Demonstrate the fundamental concepts of secure and trustworthy artificial intelligence, including robustness, fairness, privacy, and transparency.
- CO2:Identify and analyze security threats in AI systems such as adversarial attacks, data poisoning, model inversion, and model theft.
- CO3:Apply privacy-preserving techniques such as differential privacy, federated learning, and secure computation in AI applications.
- CO4:Evaluate AI models for fairness, bias, accountability, and ethical compliance using appropriate metrics.
- CO5:Design and develop secure AI systems that can operate reliably in real-world environments

Syllabus

Introduction to AI Threat Modelling : AI systems vulnerability, AI attack surface vs traditional systems, Key attack vectors: data, model, output, and interface layers AI security fundamentals , AI vs traditional cybersecurity, Definition of trustworthy, Risks and challenges in modern AI systems.

Model Security and Attacks : Threat landscape in AI, Adversarial examples and perturbation techniques, Visualizing and crafting adversarial samples, Adversarial attacks, White-box and black-box attacks, Model extraction and inversion attacks, Evasion attacks (FGSM, PGD) , Training-time attacks, Data Poisoning techniques, Trojan/backdoor attacks.

Model inversion and Privacy Leakage: Inferring training data from model output, Membership inference attacks, Privacy risks in classification and generative models.

Defense mechanisms of AI model: Different levels of defense (Data, Model, and System level), Adversarial training and data augmentation, Gradient masking and input preprocessing, Model smoothing and regularization techniques, Privacy-Preserving AI Defenses, Watermarking AI models

Explainable and Trustworthy AI: Explainable AI (XAI), Types of bias in datasets and models, Bias detection and fairness, Bias mitigation techniques, Ethical AI frameworks.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	2			2		3
CO2		2	2			3
CO3		3				3
CO4				3		2
CO5		3				3

Textbooks

1. Halstead, C., Machine Learning Security: Protecting Models from Adversarial Attacks and Data Poisoning, Kindle Direct Publishing, 2023.
2. Joseph, A. D., Nelson, B., Rubinstein, B. I. P., and Tygar, J. D., Adversarial Machine Learning, Cambridge University Press, Cambridge, UK, 2024.

3. Chang, J. M., Zhuang, D., and Samaraweera, G. D., Privacy-Preserving Machine Learning, CRC Press, Boca Raton, FL, USA, 2024.

References

1. Shokri, R. (Ed.), Trustworthy Machine Learning, Springer, Cham, Switzerland, 2025.
2. Goodfellow, I., Bengio, Y., and Courville, A., Deep Learning, MIT Press, Cambridge, MA, USA, 2016.
3. Barocas, S., Hardt, M., and Narayanan, A., Fairness and Machine Learning: Limitations and Opportunities, MIT Press, Cambridge, MA, USA, 2023.
4. Dwork, C., and Roth, A., The Algorithmic Foundations of Differential Privacy, Foundations and Trends in Theoretical Computer Science, Now Publishers, 2014.

CYE012 Quantum Computing & Post Quantum Cryptography [3-0-0-3]

Course Prerequisites

Students should have a passing grade in Mathematical Primitives of Cyber Security(CYC221) or obtain the instructor's approval.

Course Description

This course introduces the principles of quantum computing and examines its impact on modern cybersecurity and cryptographic systems. The course covers quantum computation concepts relevant to information security, quantum algorithms that threaten existing cryptographic schemes, and post-quantum cryptographic techniques designed to withstand quantum attacks. Students will explore standardized post-quantum cryptographic algorithms, quantum-safe system design, migration strategies, and emerging developments in quantum-resistant security technologies.

Course Objectives

- To introduce the fundamental concepts of quantum computing relevant to cybersecurity and cryptography.
- To understand quantum algorithms and their impact on contemporary cryptographic systems.
- To study the design principles and security foundations of post-quantum cryptographic algorithms.
- To examine standardization efforts and practical deployment considerations for quantum-safe cryptography.
- To develop an understanding of cryptographic migration strategies and future challenges in the post-quantum era.

Course Outcomes

At the end of the course, students should be able to:

- CO1: Explain the fundamental principles of quantum computing and quantum information processing.
- CO2:Analyze the impact of quantum algorithms on widely used cryptographic systems.
- CO3:Compare major families of post-quantum cryptographic algorithms and their security assumptions.
- CO4:Evaluate standardized post-quantum cryptographic solutions for practical applications.
- CO5: Assess quantum-safe migration strategies for securing real-world systems and infrastructures.

Syllabus

Fundamentals of Quantum Computing: Introduction to quantum information and computation, qubits, quantum states, superposition, entanglement, quantum measurement, Bloch sphere representation, quantum gates, quantum circuits, quantum parallelism, quantum complexity, overview of quantum computing architectures and platforms.

Quantum Algorithms and Cryptographic Impact: Introduction to quantum algorithms, Deutsch-Jozsa and Simon's algorithms, Grover's search algorithm, Shor's factoring algorithm, impact of quantum computing on RSA, Diffie-Hellman and Elliptic Curve Cryptography, quantum threat models, harvest-now-decrypt-later attacks, cybersecurity implications of large-scale quantum computers.

Foundations of Post-Quantum Cryptography: Introduction to post-quantum cryptography, security requirements and design principles, quantum-resistant cryptographic assumptions, lattice-based cryptography, code-based cryptography, hash-based cryptography, multivariate cryptography, emerging post-quantum approaches. NIST Post-Quantum Cryptography Standardization Project, ML-KEM (Kyber), ML-DSA (Dilithium), SLH-DSA (SPHINCS+), implementation challenges, side-channel security considerations.

Quantum-Safe Systems and Migration: Crypto-agility and cryptographic migration, hybrid cryptographic approaches, integration of post-quantum cryptography into TLS, VPNs and Public Key Infrastructure (PKI), quantum-safe cloud and IoT security, overview of Quantum Key Distribution (QKD), governance and standardization initiatives, future trends and challenges in quantum-safe cybersecurity

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	3					3
CO2	3	2				3
CO3	3	2				3
CO4		2		1		3
CO5		3		2		3

Textbooks

1. Michael A. Nielsen and Isaac L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press.
2. Daniel J. Bernstein, Johannes Buchmann and Erik Dahmen (Eds.), Post-Quantum Cryptography, Springer.
3. William Stallings, Cryptography and Network Security: Principles and Practice, Pearson Education.

References

1. Michael E. Whitman and Herbert J. Mattord, Principles of Information Security, Cengage Learning.
2. National Institute of Standards and Technology (NIST), Post-Quantum Cryptography Standards and Technical Reports.
3. ETSI Quantum-Safe Cryptography Working Group Reports.
4. ENISA Reports on Post-Quantum Cryptography and Migration Strategies.
5. National Cyber Security Centre (NCSC), Guidance on Preparing for Quantum-Safe Cryptography.

CYE013 Blockchain and Web3 Security [3-0-0-3]

Course Prerequisites

Student should have a knowledge of Computer Networks(CSC211) and Modern Cryptography (CYC311) or obtain the instructor's approval

Course Description

This course provides a comprehensive exploration of security principles, vulnerabilities, and defense mechanisms in blockchain and Web3 ecosystems. Students will gain a deep understanding of how decentralized systems operate, with a focus on identifying, analyzing, and mitigating security risks in smart contracts, decentralized applications (dApps), and blockchain protocols.

Course Objectives

- Introduce the architecture of blockchain technology and web security requirements in today's era.
- Enable the awareness of various consensus protocols of blockchain Technology.
- To understand the role of cryptocurrencies, smart contracts, and hyperledger.
- Provide knowledge on various applications of blockchain technologies.
- Understand the role and need of web security over insecure network.

Course Outcomes

At the end of the course, students should be able to:

- CO1: Understand the basics of blockchain technologies and their various applications.
- CO2: Understand the role of cryptocurrencies such as Bitcoin and smart contracts using Ethereum.
- CO3:Identify problems on which blockchains could be applied.
- CO4: Understand the various protocols to achieve web security over insecure network.

Syllabus

Fundamentals of Blockchain : Introduction & history of blockchain, Structure of blocks and chains, Distributed ledger technology, Types of blockchain (public, private, consortium), Consensus mechanisms (PoW, PoS, PoB, etc.), Base technologies- Docker, Cryptographic primitives for Blockchain

Cryptocurrencies : Bitcoin architecture- UTXOs, wallets, transactions, Bitcoin scripting and transaction lifecycle. Miners, 51% Attack, Selfish Mining, double spending problem.Ethereum basics- Introduction to Ethereum, Smart Contracts with Solidity, Gas, Gas limits, Tokens. Enterprise Blockchain Platform- Hyperledger Fabric architecture, Channels, chaincode.

Applications : Blockchain applications in E-governance, Smart cities and Smart Industries and Future trends

Web3 Security Infrastructure & Adversarial Vulnerabilities: Smart Contract Security and Exploits, Decentralized Finance and Protocol-Level attack Vectors, Cross-Chain & Governance, Decentralized Autonomous Organization (DAO) governance takeovers and voting manipulation vectors, Web3 Auditing and Defenses, Foundations of automated smart contract auditing, Static analysis, fuzzing, formal verification principles.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	2					3
CO2	2	2				3
CO3		2				3
CO4		3	1			1

Textbooks

1. Baxv Kevin Werbach, The Blockchain and the New Architecture of Trust, MIT Press, 2018.
2. Joseph J. Bambara and Paul R. Allen, Blockchain – A Practical Guide to Developing Business, Law, and Technology Solutions, McGraw Hill, 2018. item Jai Singh Arun, Jerry Cuomo, Nitin Gaur, Blockchain for Business, Pearson Publishers, 2019.

References

1. Joseph J. Bambara and Paul R. Allen, Blockchain, IoT, and AI: Using the Power of Three to Develop Business, Technical, and Legal Solutions, Barnes & Noble Publishers, 2018.
2. Melanie Swan, Blockchain – Blueprint for a New Economy, O'Reilly Publishers, 2018.
3. Satoshi Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System

CYE014 Lightweight Cryptograpy [3-0-0-3]

Course Prerequisites

Student should have a passing grade in Modern Cryptography (CYC311), Mathematical Primitives of Cyber Security(CYC221), IoT and Cyber-Physical Systems Security and Authentication (CYC314) or the instructor's approval.

Course Description

This course introduces the principles, design methodologies, and applications of lightweight cryptography for resource-constrained environments such as IoT devices, wireless sensor networks, RFID systems, and cyber-physical systems. It covers lightweight cryptographic primitives, key management mechanisms, and security solutions designed to achieve strong security with minimal computational, memory, and energy requirements.

Course Objectives

- To understand the design principles and security requirements of lightweight cryptographic systems.
- To analyze lightweight cryptographic primitives including block ciphers, stream ciphers, hash functions, and message authentication mechanisms.
- To study key management techniques and practical applications of lightweight cryptography in resource-constrained environments.

Course Outcomes

At the end of the course, students should be able to:

- CO1: Demonstrate an understanding of the challenges, constraints, and design strategies involved in lightweight cryptography.
- CO2: Analyse and compare lightweight cryptographic algorithms based on security, performance, and implementation requirements.
- CO3: Apply lightweight key management and authentication techniques suitable for IoT and embedded systems.
- CO4: Evaluate the use of lightweight cryptographic solutions in real-world applications such as IoT networks, cyber-physical systems, and secure communication protocols

Syllabus

Introduction: Overview of Lightweight Cryptography, security threats for resource-constrained devices, design strategies for lightweight cryptography, hardware and software implementation constraints, trade-offs among security, area, performance, memory, and energy consumption, modes of operation.

Lightweight Cryptographic Primitives Lightweight Block Ciphers: design principles and architectures, representative lightweight block ciphers: PRESENT, CLEFIA, LED,-SIMON, SPECK, RECTANGLE.

Lightweight Stream Ciphers: design goals, architectures, and representative stream cipher constructions- -RABBIT.Lightweight Hash Functions and sponge-based constructions. Lightweight Message Authentication Codes and Authenticated Encryption schemes.

Key Management and Applications of Lightweight Cryptography: Key management: challenges in designing key management protocols for IoT and embedded systems, lightweight group key management protocols, symmetric-key and asymmetric-key constructions.

Applications: RFID systems, Wireless Sensor Networks, Internet of Things (IoT), Cyber-Physical Systems, Industrial IoT, Smart Healthcare, Secure communication protocols

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	2	2				2
CO2	3	2				2
CO3		3				2
CO4		3				2

Textbooks

1. Kerry A. McKay, Larry Bassham, Meltem Sönmez Turan, Nicky Mouha, Report on lightweight Cryptography,NIST, Computer Security Division Information Technology Laboratory, https:/doi.org/10.6028/NIST.IR.8114.
2. Charalampos Manifavas, George Hatzivasilis, Konstantinos Fysarakis, Yannis Papaefstathiou, A Survey of Lightweight Stream Ciphers for Embedded Systems, 2015, https:/doi.org-/10.1002/sec.1399.
3. Mao, W., Modern Cryptography: Theory and Practice, Pearson Education, 2004

References

1. Bogdanov, A., Knudsen, L.R., Leander, G., Paar, -C.,Posch-mann, A., PRESENT: An Ultra-Lightweight Block Cipher, CHES 2007, Springer, LNCS, 4727, pp. 450–466.
2. Shirai, T., Shibutani, K., Akishita, T., Moriai, S., Iwata, T., The 128-bit Block Cipher CLEFIA, FSE 2007, Springer,-LNCS, 4593, pp. 181–195.
3. Guo, J., Peyrin, T., Poschmann, A., Robshaw, M., The LED Block Cipher, CHES 2011, Springer-Verlag, pp. 326-341.
4. Engels, D., Saarinen, M.O., Schweitzer, P., Smith, E.M., The Hummingbird-2 Lightweight Authenticated Encryption Algorithm, RFID Security and Privacy, Springer, LNCS 8.7055, pp. 19–31, 2011.
5. Albrecht, M.R., et al., Block Ciphers Focus on the Linear Layer (feat. PRIDE), CRYPTO 2014, Springer, LNCS, vol. 8616, pp. 57–76.
6. Zhang, W., et al., RECTANGLE: A Bit-slice Lightweight Block Cipher Suitable for Multiple Platforms, SCIENCE CHINA Information Sciences, 2015, Vol. 58, 122103(15), 10.1007/s11432-015-5459-7.
7. Bansod, G., et al., GRANULE: An Ultra Lightweight Cipher Design for Embedded Security, IACR Cryptology ePrint Archive, 2018.
8. Beierle, C., et al., CRAFT: Lightweight Tweakable Block Cipher with Efficient Protection Against DFA Attacks, ToSC,2019, https:/doi.org/10.13154/tosc.v2019.i1.5-45.
9. Armknecht, F., et al., A Guide to Fully HomomorphicEncryption, IACR Cryptology ePrint Archive, 2015, 1192.
10. Boneh, Dan, Xavier Boyen, Eu-Jin Goh, Hierarchical Identity-Based Encryption with Concerns and Challenges, Egyptian Informatics Journal, 2020. .
11. Siddhartha, V., Gaba, G. S., Kansal, L., A Lightweight Authentication Protocol using Implicit Certificates for Securing IoT Systems, Procedia Computer Science, 2020, 167,85–96, ELSEVIER.

12. Sun, W., et al., Security and Privacy in the Medical Internet of Things: A Review, Security and CommunicationNetworks, 2018, Wiley.
13. Hammad, B.T., Jamil, N., Rusli, M.E., Z'aba, M.R., A Survey of Lightweight Cryptographic Hash Functions, International Journal of Scientific & Engineering Research, 2017.
14. Chowdhury, A., Dasbit, S., LMAC: A Lightweight Message Authentication Code for Wireless Sensor Networks, GLOBECOM 2015.
15. Fazeldehkordi, E., Owe, O., Noll, J., Security and Privacy in IoT Systems: A Case Study of Healthcare Products, 13th International Symposium on Medical Information and Communication Technology (ISMICT)

CYE015 Secure Mobile Computing [3-0-0-3]

Course Prerequisites

Student should have a knowledge of Computer Networks(CSC211) or obtain the instructor's approval

Course Description

This course builds the concepts of the inherent vulnerabilities and cryptographic evolution of mobile telephony, localized wireless mediums and ad hoc networks and their transport protocols. An understanding of the security of data at rest, data in transit, and data processed in distributed mobile-cloud environments.

Course Objectives

- To analyze the threat models and security design considerations across cellular networks, including GSM vulnerabilities, 4G LTE architectures, and 5G network slicing.
- To evaluate vulnerabilities in wireless LANs and short-range communication, and to examine secure transport and session protocols like IPSec and WTLS.
- To understand the networking, routing, and transport layers required to move data securely across mobile interfaces
- To understand secure mobile data management, fault tolerance during disconnected operations, and privacy-preserving computations within mobile cloud and edge architectures.
- To investigate decentralized routing challenges in Mobile Ad Hoc Networks (MANETs) and study mitigation strategies against network-layer attacks using secure routing overlays and IDS.

Course Outcomes

At the end of the course, students should be able to:

- CO1: Analyze security flaws in cellular communication architectures to identify and mitigate threats such as rogue base stations, cloning, and backend signaling attacks.
- CO2: Evaluate wireless network vulnerabilities to secure Wi-Fi and Bluetooth communications
- CO3: Evaluate the secure mobility frameworks utilizing Mobile IP and IPSec.
- CO4: Apply secure transaction processing and data anonymization techniques for mobile computing operations in untrusted cloud, fog, and edge ecosystems.
- CO5: Design cryptographic routing overlays and distributed Intrusion Detection Systems (IDS) to defend MANETs against network-layer attacks.

Syllabus

Introduction to threat models of mobile telephony: and security design considerations for GSM air-interface, channel structure, and location management (HLR-VLR). Security flaws in GSM, cloning, and rogue base stations (IMSI Catchers/Stingrays). Cellular Security-Hierarchical location management and handoffs. Evolution of authentication: UMTS/3G Mutual Authentication, AKA protocol), 4G LTE security architecture, and introduction to 5G network slicing security, Exploitation and security mechanisms in backend routing protocols (SS7 and Diameter signaling threats)

Wireless LAN and Personal Area Networks Security: MAC layer issues, IEEE 802.11 architecture, and evolution of Wi-Fi security (WEP and WPA vulnerabilities). Bluetooth architecture, pairing protocols, security modes, and exploitation techniques.

Secure Mobile Transport and Network Protocols: Securing Mobile IP tunnels using IPSec and route optimization, Traditional TCP vs. Classical TCP over wireless, mitigating TCP session hijacking. WAP architecture, WTLS (Wireless Transport Layer Security), and secure WAP application deployment.

Secure Mobile Data Management: Security and fault tolerance during disconnected operations, secure transaction processing, and trusting Mobile Agents in untrusted environments. Mobile Cloud architecture models (Public, Private, Hybrid) and secure resource scheduling. Identity and access management in mobile-cloud ecosystems. Introduction to MapReduce for mobile data processing, data anonymization in the cloud, and privacy-preserving computation in Edge/Fog computing architectures.

Security in Mobile Ad Hoc Networks (MANETs): Ad hoc network architectures, localization challenges, and specific MAC issues in decentralized environments. Ad Hoc Routing Protocols and attacks- Threat models and network-layer attacks including Blackhole, Grayhole, Wormhole, Sybil, and Sinkhole attacks . Secure Routing and QoS-Cryptographic routing overlays (e.g., Secure AODV/SAODV), trust-based routing metrics, distributed Intrusion Detection Systems (IDS) for MANETs, and secure Quality of Service (QoS).

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	3	2				1
CO2	2	3				2
CO3		3	1			2
CO4	2	3		1		2
CO5		3	2		1	3

Textbooks

1. Patrick Traynor, Patrick McDaniel, and Thomas La Porta. Security for Telecommunications Networks,1st ed, Springer Publishing Company, 2010
2. Jochen Schiller. Mobile Communications, 2nd ed, Imprint Pearson Education, 2008
3. Dominic Chell, Tyrone Erasmus, Shaun Colley, Ollie Whitehouse. The Mobile Application Hacker's Handbook, John Wiley & Sons, 2015

References

1. Nikolay Elenkov. Android Security Internals: An In-Depth Guide to Android's Security Architecture,1st ed, No Starch Press, USA, 2014
2. Erdal Çayirci MS, PhD,, Chunming Rong PhD. Security in Wireless Ad Hoc and Sensor Networks, John Wiley & Sons, Ltd, 2009.
3. Stephen Fried. Mobile Device Security: A Comprehensive Guide to Securing Your Information in a Moving World, 1st ed, Auerbach Publications, USA, 2010

Open Electives

CYO001 Financial Cyber Crimes and Fraud Analytics [2-1-0-3]

Course Prerequisites

None

Course Description

This course introduces the concepts, techniques, and challenges associated with financial cybercrimes, digital frauds, and anti money laundering practices in modern financial ecosystems. It covers fraud analytics, network based investigation techniques, phishing and social engineering attacks, and fraud prevention using data driven approaches. The course also explores cryptocurrency related crimes, blockchain security issues, digital forensics for financial investigations, and emerging threats in digital financial systems. Emphasis is placed on real time fraud detection, regulatory compliance and ethical/legal considerations.

Course Objectives

- To understand the nature and ecosystem of financial cybercrimes and digital frauds.
- To study anti money laundering regulations, suspicious transaction monitoring, and financial intelligence frameworks.
- To introduce fraud analytics including anomaly detection, behavioural analysis, and real time risk scoring.
- To explore phishing, social engineering, and fraud prevention mechanisms in digital financial systems.
- To understand cryptocurrency related crimes, wallet forensics, and security issues in digital financial ecosystems.
- To apply basic digital forensics and log analysis to financial crime case studies.

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Course Outcomes

At the end of the course, students should be able to:

- CO1: Understand major categories of financial cybercrimes and digital fraud schemes.
- CO2:Understand anti money laundering frameworks and suspicious transaction monitoring approaches.
- CO3: Apply fraud analytics and graph based techniques to identify suspicious financial activities.
- CO4: Analyze phishing, social engineering attacks, and fraud prevention strategies.
- CO5:Understand cryptocurrency related frauds and perform basic blockchain transaction tracing.
- CO6:Identify legal, regulatory, and ethical boundaries in financial fraud investigation.

Syllabus

Foundations of Financial Cyber Crimes & AML Essentials : Overview of financial cybercrimes and digital fraud ecosystems: Banking frauds, payment frauds, identity theft, insider threats, online scams. Money laundering concepts: stages, cyber enabled laundering. AML regulations: RBI guidelines, FATF recommendations, suspicious transaction reporting (STR), and Financial Intelligence Units (FIU). Red flag indicators and rule based transaction monitoring.

Fraud Analytics – Detection & Real Time Systems : Fraud analytics concepts: anomaly detection, behavioural profiling, transaction pattern analysis, and risk scoring. Rule-based and data-driven approaches for real-time fraud detection. Dashboard and reporting concepts.

Network Based Investigation & Graph Analytics : Introduction to digital forensics for financial crimes: log analysis, evidence handling, and case documentation. Graph analytics, link analysis, and graph-based investigation concepts for fraud detection. Network centric fraud patterns – money mules, circular transactions, collusive cliques. Applications in banking, UPI/digital payments, and e commerce platforms.

Social Engineering, Phishing & Fraud Prevention: Social engineering principles – authority, urgency, familiarity, and attack lifecycle. Phishing variants – spear phishing, vishing, smishing, business email compromise, account takeover, AI-enabled frauds and impersonation attacks. Deception strategies and fraud prevention frameworks – multi factor authentication, user awareness, and incident response basics. Regulatory and ethical considerations – privacy laws, customer due diligence, and ethical boundaries in fraud monitoring.

Cryptocurrency Crimes & Digital Ecosystem Security: Introduction to cryptocurrencies, blockchain structure, wallets, and exchanges. Cryptocurrency frauds – Ponzi schemes, fake ICOs, rug pulls, ransomware payments, and wallet attacks. Blockchain forensics basics – transaction tracing using public ledgers (Bitcoin/Ethereum), clustering heuristics, and mixing services. Security risks in digital assets and online financial platforms – exchange hacks, DeFi risks.Trust, privacy, and disclosure issues in digital financial ecosystems.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	1				1	
CO2				3		
CO3		2	3		2	
CO4		2	2		2	
CO5			3		3	2
CO6				3		

Textbooks / References

1. Tracy L. Coenen, Expert Fraud Investigation: A Step-by-Step Guide, 2020.
2. George A. Manning, Financial Investigation and Forensic Accounting, Routledge, 2010.
3. Delena D. Spann, Forensic Analytics: Methods and Techniques for Forensic Accounting Investigations, Wiley, 2020.

References

1. Arun Srivastava and Andrew Keltie, A Practitioner's Guide to the Law and Regulation of Financial Crime, 2010.
2. Markus Jakobsson and Steven Myers (Eds.), Phishing and Countermeasures: Understanding the Increasing Problem of Electronic Identity Theft, Wiley, 2006.
3. Yuval Elovici and Yaniv Altshuler, Security and Privacy in Social Networks, Springer, 2019.
4. Frunza, Marius-Cristian. Solving modern crime in financial markets: Analytics and case studies. Academic Press, Elsevier, 2015.
5. Scharfman, Jason. The cryptocurrency and digital asset fraud casebook. Palgrave Macmillan, 2023.
6. Cong, Lin William, et al. "Blockchain forensics and crypto-related cybercrimes." Available at SSRN 4358561 (2023).

CYO002 Malware Analysis and Reverse Engineering [3-0-0-3]

Course Prerequisites

Student should have a basic understanding about Operating Systems (CSC214) or the instructor's approval.

Course Description

This course introduces the concepts related to malwares and its characteristics. It also provides fundamental understanding of the malware analysis process and its importance in the defensive, offensive and forensic domains of cyber security. Through this course students can gain knowledge about the malware characteristics and tools used for analysis and also how to debug, dissect a malware with reversing. In addition to this, students can understand the applicability of AI/ML in the context of malware analysis.

Course Objectives

- To learn to architect secure, multi-OS isolated analysis labs to contain sophisticated threats during execution.
- To gain a thorough understanding of x86, x64, and ARM architectures and how they translate to machine-level instructions.
- To familiarize the methodologies for reverse engineering proprietary and malicious software across Windows, Linux, and Mobile platforms.
- To identify and bypass complex obfuscation, anti-debugging, and anti-VM triggers used by high-level threat actors.
- To explore the application of machine learning and deep learning for automated malware classification and large-scale detection.
- Learn to document Indicators of Compromise (IoCs) and TTPs to support enterprise incident response teams

Course Outcomes

At the end of the course, students should be able to:

- CO1: Set up analysis environemts with multi-AV scanners and sandboxes.
- CO2:Conduct static triage and dynamic monitoring to map malware behavior.
- CO3: Understand assembly coding and analyze code for x86, x64, and ARM systems to identify malicious logic
- CO4: Understand the decompilation and analysis process of bytecode for various applications
- CO5:Understand and evaluate ML and DL-based malware detectors using extracted behavioral features.
- CO6: Apply the knowledge of malware development to create custom detection scripts or rules for proactive hunting.

Syllabus

Introduction: Types of Malware, Malware Functionalities and Persistence,Code Injection and and Hooking, Malware Obfuscation Techniques, Malware Classification Methods, Setting up the environment.
Basic analysis: Assembly language and Disassembly, Debugging malicious binaries, Sandboxes and Multi AV Scanners.
Reverse Engineering Foundations, comparing x86, x64, ARM, windows and linux fundamentals, Applied Reversing, Beyond the Documentation, Deciphering File Formats, Auditing Program Binaries,Reversing malware, Mo-

bile and linux malware analysis. The role of AI/ML in malware analysis, ML based malware detectors, Build and evaluate malware detectors, Application of deep learning in malware detection and analysis, Malware development.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1			3		3	
CO2			3		3	
CO3	2		3		3	
CO4			3		3	
CO5		2	3		2	2
CO6		3	3		3	

Textbooks

1. Monnappa, K. A. (2018). Learning Malware Analysis: Explore the concepts, tools, and techniques to analyze and investigate Windows malware. Packt Publishing.
2. Sikorski, M., & Honig, A. (2012). Practical Malware Analysis: The Hands-On Guide to Dissecting Malicious Software. No Starch Press
3. Joshua Saxe and Hillary Sanders. 2018. Malware Data Science: Attack Detection and Attribution. No Starch Press, USA.

References

1. Ligh, M., Adair, S., Hartstein, B., & Richard, M. (2010). Malware Analyst's Cookbook and DVD: Tools and Techniques for Fighting Malicious Code. Wiley Publishing. ISBN: 978-0-470-61303-
2. Dang, B., Gazet, A., Bachaalany, E., & Josse, S. (2014). Practical reverse engineering: x86, x64, ARM, windows kernel, reversing tools, and obfuscation. John Wiley & Sons
3. Eilam, E. (2005). Reversing: Secrets of reverse engineering. John Wiley & Sons.
4. Andriesse, D. (2018). Practical Binary Analysis: Build Your Own Linux Tools for Binary Instrumentation, Analysis, and Disassembly. No Starch Press.
5. Han, Q., Mandujano, S., Porst, S., Subrahmanian, V. S., & Tetali, S. D. (2023). The Android Malware Handbook: Detection and Analysis by Human and Machine. No Starch Press

CYO003 Dark Web Intelligence and Investigation [3-0-0-3]

Course Prerequisites

Student should have a basic understanding about Operating Systems (CSC214), Computer Networks (CSC211) or the instructor's approval.

Course Description

This course introduces the technologies, ecosystems, and investigative methodologies associated with the dark web and anonymous communication platforms. The course covers anonymising networks, cybercriminal marketplaces, cryptocurrency-enabled underground economies,cyber threat intelligence, digital forensics, and investigation techniques relevant to dark web environments. It also explores operational security, legal, ethical, and privacy considerations associated with dark web intelligence and cybercrime investigations.

Course Objectives

- To understand the architecture and operational principles of anonymous communication networks and dark web ecosystems.
- To study cybercriminal activities, underground marketplaces, and cryptocurrency-enabled transactions in dark web environments.
- To explore dark web intelligence collection, and cyber threat monitoring techniques.
- To analyze digital investigation methodologies, blockchain forensics, and digital forensic concepts relevant to dark web investigations.
- To understand legal, ethical, privacy, and operational challenges associated with dark web intelligence operations.

Course Outcomes

At the end of the course, students should be able to:

- CO1: Understand the technologies and operational principles underlying dark web platforms and anonymous communication systems.
- CO2:Analyze cybercriminal ecosystems, underground marketplaces, and illicit online economies.
- CO3: Apply dark web intelligence collection concepts in cyber threat investigations.
- CO4: Understand cryptocurrency tracing, digital forensic analysis, and investigation concepts in dark web environments.
- CO5: Evaluate legal, ethical, operational security, and privacy considerations associated with dark web investigations.

Syllabus

Foundations of the Dark Web and Anonymous Networks: Surface web, deep web, and dark web concepts. Evolution of anonymous communication systems and dark web ecosystems. Onion routing, hidden services, anonymous browsing, and anonymity concepts. Introduction to Tor, I2P, and decentralized anonymous platforms. Operational security considerations in dark web environments.

Cybercrime Ecosystems and Dark Web Intelligence : Dark web marketplaces, underground forums, and cybercriminal communities. Cybercrime business models including ransomware-as-a-service, malware services, credential markets, cyber fraud, and illicit trade ecosystems. Intelligence lifecycle, OSINT methodologies, dark web intelligence collection, threat monitoring, persona analysis, and cross-platform identity correlation concepts. Cryptocurrency Forensics and Investigation Techniques : Blockchain fundamentals and cryptocurrency transaction analysis concepts. Cryptocurrency-enabled underground economies, transaction tracing, address clustering, and blockchain analysis concepts. Privacy-enhancing cryptocurrencies, obfuscation mechanisms, laundering techniques, deanonymization approaches, and link analysis concepts.

Digital Forensics in Dark Web Investigations: Digital evidence acquisition and preservation in dark web investigations. Browser artefacts, network traces, cryptocurrency wallet artefacts, and forensic analysis concepts. Investigation methodologies for dark web-enabled cybercrimes. Evidence integrity, chain-of-custody, and operational considerations in cyber investigations.

Legal, Ethical, and Emerging Challenges: Legal and jurisdictional challenges in dark web investigations. Privacy, anonymity, lawful interception, and ethical considerations in cyber intelligence operations. Undercover investigations, responsible disclosure, and operational risks. Emerging trends in dark web intelligence, decentralized cybercrime ecosystems, and AI-assisted cyber threat investigations.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	2				2	
CO2			2		3	
CO3			3		3	
CO4			3	3	2	

Textbooks

1. Akhgar, Babak, et al., eds. Dark web investigation. Berlin: Springer, 2021.
2. Sario, Azhar Ul Haque. Dark Web Intelligence and OSINT Techniques. LAP Lambert Academic Publishing, 2025.
3. Prasad, Prakash. Cryptocurrency Forensics and Investigation Using Open Source Intelligence Techniques (OSINT): Volume II. CRC Press, 2026.

References

1. Bartlett, Jamie. The dark net: Inside the digital underworld. Melville House, 2015.
2. Greenberg, Andy. Tracers in the dark: The global hunt for the crime lords of cryptocurrency. Vintage, 2024.
3. Furneaux, Nick. Investigating cryptocurrencies: understanding, extracting, and analyzing blockchain evidence. John Wiley & Sons, 2018.
4. Meiklejohn, Sarah, et al. "A fistful of bitcoins: characterizing payments among men with no names." Proceedings of the 2013 conference on Internet measurement conference. 2013.
5. Dingledine, Roger, Nick Mathewson, and Paul Syverson. "Tor: The Second-Generation Onion Router." SSYM'04: Proceedings of the 13th Conference on USENIX Security Symposium, vol. 13, 2004, p. 21. USENIX
6. Chen, Hsinchun. Dark Web: Exploring and Data Mining the Dark Side of the Web. Springer Science+Business Media, 2012.

CYO004 Web Application Security [3-0-0-3]

Course Prerequisites

Student should have a passing grade in Offensive Security and Penetration Testing(CYC321) or the instructor's approval

Course Description

This course provides a comprehensive study of web application security architectures and the methodologies for identifying vulnerabilities within modern web environments. It covers the complete assessment lifecycle, including protocol analysis, systematic reconnaissance, and the exploitation of both server-side and client-side flaws. Students will evaluate advanced attack vectors such as LLM-based threats and request smuggling, culminating in the development of industry-standard security reports.

Course Objectives

- To understand the architectural foundations of web applications and the industry-standard OWASP testing frameworks.
- To master systematic reconnaissance and enumeration techniques to map complex web attack surfaces.
- To perform technical analysis and exploitation of server-side and client-side vulnerabilities.
- To analyze emerging web attack vectors and master the art of professional security reporting.

Course Outcomes

At the end of the course, students should be able to:

- CO1: Apply the OWASP testing methodology and interception tools to analyze web traffic and session management.
- CO2: Execute multi-level reconnaissance and enumeration to discover hidden assets and fingerprint web frameworks.
- CO3: Analyze server-side vulnerabilities to validate flaws in authentication, injection points, and resource handling.
- CO4:Evaluate client-side security controls and business logic to identify weaknesses in browser-based environments and APIs.
- CO5:Synthesize modern attack vectors and technical findings into a professional-grade penetration testing report.

Syllabus

Introduction to Web applications: Static Vs Dynamic web applications - Session Management and Cookies - HTTP Methods and Response Codes - OWASP TOP 10 - OWASP Testing Guide - Intercepting web traffic using Proxies

Reconnaissance and Enumeration: DNS Recon - Sub Domain Enumeration - End Point Discovery - Directory Brute forcing - Dictionary Attacks - Detecting the web frameworks - overview of web archives

Server-Side Attacks: Authentication Bypass - Path Traversal - Command Injection - SQL Injections - Access Control Vulnerability - File Upload Vulnerability - Server-Side Request Forgery (SSRF) - Race Conditions - Web Cache Deception and Poisoning

Client-Side Attacks: Cross Site Scripting (XSS) - Cross Site Request Forgery (CSRF) - Click Jacking - Cross Origin Resource Sharing (CORS) Misconfigurations - API Security - Business Logic Attacks - Insecure Deserialization

Modern web attacks and Reporting: Web LLM Attacks - HTTP Request Smuggling - OAuth Authentication - JWT Attacks - Prototype Pollution - Professional Reporting

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1		2	3		3	
CO2			3		3	
CO3			3		3	
CO4			3		3	
CO5			2	3	3	

Textbooks References

1. Samir Kumar Rakshit, Ethical Hacker's Penetration Testing Guide , 2022, ISBN 978-93-55512-154
2. Andrew Hoffman,Web Application Security - Exploitation and Counter Measures for Modern Web Applications , 2nd Edition, OREILLY,2024

References

1. Gus Khawaja, Practical Web Penetration Testing , PACKT Publications, 2018
2. Prakhar Prasad, Mastering Modern Web Penetration Testing , PACKT Publications, 2016

CYO005 Agentic AI for Cyber Security [3-0-0-3]

Course Prerequisites

Student should have a passing grade in Machine Learning and Data Analysis(CAC221) or the instructor's approval.

Course Description

This course provides a comprehensive study of Agentic Artificial Intelligence and its applications in contemporary cybersecurity. It introduces the fundamental concepts, architectures, and operational principles of intelligent agents, including autonomous decision-making, memory management, retrieval-augmented systems, and multi-agent coordination. Emphasis is placed on the design, development, and deployment of AI-driven security solutions using modern agentic frameworks and tools. The course combines theoretical foundations with practical implementation approaches to prepare students for the evolving landscape of AI-enabled cyber defense and autonomous security operations.

Course Objectives

- To provide a comprehensive understanding of Agentic Artificial Intelligence, its architectures, operational principles, and its role in transforming modern cybersecurity practices.
- To develop knowledge of autonomous and multi-agent systems, including decision-making, memory management, retrieval-augmented generation, and agent orchestration techniques.
- To examine the application of AI agents in cyber threat intelligence, security operations, incident response, threat hunting, and real-time cyber defense mechanisms.
- To equip students with the ability to analyze security, privacy, and ethical challenges in agentic systems and design, develop, and deploy secure AI-driven cybersecurity solutions using contemporary agentic frameworks.

Course Outcomes

At the end of the course, students should be able to:

- CO1: Explain the concepts, architectures, and operational characteristics of Agentic AI systems and their significance in modern cybersecurity environments.
- CO2: Apply agentic AI architectures, memory mechanisms, retrieval-augmented generation, and orchestration techniques to design autonomous and multi-agent systems.
- CO3: Analyze the use of AI-driven approaches for cyber threat intelligence, security operations, incident response, and anomaly detection within cyber defense ecosystems.
- CO4 :Evaluate the security, privacy, ethical, and governance challenges associated with autonomous AI agents, including emerging attack vectors and threat modeling methodologies.
- CO5:Design and deploy AI-powered cybersecurity agents using contemporary agentic frameworks to address real-world security challenges.

Syllabus

Introduction to Agentic AI and Cyber Security: Definition and evolution of AI agents - Agents vs. traditional AI and chatbots - Types of AI agents including reactive, deliberative, and hybrid - AI as a cyber security game changer - The cyber threat landscape and hybrid warfare

Foundations of AI Agent Architectures: Core components of agent systems including foundation models, tools, and memory - Single-agent vs. multi-agent architectures - Decision-making and orchestration in autonomous systems - Memory management and Retrieval-Augmented Generation (RAG) - Multi-agent system coordination

AI-Driven Cyber Threat Intelligence and Defense: Enhancing the CIA Triad for confidentiality, integrity, and availability - Cyber Threat Intelligence (CTI) gathering using HUMINT, SIGINT, and OSINT - Automated incident response and threat hunting - Integrating Security Operations Center (SOC) agents - Real-time monitoring and anomaly detection

Security, Privacy, and Threat Modeling for AI Agents: Unique risks of agentic systems including goal misalignment and automation bias - Emerging threat vectors such as prompt injection, jailbreaking, and evasion attacks - Threat modelling with the MAESTRO framework - Securing agents and data privacy - Ethical AI considerations and human-in-the-loop oversight

Building and Deploying Real-World Security Agents: Setting up the Python development environment - Utilizing agentic frameworks like LangGraph, AutoGen, and CrewAI

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	3	2				1
CO2	2	3	1			2
CO3	1	3	3		2	1
CO4		1		3		2
CO5	1	3	2	1	1	3

Textbooks / References

1. Albada, M. (2025). Building Applications with AI Agents: Designing and Implementing Multiagent Systems. O'Reilly Media, Inc.
2. Bhagwat, S. (2025). Principles of Building AI Agents (2nd ed.). Mastra.ai.
3. Van Der Post, H. (2025). AI Agents with Python: Build Autonomous Systems That Think, Learn, and Act. Reactive Publishing.

References

1. Lighthaven, M. (2025). Mastering AI Agents: A Practical Handbook for Understanding, Building, and Leveraging LLM-Powered Autonomous Systems to Automate Tasks, Solve Complex Problems, and Lead the AI Revolution.
2. Nika, M. (2025). Building AI-Powered Products: The Essential Guide to AI and GenAI Product Management. O'Reilly Media, Inc.
3. Tjioe, A., Solihin, D., Mulyono, H., Mangindaan, R., Tobing, R. L., & Prihartanto, H. (2025). AI-Driven Intelligence and Cyber Security: Strategi Stabilitas Keamanan untuk Ketahanan Nasional. UNJ Press.